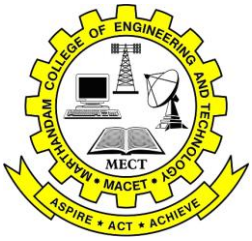
MARTHANDAM COLLEGE OF ENGINEERING & TECHNOLOGY

P.M VILLAGE ROAD, KUTTAKUZHI,
VEEYANNOR-629 177,
K.K.DISTRICT



LABORATORY RECORD

Reg.No.

Name : _____

Semester/Year : _____

Department : _____

Subject : _____

MARTHANDAM COLLEGE OF ENGINEERING & TECHNOLOGY

P.M VILLAGE ROAD, KUTTAKUZHI,
VEEYANNOR-629 177,
K.K.DISTRICT

Certified that this the bonafide record of work done by Mr/Ms - _____ of the _____ Year/Semester in _____of this college, in the _____Laboratory during the year _____ , in partial fulfillment of the requirements of the B.E/B.TECH Degree Course of the Anna University Chennai.

Staff-in-charge **Head of the Department**

University Reg. No:

University Examination held on _____

Internal examiner **External Examiner**

INDEX PAGE
LIST OF EXPERIMENT

Ex. No	Date	Name of Experiment	Page no.	Marks	Signature

EXP.NO:1 DATE:	INSTALL VIRTUALBOX WITH DIFFERENT FLAVOURSOF WINDOWS OS ON TOP OF WINDOWS

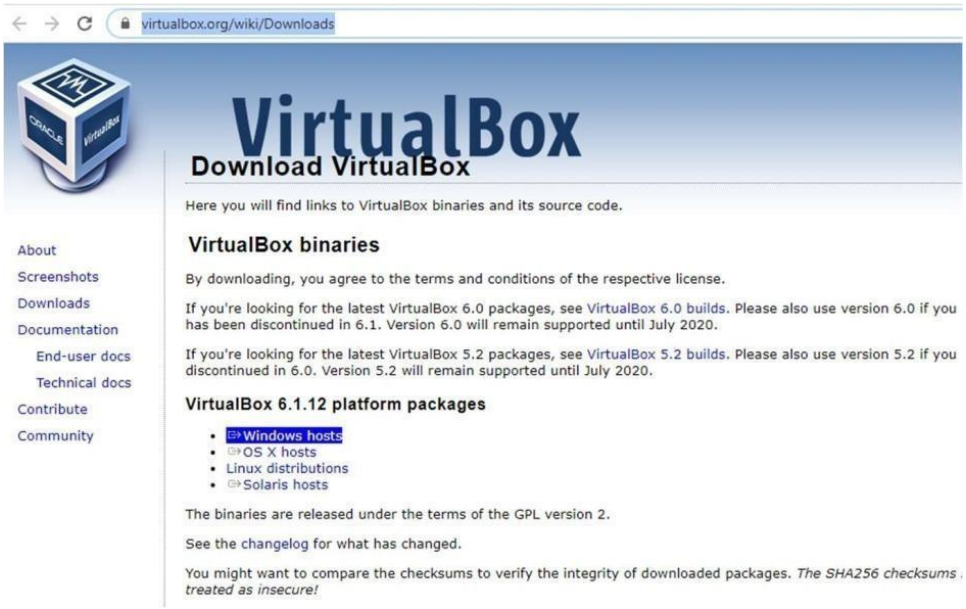
Aim :

To install Virtualbox woith windows OS and create Virtual Machines on it.

PROCEDURE:

Steps to install VirtualBox

1. Download VirtualBox installer for windows
2. The installer can be downloaded from the link (https://www.virtualbox.org/wiki/Downloads)



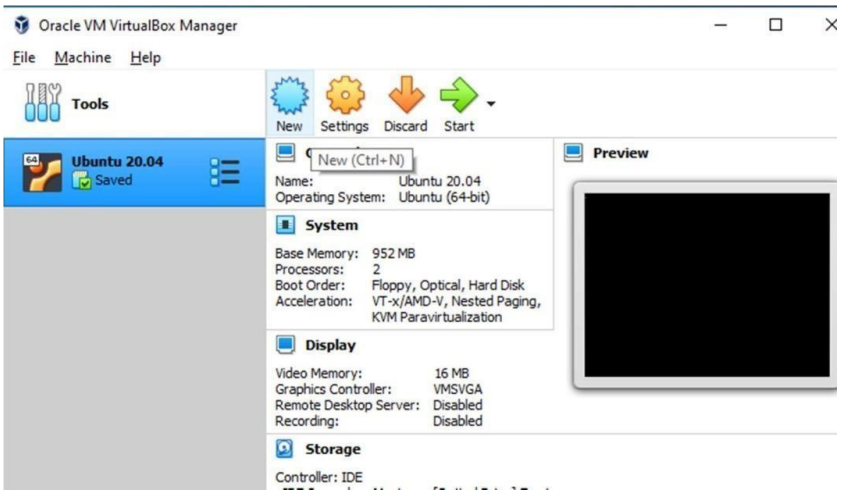
3. Click “Windows host” to download the binary version for windows host
4. The installer file downloaded will have the file name format like VirtualBox- VersionNumber-BuildNumber-Win.exe.
   Example: VirtualBox-6.1.12-139181-Win.exe
5. Double click on the installer to launch the setup Wizard. Click on Next to continue.



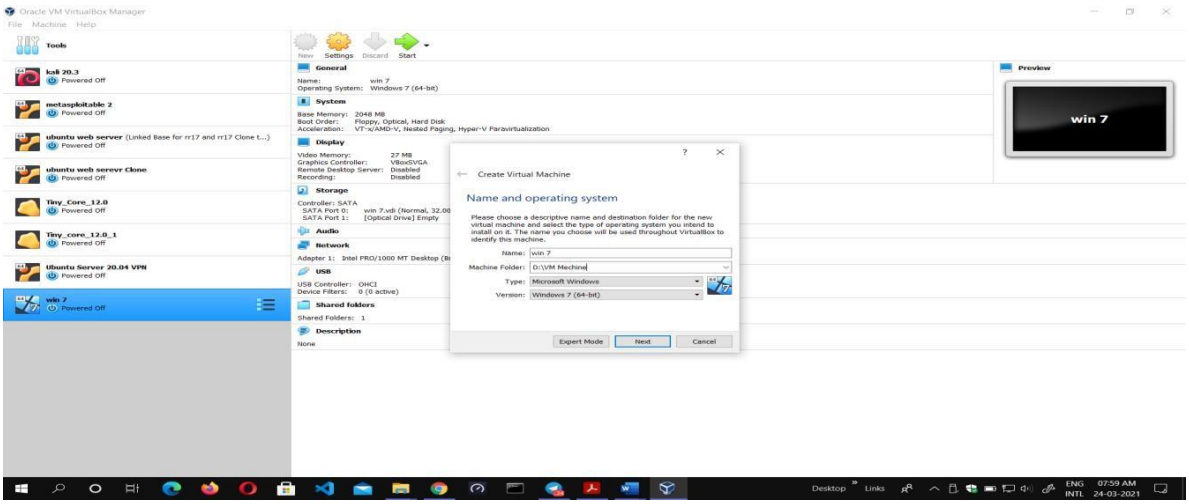
6. Custom setup dialog box will be opened. Accept the default settings and click next
7. Select the way you want the features to be installed. You can accept the default and click next.
8. A dialog box opens with Network Interfaces warning. Click Yes to proceed
9. Click install to begin the installation process
10. When prompted with a message to install (Trust) Oracle Universal Serial Bus, click Install to continue
11. After the installation completes, click finish to exit the setup wizard
12. Launch the Oracle VM VirtualBox

Steps to create a virtual machine [Windows] in VirtualBox

1. Open the Oracle VM VirtualBox
2. Click New icon or „Ctrl + N" to create a new virtual machine

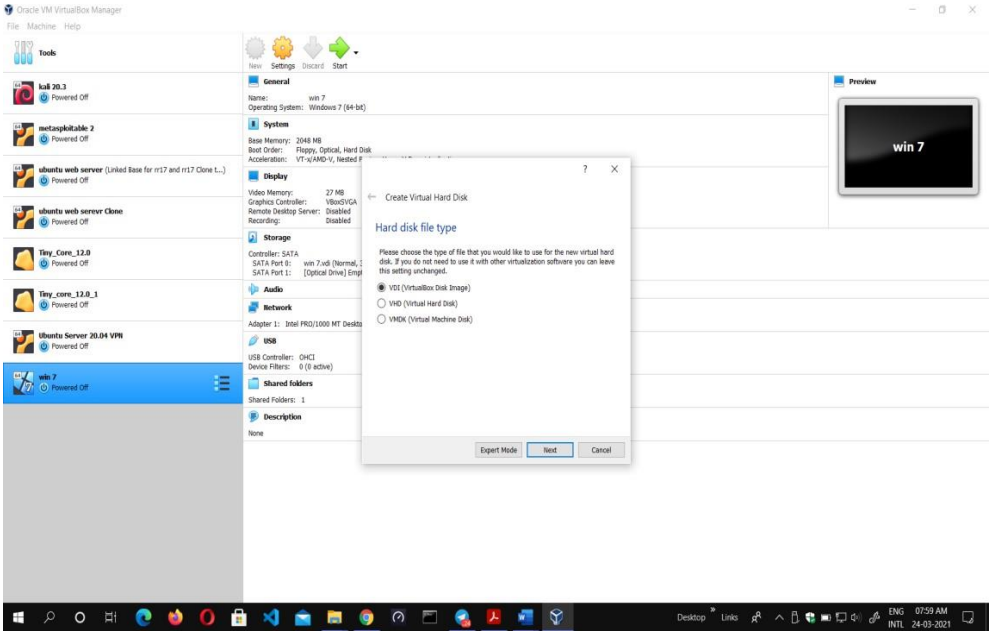


3. Enter a name for the new virtual machine. Choose the Type and Version. Note that VirtualBox automatically changes 'Type' to Linux and 'Version' to 'Windows (64 bit)' if the name is given as „Window". Click Next

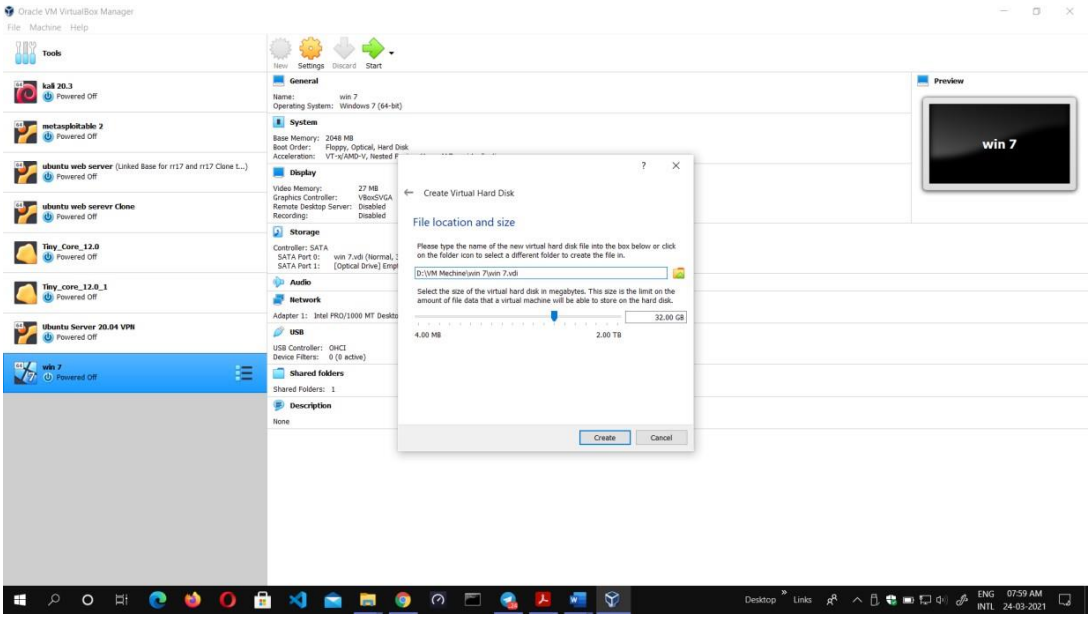


4. Select the amount of RAM to use. The ideal amount of RAM will automatically be selected. Do not increase the RAM into the red section of the slider; keep the slider in the green section
5. Accept the default 'Create a virtual hard drive now' and click 'Create' button.
    - Choose the hard disk file type as VDI (VirtualBox Disk Image). Click Next
6. Click Next to accept the default option „Dynamically allocated" for storage on physical hard drive
7. Select the size of the virtual hard disk and click create
8. The newly created virtual machine will be displayed in the dashboard

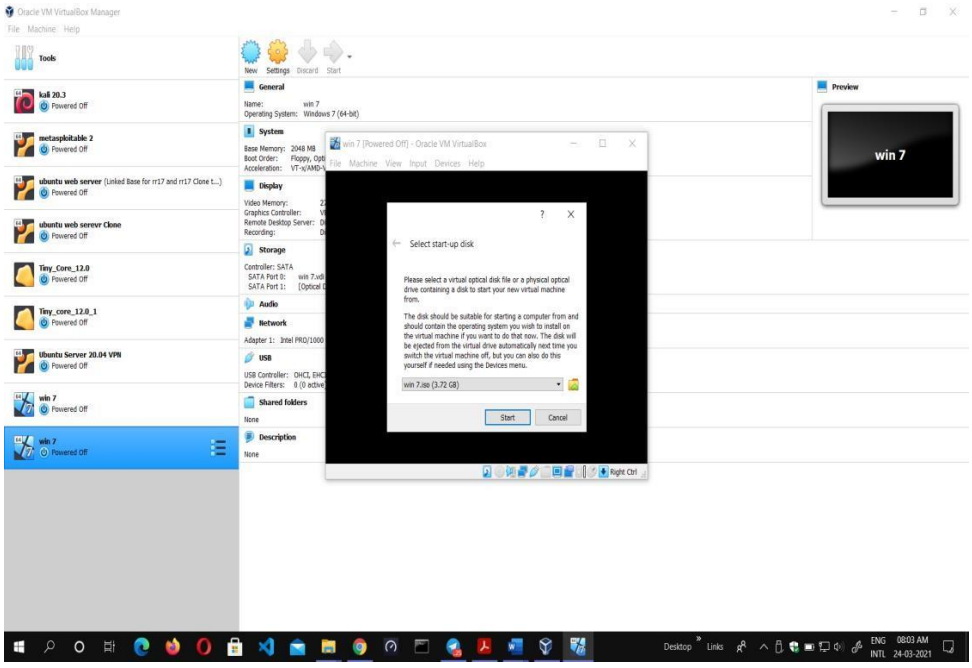
9. Download the ISO file [Window disk image file]. Latest version of Window iso file can be downloaded from the link
10. For previous versions, goto http://releases.ubuntu.com. Choose the preferred version of Windows and download the iso file
11. To setup the Windows disk image file (iso file) goto settings. Click Storage. Under „Storage Devices" section click „Empty
12. In Attributes section, click the disk image and then "Choose Virtual Optical Disk File".



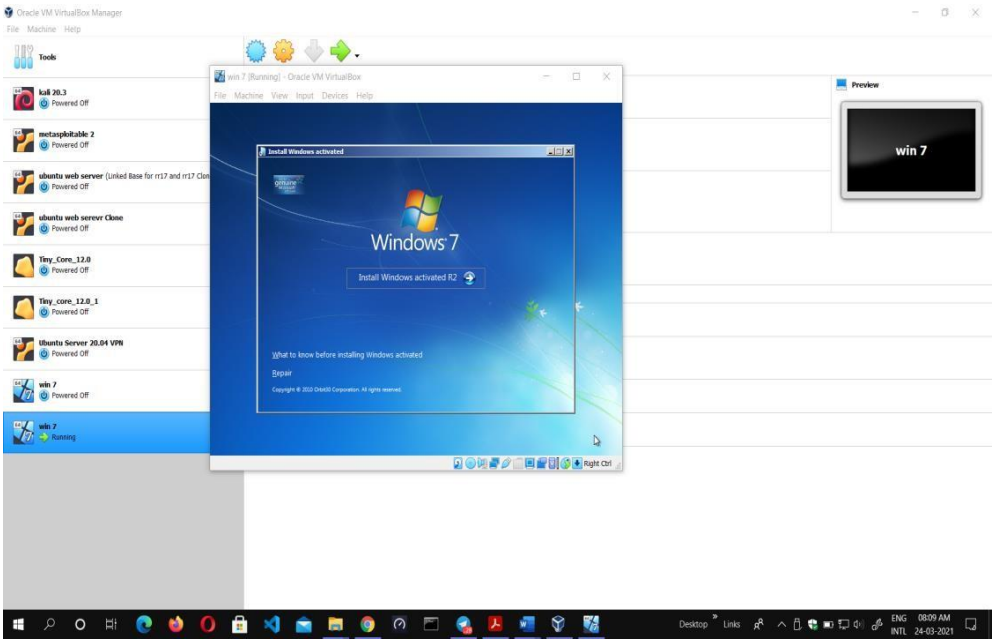
13. Browse and select the downloaded iso file. Click ok.



14. Select the newly created virtual machine in the dashboard and click start button



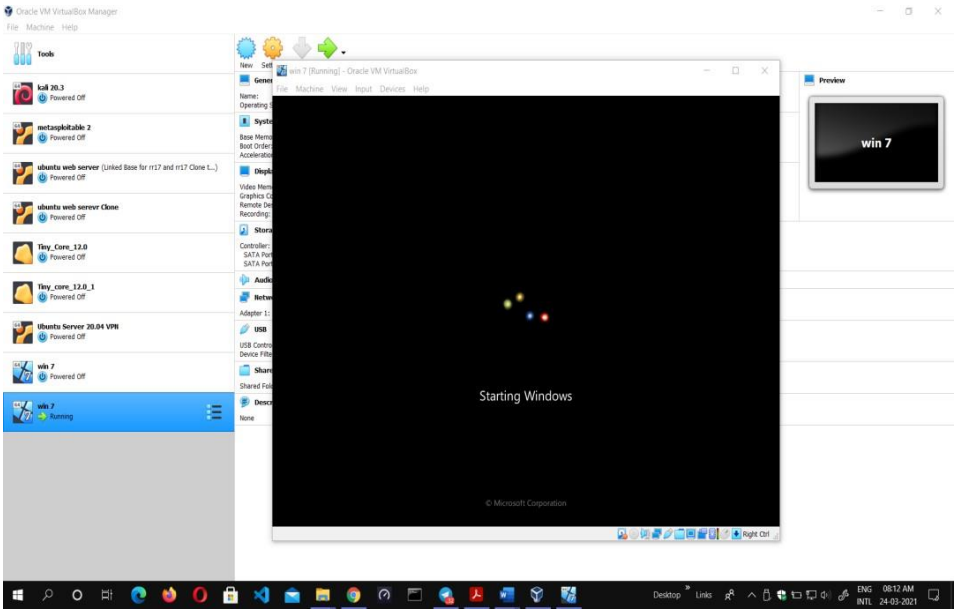
15. In the welcome screen, click „Install Windows‟ button



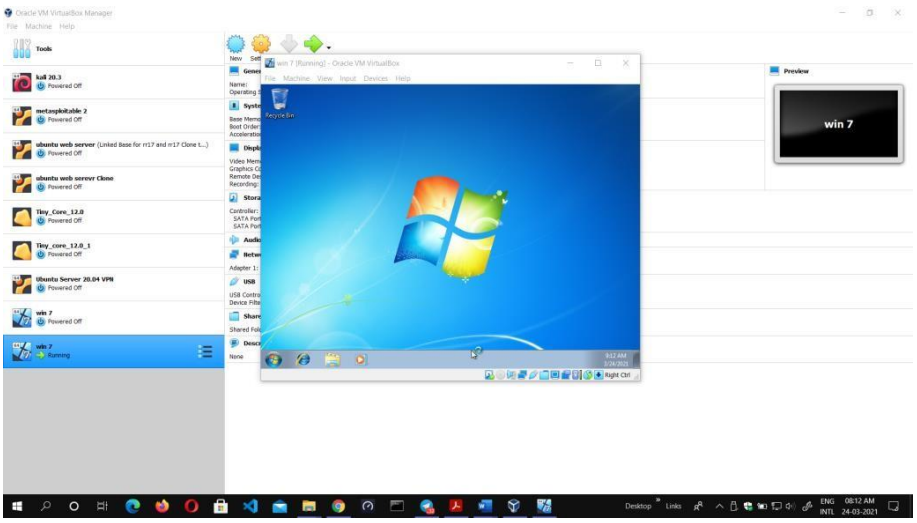
16. Click 'Continue' button
17. Make sure 'Erase disk and install Window' option is selected and click 'Install Now' button
18. Choose the default and click continue
19. Setup up your profile by creating username and password
20. After installation is complete, click 'Restart Now' button and follow the instructions
21. The Window OS is ready to use. Login with the username and password

OUTPUT / SCREENSHOT:

- Virtualbox on top of windows.



- Installation of Virtual box with **Window OS (Guest OS/VM)** on top of windows Host.



RESULT:

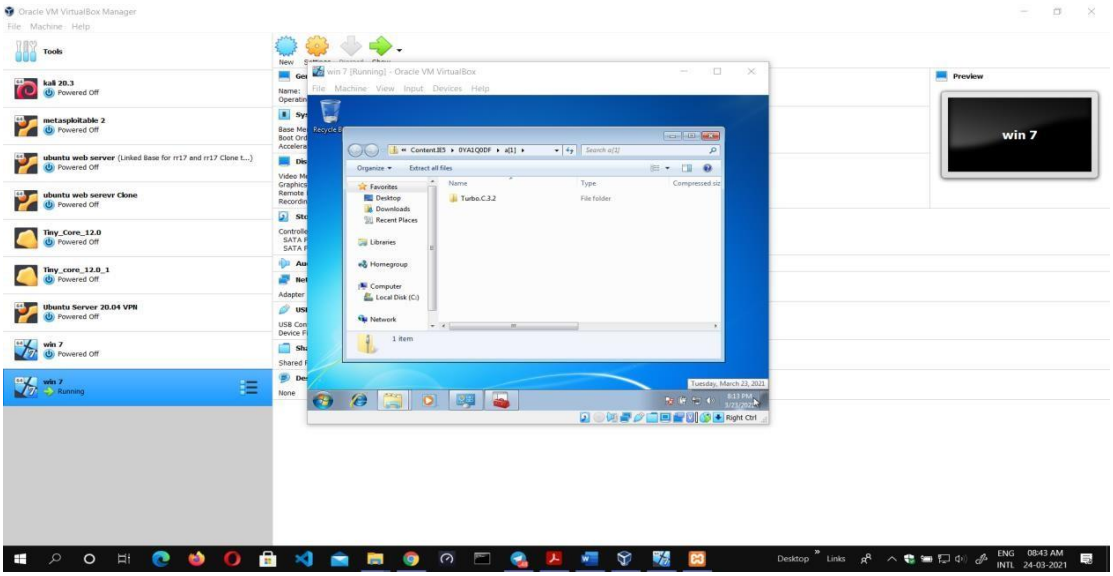
EXP.NO:2 DATE:	INSTALL A C COMPILER IN THE VIRTUAL MACHINE CREATED USING VIRTUAL BOX AND EXECUTE SIMPLE PROGRAMS

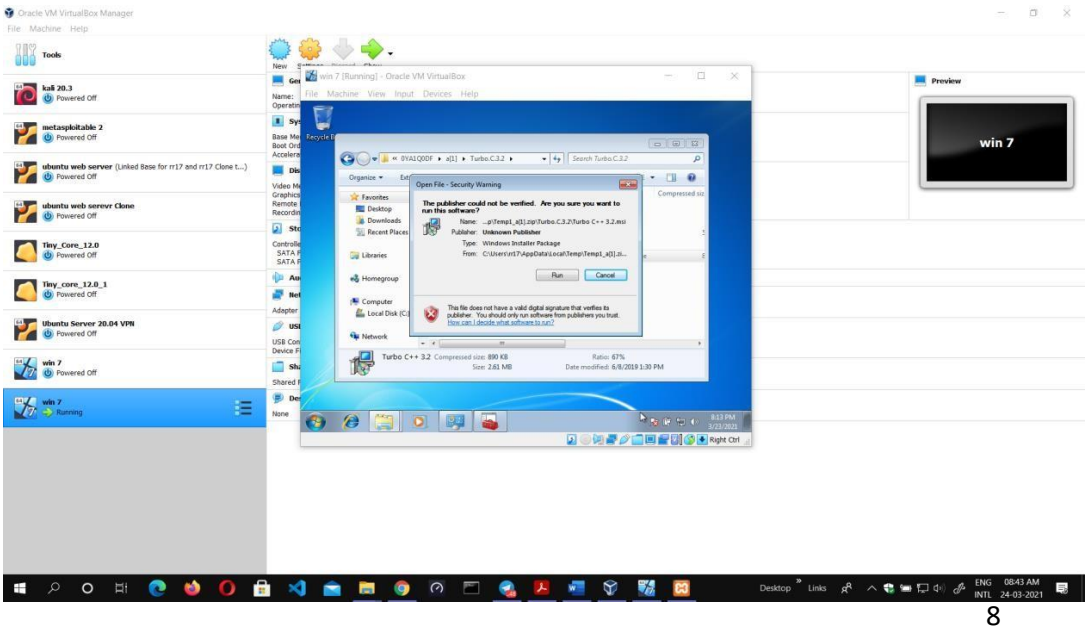
AIM

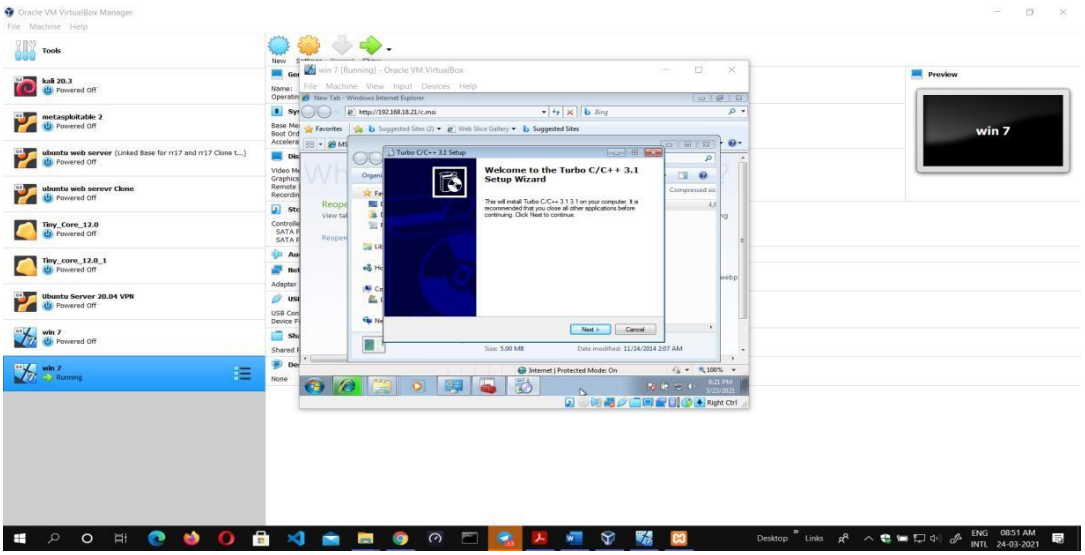
To install a C compiler in the virtual machine and execute a sample program

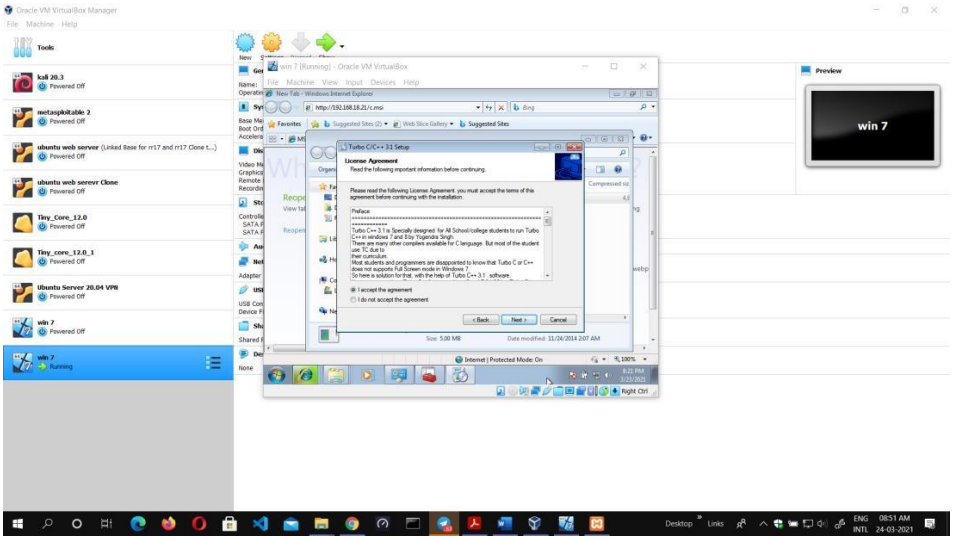
PROCEDURE:

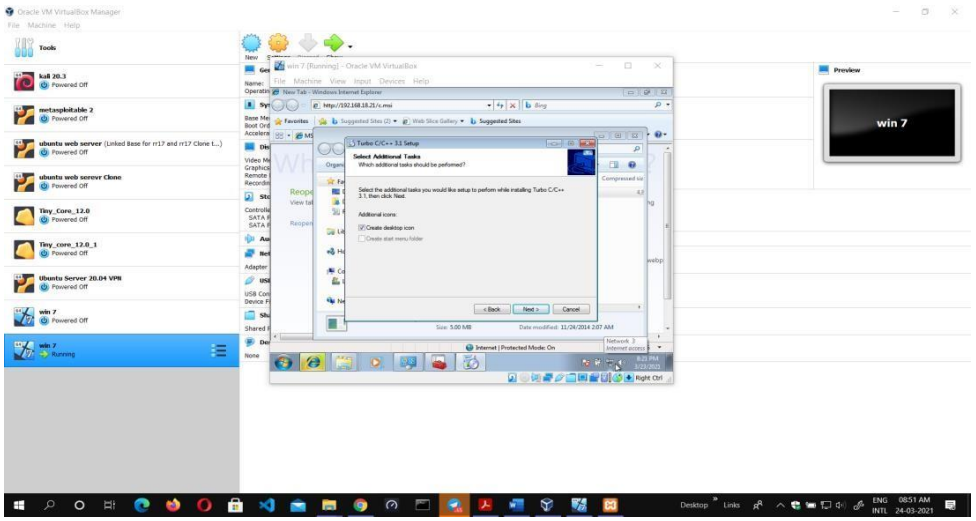
1. Install VirtualBox.
2. Create a Virtual Machine using VirtualBox.
3. Download C Compiler from Web using Virtual Machine / Copy from network.
4. Install the compiler on virtual machine.

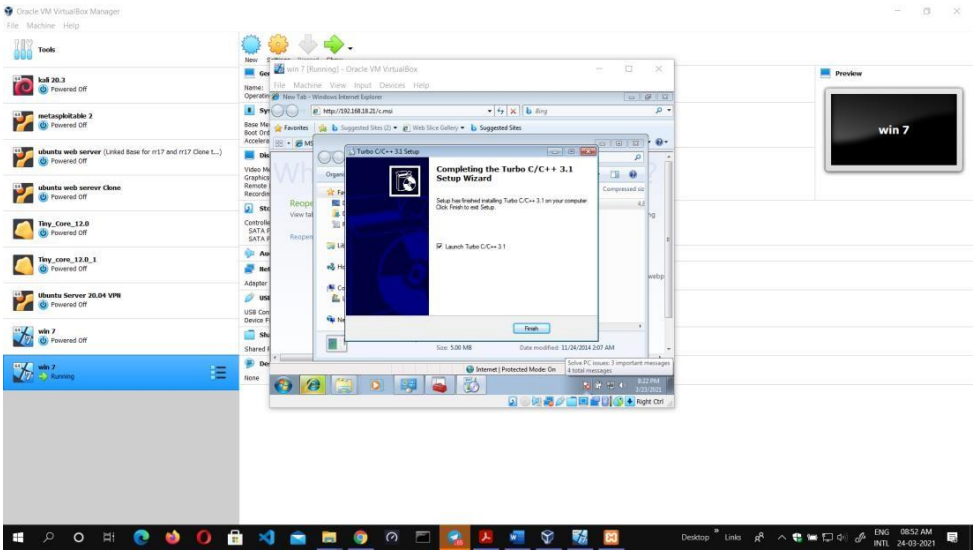





Oracle VM VirtualBox Manager
Tools
kali 20.3
metasploitable 2
ubuntu web server
ubuntu web serevr Clone
Tiny_Core_12.0
Tiny_core_12.0_1
Ubuntu Server 20.04 VPN
win 7
win 7 [Running] - Oracle VM VirtualBox
Turbo C/C++ 3.1 Setup
Welcome to the Turbo C/C++ 3.1 Setup Wizard
Next >
Cancel
Preview
win 7


Oracle VM VirtualBox Manager
Tools
kali 20.3
metasploitable 2
ubuntu web server
ubuntu web serevr Clone
Tiny_Core_12.0
Tiny_core_12.0_1
Ubuntu Server 20.04 VPN
win 7
win 7 [Running] - Oracle VM VirtualBox
Turbo C/C++ 3.1 Setup
License Agreement
I accept the agreement
I do not accept the agreement
< Back
Next >
Cancel
Preview
win 7

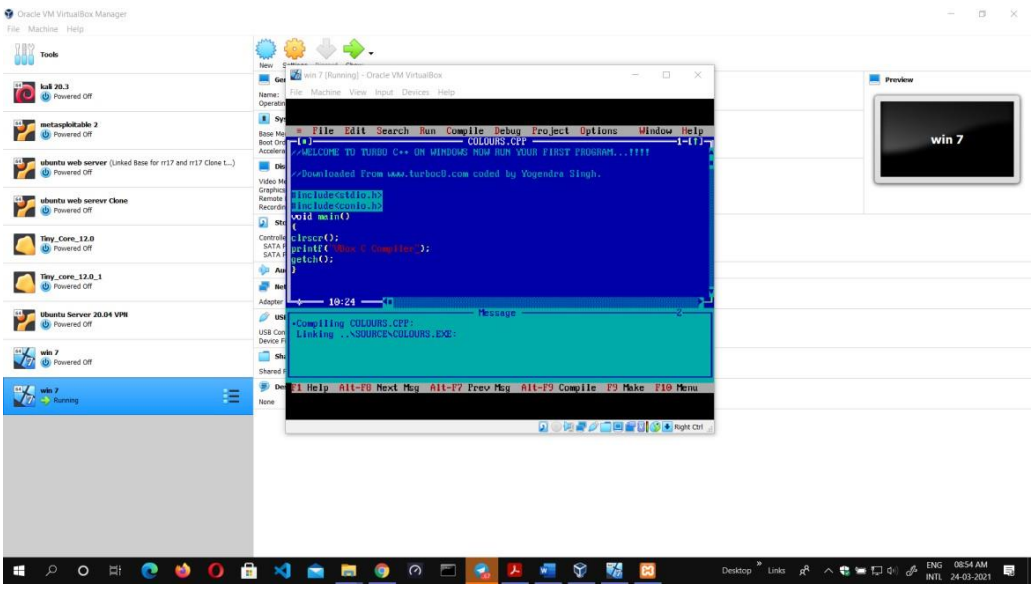




5. Type the C Program

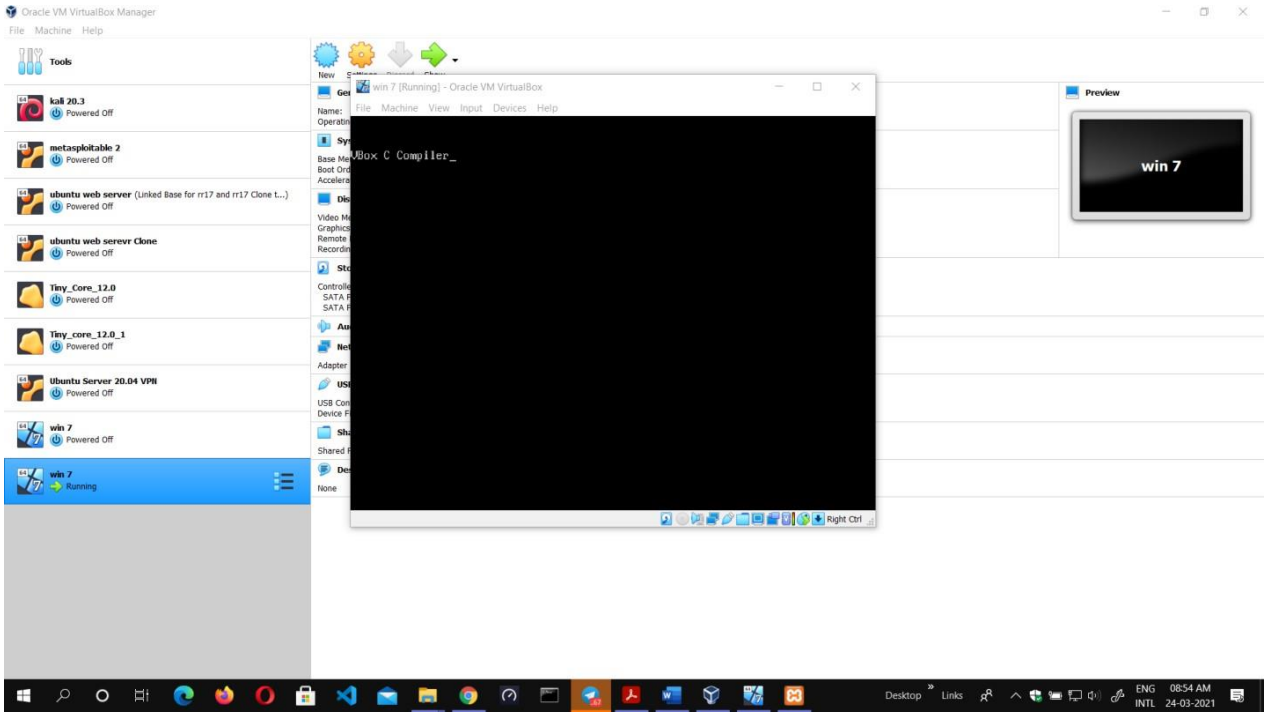
demo.c

```
#include<stdio.h>
#include<conio.h>
void main()
        {
        clrsrc();
        printf("VBox C Compiler");
        getch();
        }
```



6. Execute / Run the program.

OUTPUT / SCREENSHOT:



RESULT:

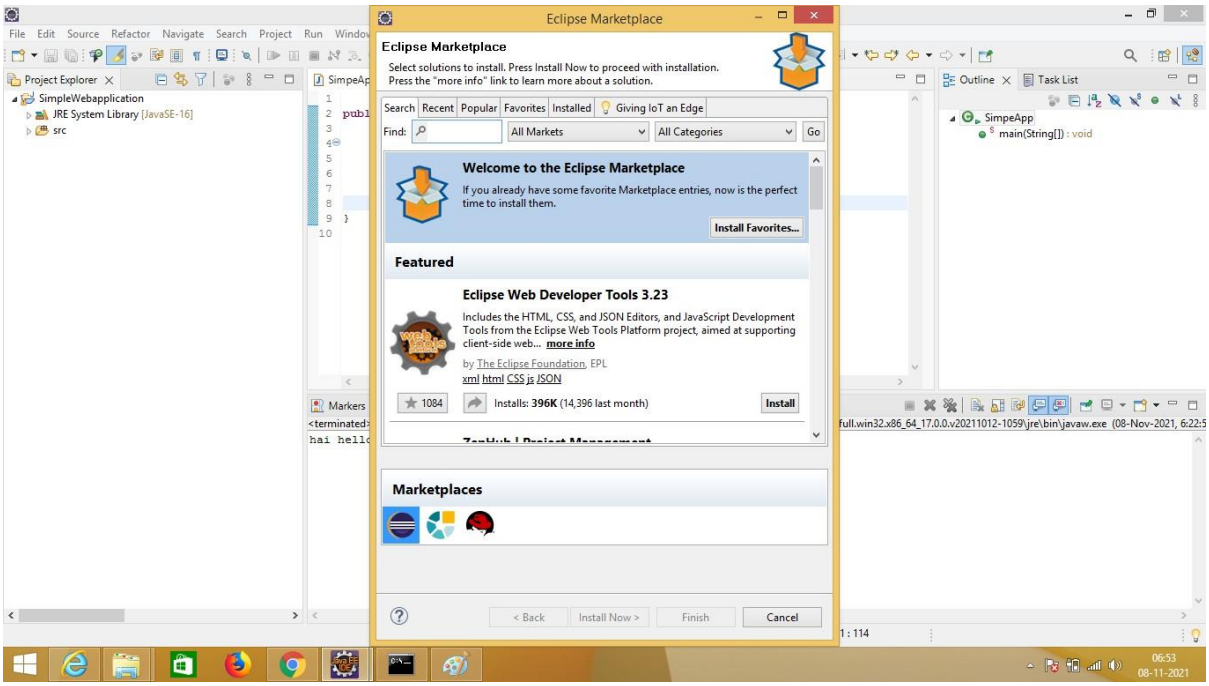
EXP.NO:3 DATE:	INSTALL GOOGLE APP ENGINE AND CREATE SIMPLE WEB APPLICATIONS USING JAVA

AIM

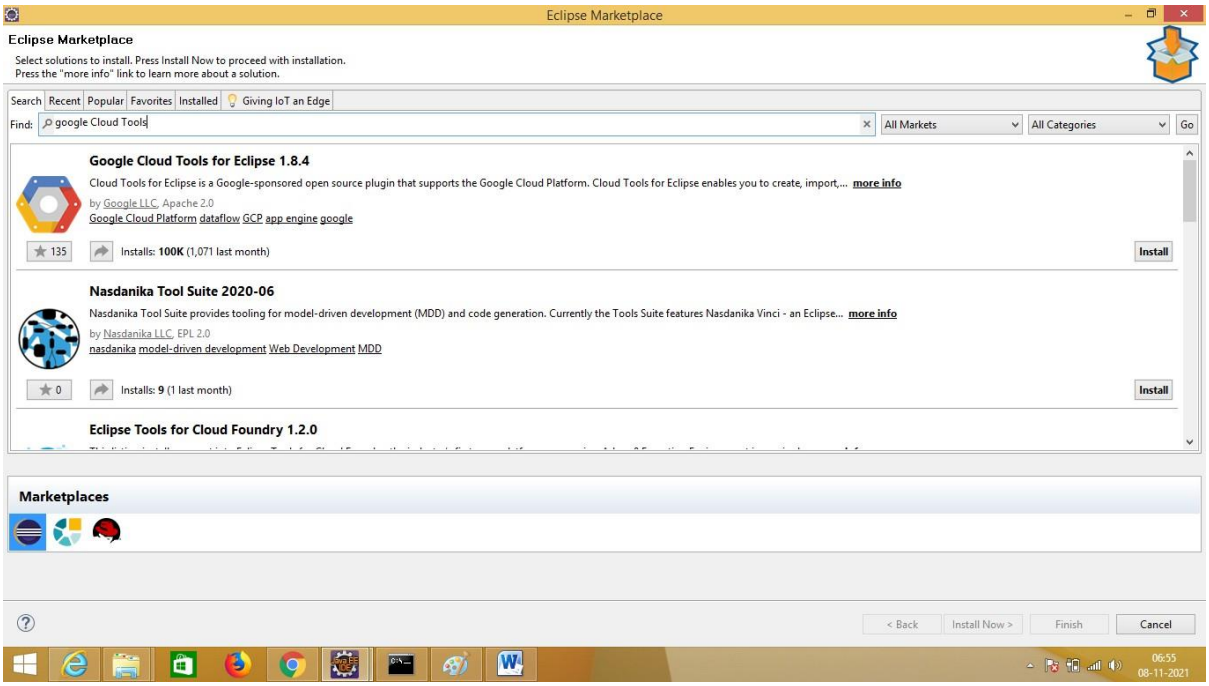
To install Google App Engine and create simple web application using Java.

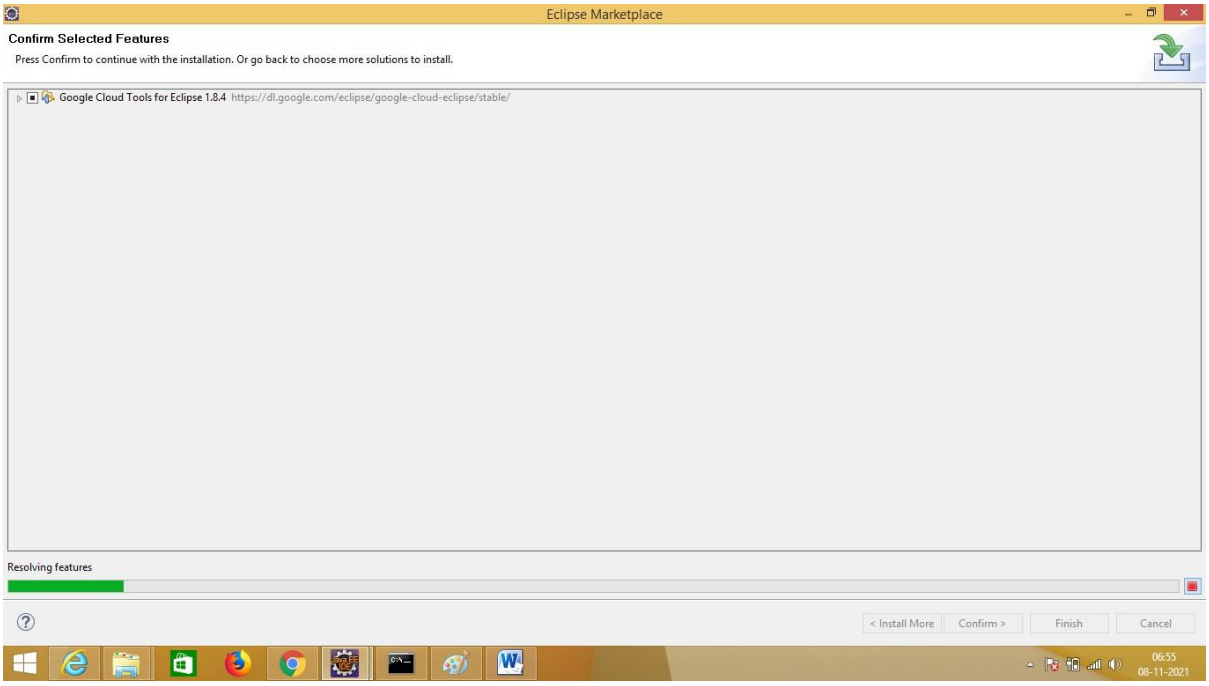
PROCEDURE

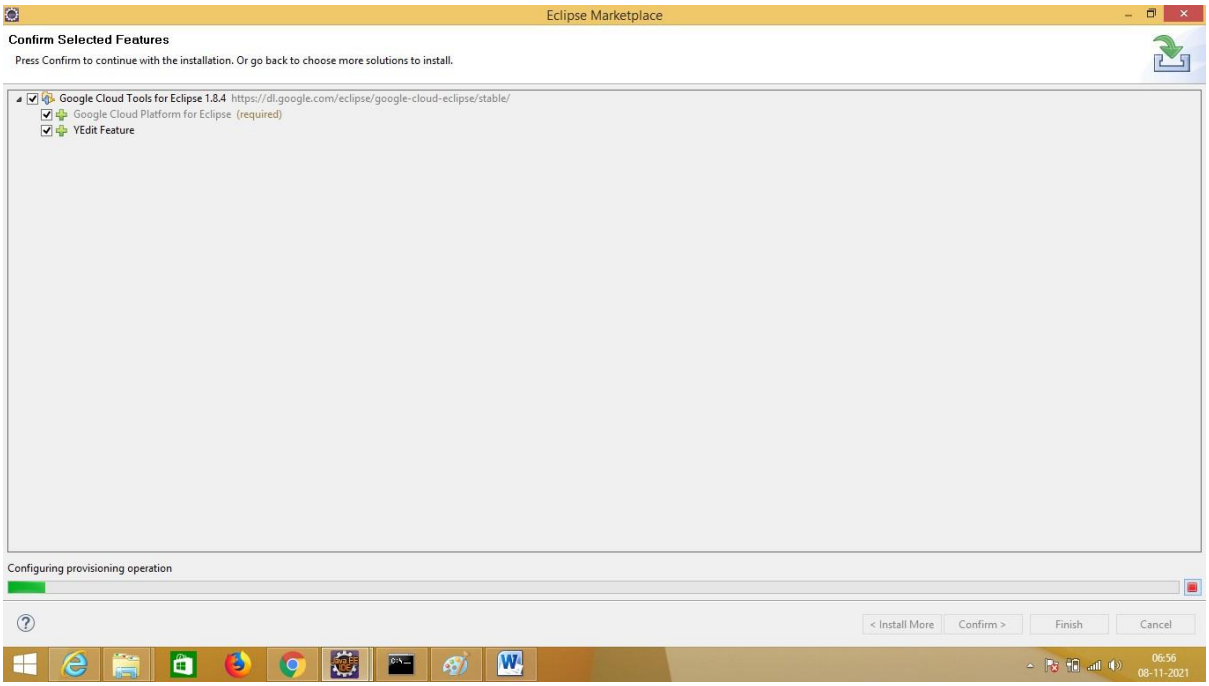
1. Download Java and install.
2. Download Eclipse IDE and install
3. Go to help in Eclipse IDE and click Eclipse Market Place.

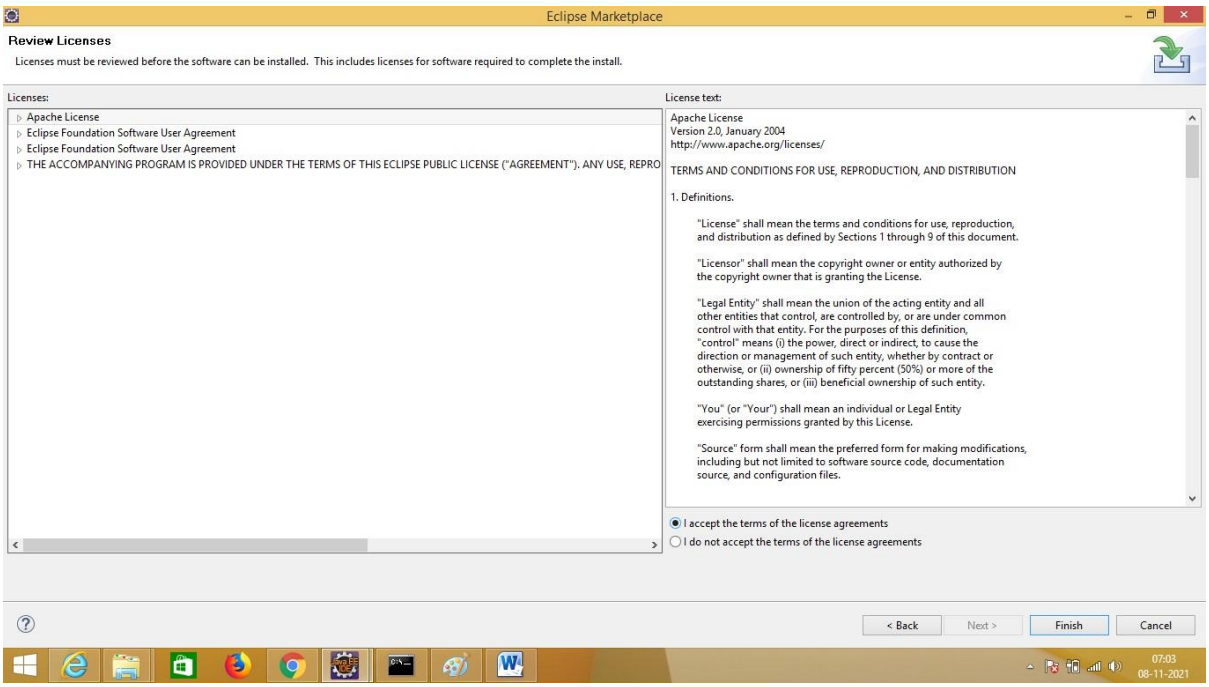


4. Type Google Cloud Tools for Eclipse and click install.

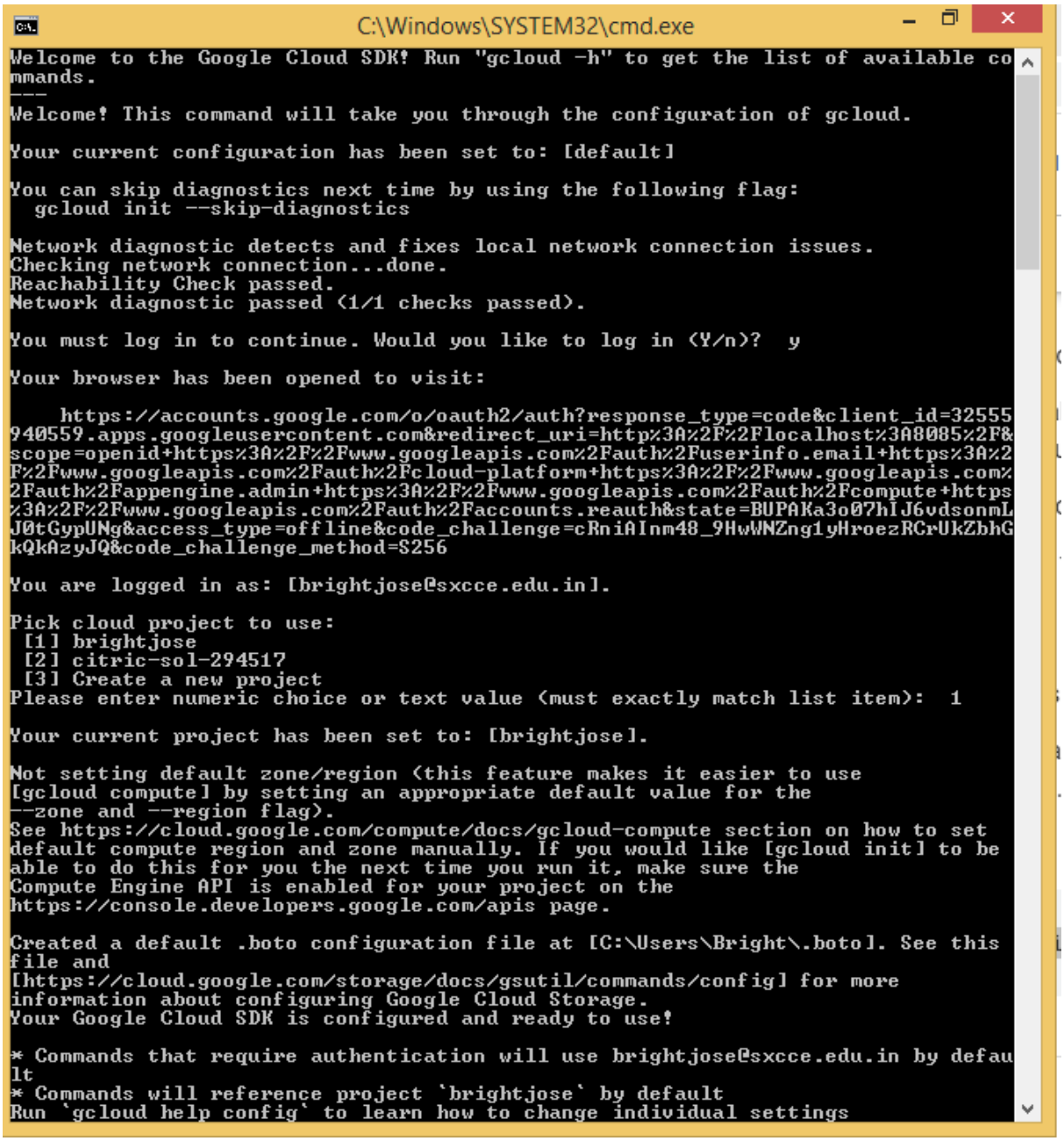

Eclipse Marketplace
Eclipse Marketplace
Select solutions to install. Press Install Now to proceed with installation.
Press the "more info" link to learn more about a solution.
Search
Recent
Popular
Favorites
Installed
Giving IoT an Edge
Find:
google Cloud Tools
All Markets
All Categories
Go
Google Cloud Tools for Eclipse 1.8.4
Cloud Tools for Eclipse is a Google-sponsored open source plugin that supports the Google Cloud Platform. Cloud Tools for Eclipse enables you to create, import,... more info
by Google LLC, Apache 2.0
Google Cloud Platform dataflow GCP app engine google
135
Installs: 100K (1,071 last month)
Install
Nasdanika Tool Suite 2020-06
Nasdanika Tool Suite provides tooling for model-driven development (MDD) and code generation. Currently the Tools Suite features Nasdanika Vinci - an Eclipse... more info
by Nasdanika LLC, EPL 2.0
nasdanika model-driven development Web Development MDD
0
Installs: 9 (1 last month)
Install
Eclipse Tools for Cloud Foundry 1.2.0
Marketplaces
< Back
Install Now >
Finish
Cancel
06:55
08-11-2021


Eclipse Marketplace
Confirm Selected Features
Press Confirm to continue with the installation. Or go back to choose more solutions to install.
Google Cloud Tools for Eclipse 1.8.4 https://dl.google.com/eclipse/google-cloud-eclipse/stable/
Resolving features
< Install More
Confirm >
Finish
Cancel
06:55
08-11-2021

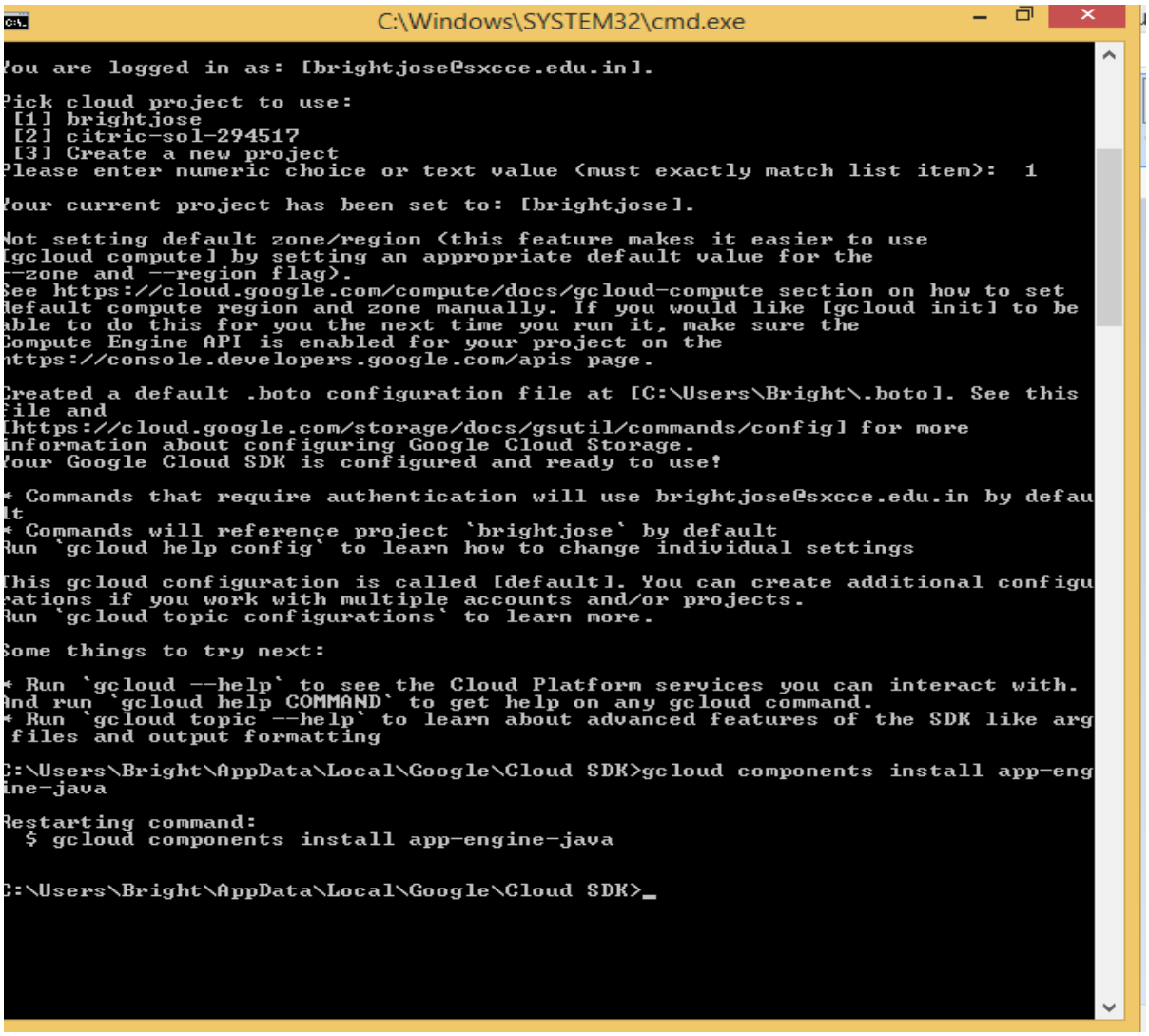


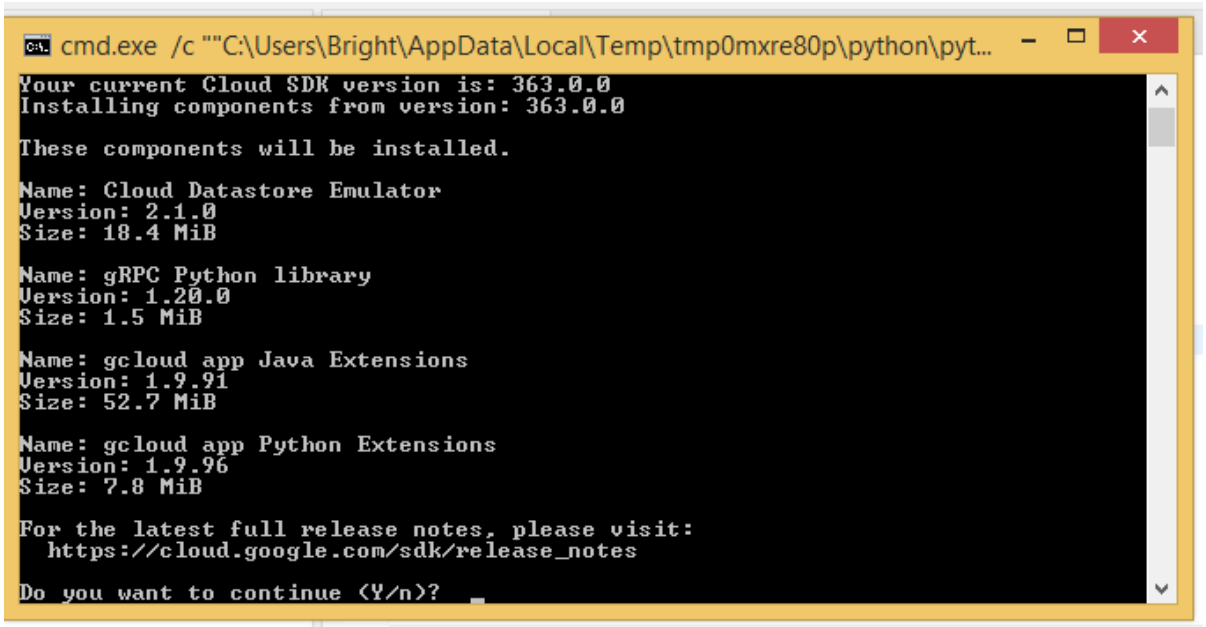


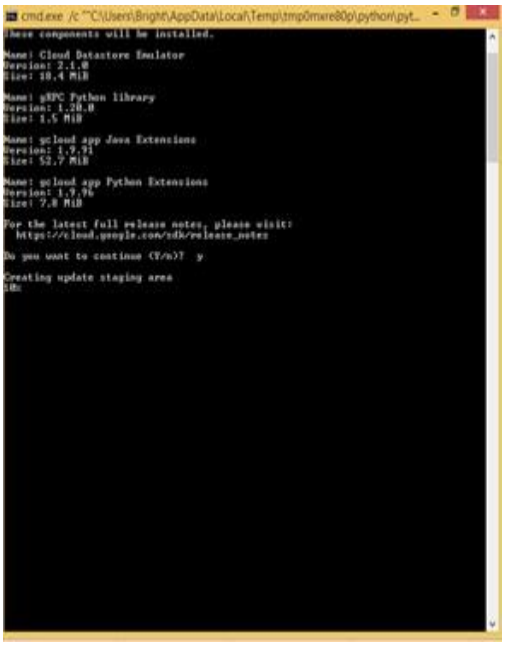
5. Download Google Cloud SDK Shell from https://cloud.google.com/sdk/docs/install and download Java version for Windows and install.
6. Click Google Cloud SDK Shell and type gcloud init command and make configuration and authenticate with Google Account.

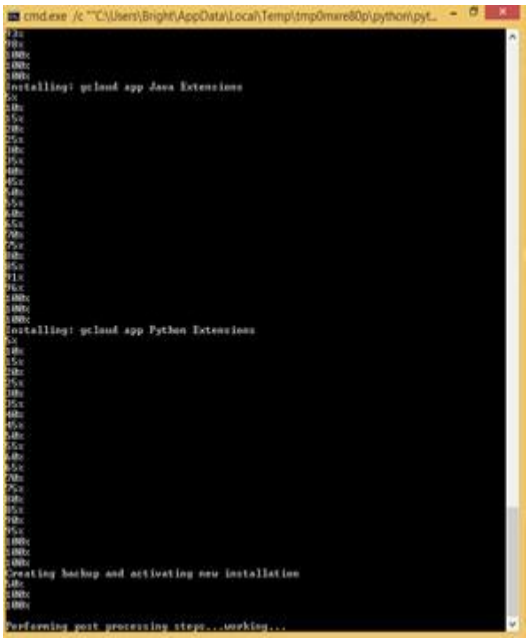

C:\Windows\SYSTEM32\cmd.exe
Welcome to the Google Cloud SDK! Run "gcloud -h" to get the list of available commands.

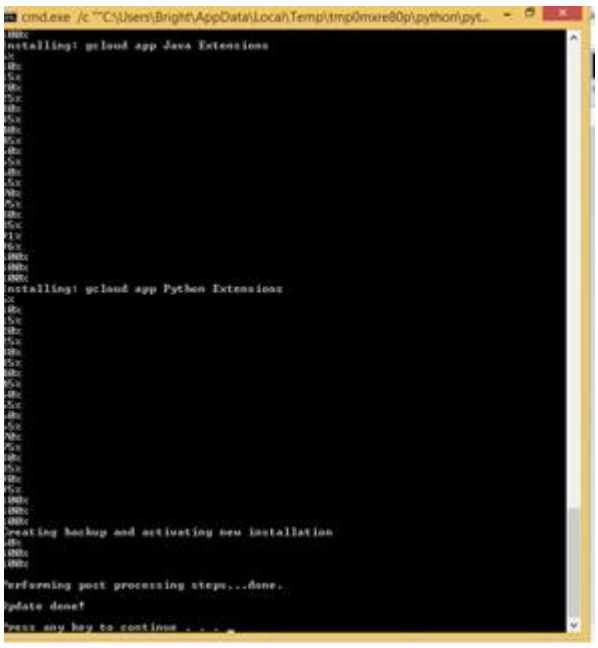
Welcome! This command will take you through the configuration of gcloud.
Your current configuration has been set to: [default]
You can skip diagnostics next time by using the following flag:
gcloud init --skip-diagnostics
Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability Check passed.
Network diagnostic passed (1/1 checks passed).
You must log in to continue. Would you like to log in (Y/n)? y
Your browser has been opened to visit:
https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=http%3A%2F%2Flocalhost%3A8085%2F&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=BUPAKa3o07hIJ6vdsonmLJ0tGypUNg&access_type=offline&code_challenge=cRniAInm48_9HwWNZng1yHroezRCrUkZbhGkQkAzyJQ&code_challenge_method=S256
You are logged in as: [brightjose@sxcce.edu.in].
Pick cloud project to use:
[1] brightjose
[2] citric-sol-294517
[3] Create a new project
Please enter numeric choice or text value (must exactly match list item): 1
Your current project has been set to: [brightjose].
Not setting default zone/region (this feature makes it easier to use
[gcloud compute] by setting an appropriate default value for the
--zone and --region flag).
See https://cloud.google.com/compute/docs/gcloud-compute section on how to set
default compute region and zone manually. If you would like [gcloud init] to be
able to do this for you the next time you run it, make sure the
Compute Engine API is enabled for your project on the
https://console.developers.google.com/apis page.
Created a default .boto configuration file at [C:\Users\Bright\.boto]. See this
file and
[https://cloud.google.com/storage/docs/gsutil/commands/config] for more
information about configuring Google Cloud Storage.
Your Google Cloud SDK is configured and ready to use!
* Commands that require authentication will use brightjose@sxcce.edu.in by default
* Commands will reference project `brightjose` by default
Run `gcloud help config` to learn how to change individual settings


C:\Windows\SYSTEM32\cmd.exe
You are logged in as: [brightjose@sxcce.edu.in].
Pick cloud project to use:
[1] brightjose
[2] citric-sol-294517
[3] Create a new project
Please enter numeric choice or text value (must exactly match list item): 1
Your current project has been set to: [brightjose].
Not setting default zone/region (this feature makes it easier to use
[gcloud compute] by setting an appropriate default value for the
--zone and --region flag).
See https://cloud.google.com/compute/docs/gcloud-compute section on how to set
default compute region and zone manually. If you would like [gcloud init] to be
able to do this for you the next time you run it, make sure the
Compute Engine API is enabled for your project on the
https://console.developers.google.com/apis page.
Created a default .boto configuration file at [C:\Users\Bright\.boto]. See this
file and
[https://cloud.google.com/storage/docs/gsutil/commands/config] for more
information about configuring Google Cloud Storage.
Your Google Cloud SDK is configured and ready to use!
* Commands that require authentication will use brightjose@sxcce.edu.in by defau
lt
* Commands will reference project `brightjose` by default
Run `gcloud help config` to learn how to change individual settings
This gcloud configuration is called [default]. You can create additional configu
rations if you work with multiple accounts and/or projects.
Run `gcloud topic configurations` to learn more.
Some things to try next:
* Run `gcloud --help` to see the Cloud Platform services you can interact with.
And run `gcloud help COMMAND` to get help on any gcloud command.
* Run `gcloud topic --help` to learn about advanced features of the SDK like arg
files and output formatting
C:\Users\Bright\AppData\Local\Google\Cloud SDK>gcloud components install app-eng
ine-java
Restarting command:
$ gcloud components install app-engine-java
C:\Users\Bright\AppData\Local\Google\Cloud SDK>

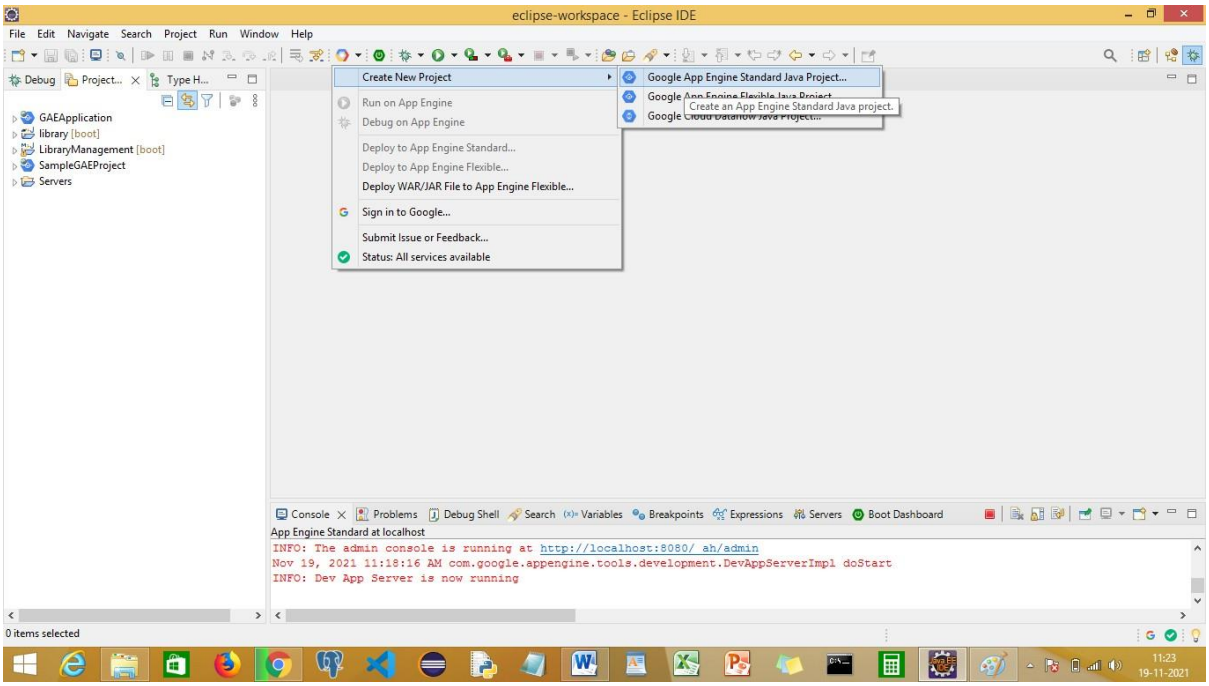

cmd.exe /c ""C:\Users\Bright\AppData\Local\Temp\tmp0mxre80p\python\pyt...
Your current Cloud SDK version is: 363.0.0
Installing components from version: 363.0.0
These components will be installed.
Name: Cloud Datastore Emulator
Version: 2.1.0
Size: 18.4 MiB
Name: gRPC Python library
Version: 1.20.0
Size: 1.5 MiB
Name: gcloud app Java Extensions
Version: 1.9.91
Size: 52.7 MiB
Name: gcloud app Python Extensions
Version: 1.9.96
Size: 7.8 MiB
For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes
Do you want to continue (Y/n)?

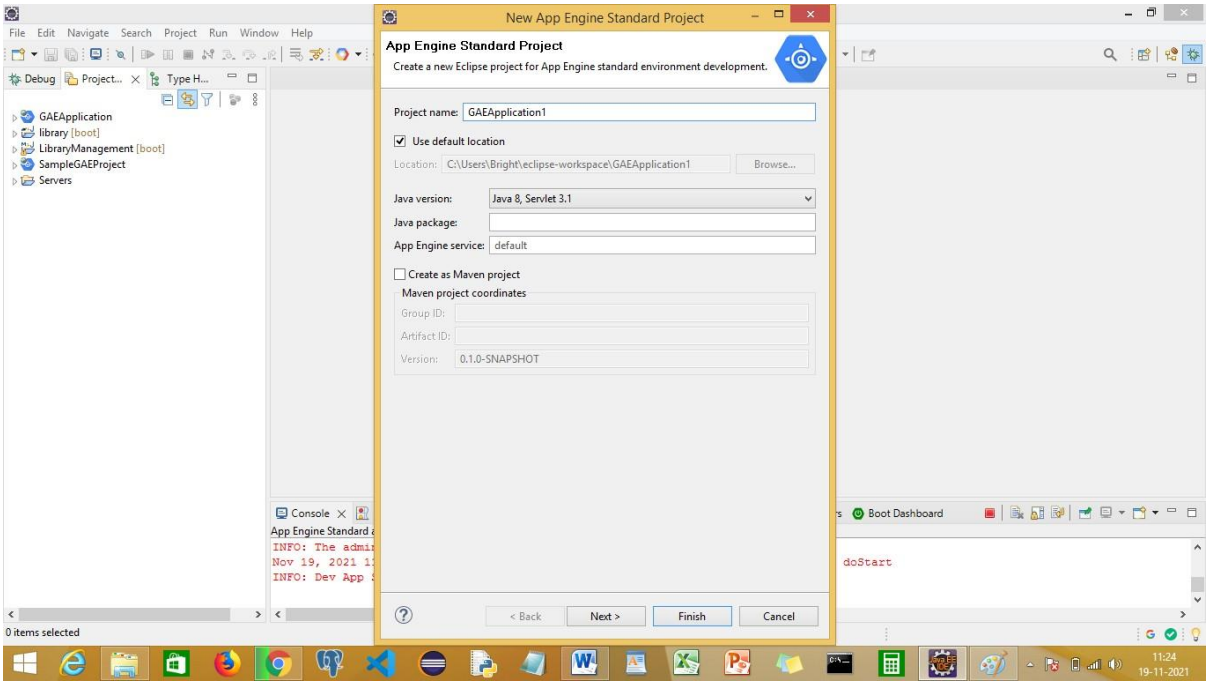






7. Create a project using Google Cloud Platform add on in the Eclipse IDE and add the necessary libraries.
8. Create necessary class and html files and jsp files for web application.
9. Run the application using Google App Engine.





SOURCE CODE

Application 1: (Hello World)

HelloAppEngine.java

```
import java.io.IOException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
```

```
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(name = "HelloAppEngine", urlPatterns = { "/hello"
})
public class HelloAppEngine extends HttpServlet {

     @Override
     public void doGet(HttpServletRequest request,
HttpServletResponse response) throws IOException {

          response.setContentType("text/plain");
          response.setCharacterEncoding("UTF-8");

          response.getWriter().print("Hello World!\r\n");

     }
}
```

Index.html

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml" lang="en">
<head>
<meta http-equiv="content-type"
     content="application/xhtml+xml; charset=UTF-8" />
<title>GAE - Hello World</title>
</head>
<body>
     <table>
          <tr>
     <td colspan="2" style="font-weight: bold;">Hello World
Servlet</td>
          </tr>
          <tr>
               <td><a href='/hello'>Click Here!!! </a></td>
          </tr>
     </table>
</body>
</html>
```

Application 2: (Add Two Numbers)

HelloAppEngine.java

```
import java.io.IOException;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
```

```
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(name = "HelloAppEngine", urlPatterns = { "/hello"
})
public class HelloAppEngine extends HttpServlet {

     @Override
     public void doGet(HttpServletRequest request,
HttpServletResponse response) throws IOException,
ServletException {

          RequestDispatcher view =
request.getRequestDispatcher("sample.html");
          view.forward(request, response);

     }
}
```

index.html

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml" lang="en">
  <head>
    <meta http-equiv="content-type"
content="application/xhtml+xml; charset=UTF-8" />

<title>Add Two Numbers</title>
  </head>

  <body>
    <h1>GAE - Sum of Two Numbers !</h1>

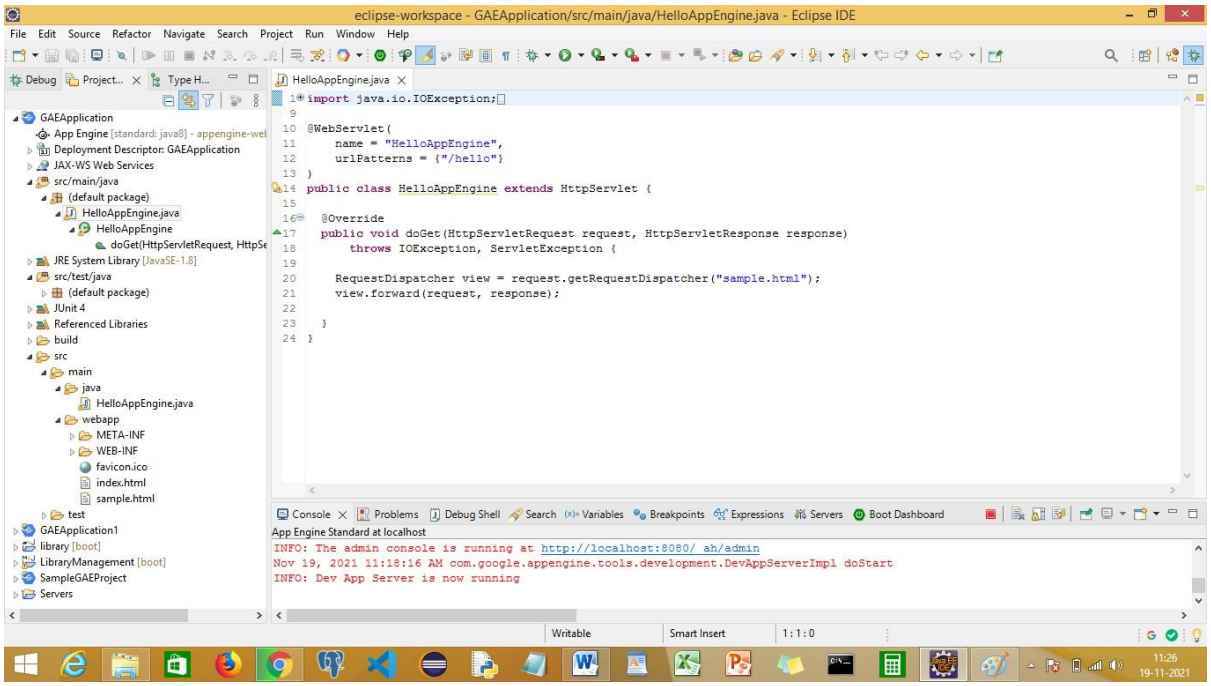
    <table>
      <tr>
        <td colspan="2" style="font-weight:bold;">Sum
Here:</td>
      </tr>
      <tr>
        <td><a href='/hello'>Input Here !!!</a></td>
      </tr>
    </table>
  </body>
</html>
```

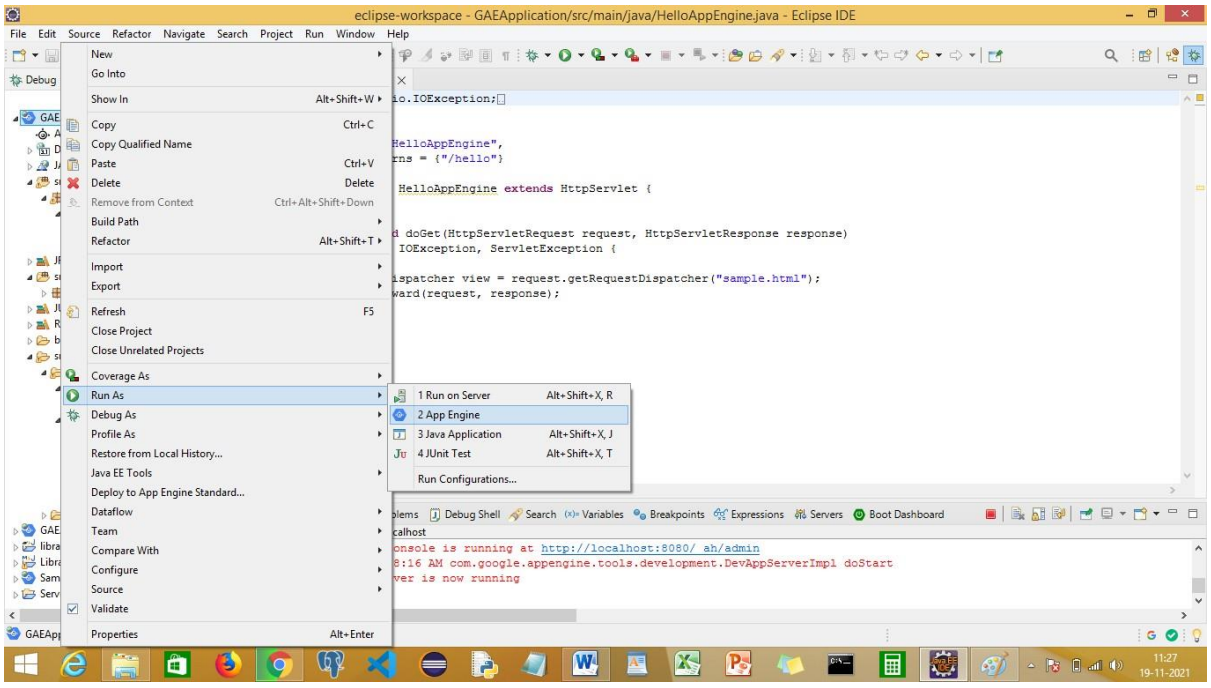
sample.html

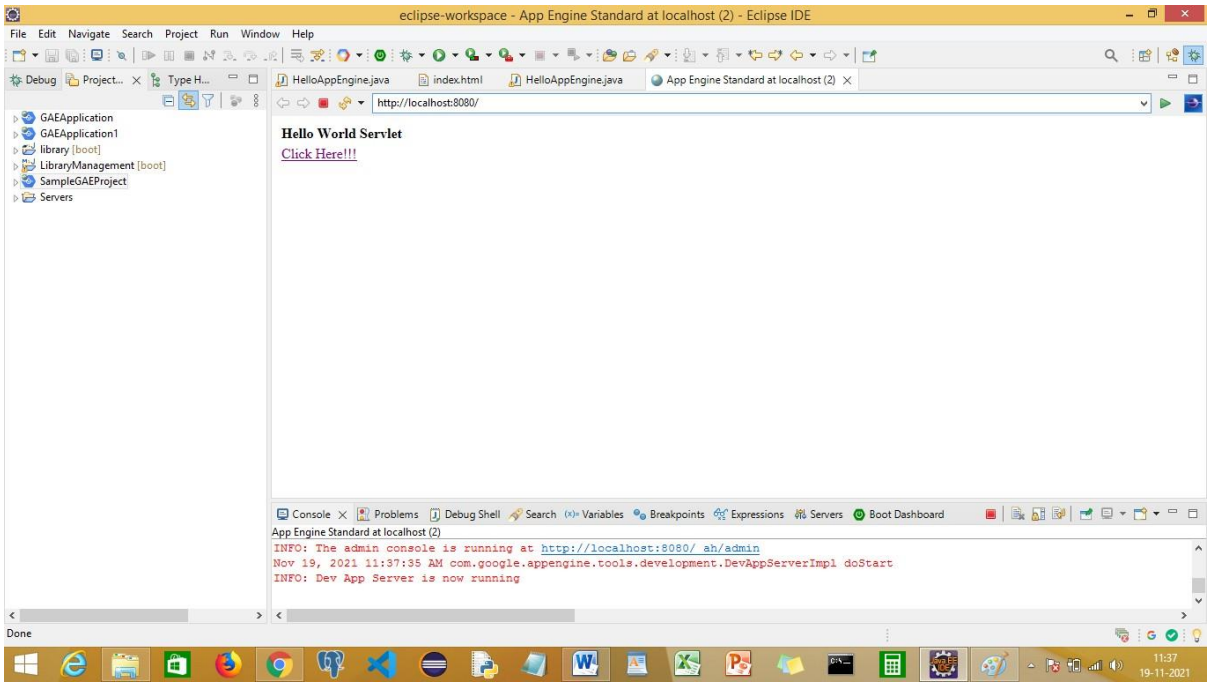
```
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Add Operation</title>
</head>
<body>
    <div>
        <div>
            <h1>Add two number using text box as input using
javascript</h1>
        </div>
          Enter First Number : <br>
          <input type="text" id="Text1" name="TextBox1"><br>
          Enter Second Number : <br>
          <input type="text" id="Text2" name="TextBox2"><br>
          <input type="button" name="clickbtn" value="Display
Result" onclick="add_number()"><br>
          Result : <br>
          <input type="text" id="txtresult" name="TextBox3">

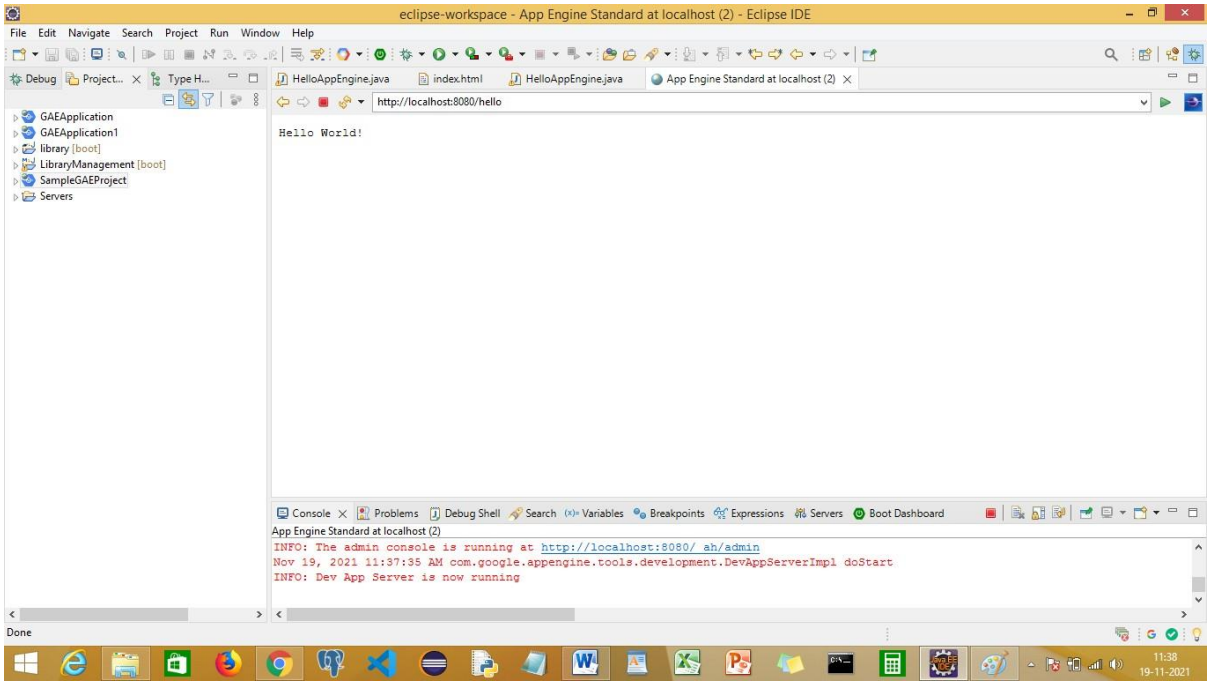
        <script type="text/javascript">
            function add_number() {
            var first_number =
parseInt(document.getElementById("Text1").value);
            var second_number =
parseInt(document.getElementById("Text2").value);
            var result = first_number + second_number;
            document.getElementById("txtresult").value =
result;
        }
        </script>
     </div>
</body>
</html>
```

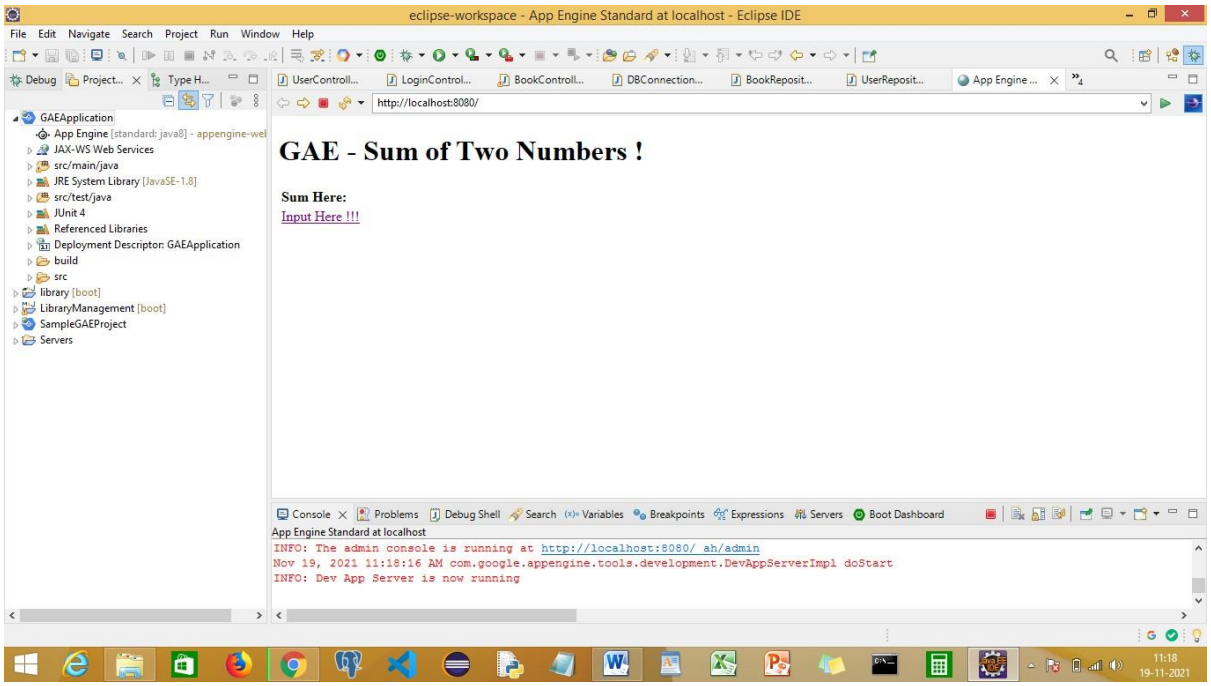
OUTPUT / SCREENSHOTS:

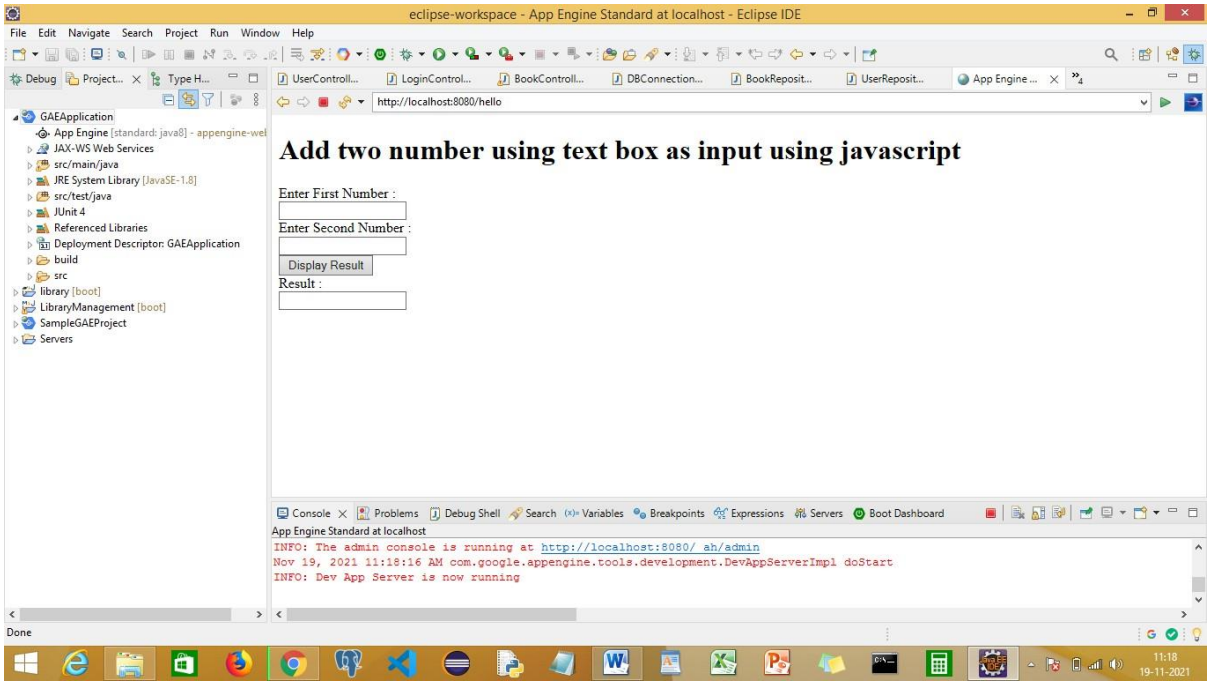


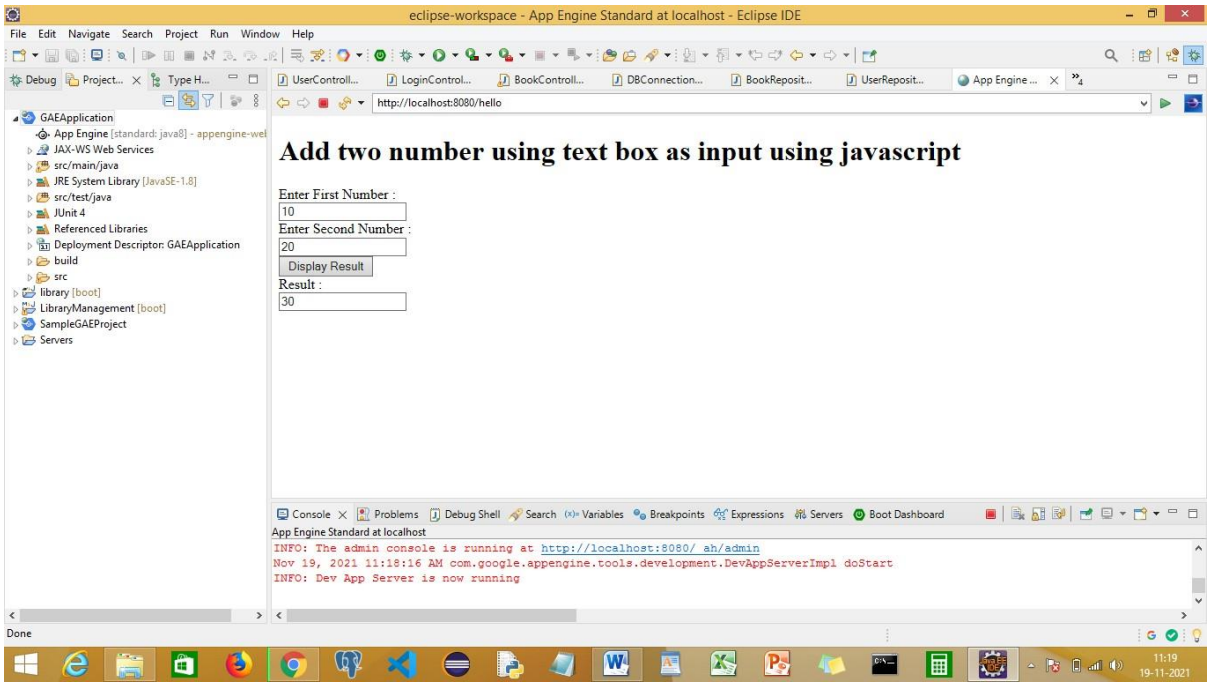



eclipse-workspace - App Engine Standard at localhost (2) - Eclipse IDE
File Edit Navigate Search Project Run Window Help
Debug Project... Type H...
GAEApplication
GAEApplication1
library [boot]
LibraryManagement [boot]
SampleGAEProject
Servers
HelloAppEngine.java
index.html
HelloAppEngine.java
App Engine Standard at localhost (2)
http://localhost:8080/
Hello World Servlet
Click Here!!!
Console Problems Debug Shell Search Variables Breakpoints Expressions Servers Boot Dashboard
App Engine Standard at localhost (2)
INFO: The admin console is running at http://localhost:8080/_ah/admin
Nov 19, 2021 11:37:35 AM com.google.appengine.tools.development.DevAppServerImpl doStart
INFO: Dev App Server is now running
Done
11:37
19-11-2021


eclipse-workspace - App Engine Standard at localhost (2) - Eclipse IDE
File Edit Navigate Search Project Run Window Help
Debug Project... Type H...
GAEApplication
GAEApplication1
library [boot]
LibraryManagement [boot]
SampleGAEProject
Servers
HelloAppEngine.java
index.html
HelloAppEngine.java
App Engine Standard at localhost (2)
http://localhost:8080/hello
Hello World!
Console Problems Debug Shell Search Variables Breakpoints Expressions Servers Boot Dashboard
App Engine Standard at localhost (2)
INFO: The admin console is running at http://localhost:8080/_ah/admin
Nov 19, 2021 11:37:35 AM com.google.appengine.tools.development.DevAppServerImpl doStart
INFO: Dev App Server is now running
Done
11:38
19-11-2021


eclipse-workspace - App Engine Standard at localhost - Eclipse IDE
File Edit Navigate Search Project Run Window Help
http://localhost:8080/
GAE - Sum of Two Numbers !
Sum Here:
Input Here !!!
Console Problems Debug Shell Search Variables Breakpoints Expressions Servers Boot Dashboard
App Engine Standard at localhost
INFO: The admin console is running at http://localhost:8080/_ah/admin
Nov 19, 2021 11:18:16 AM com.google.appengine.tools.development.DevAppServerImpl doStart
INFO: Dev App Server is now running


eclipse-workspace - App Engine Standard at localhost - Eclipse IDE
File Edit Navigate Search Project Run Window Help
http://localhost:8080/hello
Add two number using text box as input using javascript
Enter First Number :
Enter Second Number :
Display Result
Result :
Console Problems Debug Shell Search Variables Breakpoints Expressions Servers Boot Dashboard
App Engine Standard at localhost
INFO: The admin console is running at http://localhost:8080/_ah/admin
Nov 19, 2021 11:18:16 AM com.google.appengine.tools.development.DevAppServerImpl doStart
INFO: Dev App Server is now running
Done


eclipse-workspace - App Engine Standard at localhost - Eclipse IDE
File Edit Navigate Search Project Run Window Help
GAEApplication
App Engine [standard: java8] - appengine-wel
JAX-WS Web Services
src/main/java
JRE System Library [JavaSE-1.8]
src/test/java
JUnit 4
Referenced Libraries
Deployment Descriptor: GAEApplication
build
src
library [boot]
LibraryManagement [boot]
SampleGAEProject
Servers
http://localhost:8080/hello
Add two number using text box as input using javascript
Enter First Number :
10
Enter Second Number :
20
Display Result
Result :
30
Console Problems Debug Shell Search Variables Breakpoints Expressions Servers Boot Dashboard
App Engine Standard at localhost
INFO: The admin console is running at http://localhost:8080/_ah/admin
Nov 19, 2021 11:18:16 AM com.google.appengine.tools.development.DevAppServerImpl doStart
INFO: Dev App Server is now running
Done
11:19
19-11-2021

RESULT:

EXP.NO: 4 DATE:	USE GAE LAUNCHER TO LAUNCH THE WEBAPPLICATIONS

AIM

To run a web application using Google App Engine.

PROCEDURE

1. Create an HTML and CSS files to design the web page
2. Create a python file and save it as todo.py
3. Create a YAML file for configuration and name it as app.yaml
4. Open the Cloud SDK window and type the comment
   Google-cloud-sdk\bin\dev-appserver.py "<<path_address>>"
5. In web browser obtain the result from address localhost:8080

SOURCE CODE

index.py

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html>
   <head>
      <title>Todos</title>
      <link rel="stylesheet" type="text/css" href="css/main.css"/>
      <meta http-equiv="Content-Type" content="text/html; charset=UTF-8"></meta>
   </head>
   <body>
   <div style="width: 100%;">
      <div class="topLine">
         <div style="float: left;"><img src="images/todo.png" /></div>
         <div style="float: left;" class="headline">Todos</div>
         <div style="float: right;"><a href="{{ url }}">{{ url_linktext }}</a>
{{user.nickname}}</div>
      </div>
   </div>
<div style="clear: both;"/>
{# Check if we have any todos, only in this case render the table #}
{% if numbertodos %}
You have a total number of {{numbertodos}} Todos.
<table>
  <tr>
     <th>Short description </th>
     <th>Due Date</th>
     <th>Long Description</th>
     <th>URL</th>
     <th>Created</th>
```

```
    <th>Updated</th>
    <th>Done</th>
    <th>Send Email reminder</th>
  </tr>
{% for todo in todos %}
<tr>
<td>
{{todo.shortDescription}}
</td>
<td>
{{todo.dueDate}}
</td>
<td>
{{todo.longDescription}}
</td>
<td>
{% if todo.url %}
<a  href="{{todo.url}}" > {{todo.url}}</a>
{% endif %}
</td>
<td>
{{todo.created|date:"d.m.Y"}}
</td>
<td>
{{todo.updated|date:"d.m.Y"}}
</td>
<td>
<a class="done" href="/done?id={{todo.key.id}}" >Done</a>
</td>
<td>
<a class="email" href="/email?id={{todo.key.id}}" >Email</a>
</td>
</tr>
{% endfor %}
</table>
{% endif %}
<hr />
<div class="headline">Create new Todo?</div>
{% if user %}
<form action="/new" method="post">
  <table>
    <tr>
      <td><label for="shortDescription">Summary</label></td>
      <td><input type="text" name="shortDescription" id="shortDescriptions" size="80"/></td>
    </tr>
    <tr>
      <td><label for="dueDate">Due Date</label></td>
      <td><input type="dueDate" name="dueDate" id="dueDate"/></td>
```

```
      </tr>
      <tr>
        <td valign="top"><label for="longDescription">Description</label></td>
        <td><textarea rows="4" cols="80" name="longDescription"
id="longDescription"></textarea></td>
      </tr>
      <tr>
        <td><label for="url">URL</label></td>
        <td><input type="text" name="url" id="url" size="80"/></td>
      </tr>
      <tr>
        <td colspan="2" align="right"><input type="submit" value="Create"/></td>
      </tr>
  </table>
</form>
{% else %}
Login with your Google Account to create and review todos.
{% endif%}
</body>
</html>
```

main.css

```
body {
  margin: 5px;
  font-family: Arial, Helvetica, sans-serif;
}
hr {
  border: 0;
  background-color: #DDDDDD;
  height: 1px;
  margin: 5px;
}
table th {
  background:#EFEFEF none repeat scroll 0 0;
  border-top:1px solid #CCCCCC;
  font-size:small;
  padding-left:5px;
  padding-right:4px;
  padding-top:4px;
  vertical-align:top;
  text-align:left;
}
table tr {
  background-color: #e5ecf9;
```

```
    font-size:small;
}
.topLine {
    height: 1.25em;
    background-color: #e5ecf9;
    padding-left: 4px;
    padding-right: 4px;
    padding-bottom: 3px;
    padding-top: 2px;
}
.headline {
    font-weight: bold;
    color: #3366cc;
}
.done {
    font-size: x-small;
    vertical-align: top;
}
.email {
    font-size: x-small;
    vertical-align: top;
}
form td {
    font-size: smaller;
}
```

todo.py

```
from google.appengine.api import users
from google.appengine.ext import webapp
from google.appengine.ext.webapp.util import run_wsgi_app
from google.appengine.ext import db
from google.appengine.ext.webapp import template
from google.appengine.api import mail
# Todo defines the data model for the Todos
# as it extends db.model the content of the class will automatically stored
class TodoModel(db.Model):
  author       = db.UserProperty(required=True)
  shortDescription = db.StringProperty(required=True)
  longDescription  = db.StringProperty(multiline=True)
  url          = db.StringProperty()
  created          = db.DateTimeProperty(auto_now_add=True)
  updated      = db.DateTimeProperty(auto_now=True)
```

```
  dueDate          = db.StringProperty(required=True)
  finished         = db.BooleanProperty()
# The main page where the user can login and logout
# MainPage is a subclass of webapp.RequestHandler and overwrites the get method
class MainPage(webapp.RequestHandler):
    def get(self):
        user = users.get_current_user()
        url = users.create_login_url(self.request.uri)
        url_linktext = 'Login'
        if user:
            url = users.create_logout_url(self.request.uri)
            url_linktext = 'Logout'
# GQL is similar to SQL
        todos = TodoModel.gql("WHERE author = :author and finished=false",
              author=users.get_current_user())
        values = {
            'todos': todos,
        'numbertodos' : todos.count(),
            'user': user,
            'url': url,
            'url_linktext': url_linktext,
        }
        self.response.out.write(template.render('index.html', values))
# This class creates a new Todo item
class New(webapp.RequestHandler):
    def post(self):
        user = users.get_current_user()
        if user:
          testurl = self.request.get('url')
        if not testurl.startswith("http://") and testurl:
          testurl = "http://"+ testurl
          todo = TodoModel(
                author  = users.get_current_user(),
                shortDescription = self.request.get('shortDescription'),
                longDescription = self.request.get('longDescription'),
                dueDate = self.request.get('dueDate'),
                url = testurl,
        finished = False)
        todo.put();
        self.redirect('/')
# This class deletes the selected Todo
class Done(webapp.RequestHandler):
    def get(self):
        user = users.get_current_user()
```

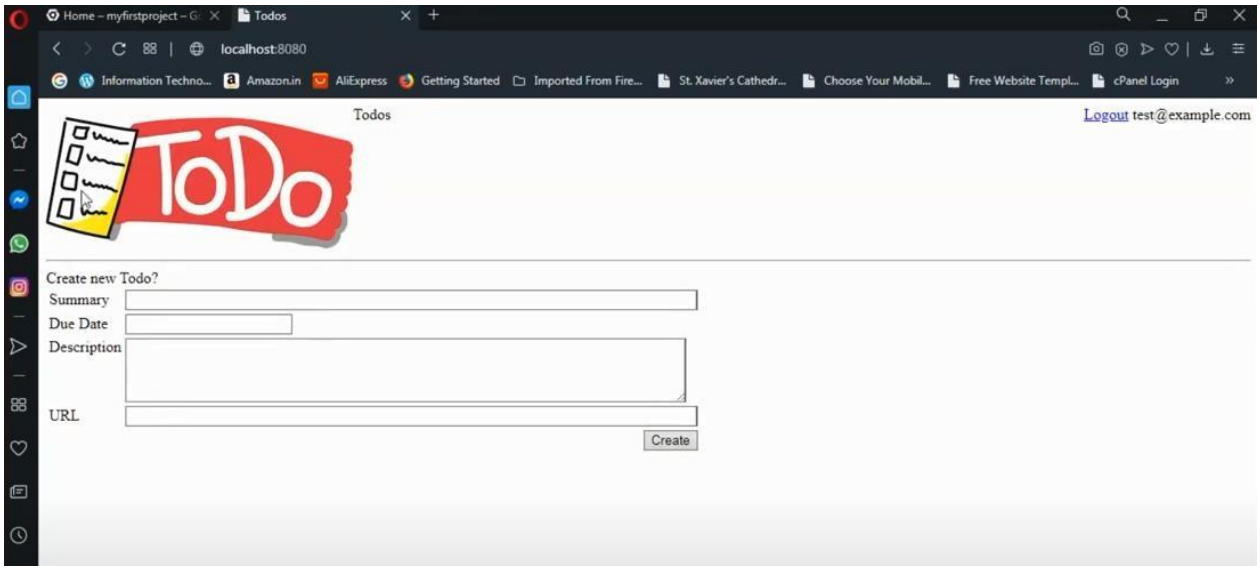
```
        if user:
            raw_id = self.request.get('id')
            id = int(raw_id)
            todo = TodoModel.get_by_id(id)
            todo.delete()
            self.redirect('/')
#This class emails the task to yourself
class Email(webapp.RequestHandler):
    def get(self):
      user = users.get_current_user()
      if user:
            raw_id = self.request.get('id')
            id = int(raw_id)
            todo = TodoModel.get_by_id(id)
      message = mail.EmailMessage(sender=user.email(), subject=todo.shortDescription)
      message.to = user.email()
      message.body = todo.longDescription
      message.send()
      self.redirect('/')
# Register the URL with the responsible classes
application = webapp.WSGIApplication(
                                  [('/', MainPage),
                                   ('/new', New),
                    ('/done', Done),
                    ('/email', Email)],
                                  debug=True)
# Register the wsgi application to run
def main():
run_wsgi_app(application)
if__name__== "_main_":
  main()
```

app.yaml

```
application: googleappengine01
version: 1
runtime: python
api_version: 1

handlers:
- url: /css
  static_dir: css
- url: /images
  static_dir: images
- url: /.*
  script: todo.py
```

OUPUT/ SCREENSHOTS:



RESULT:

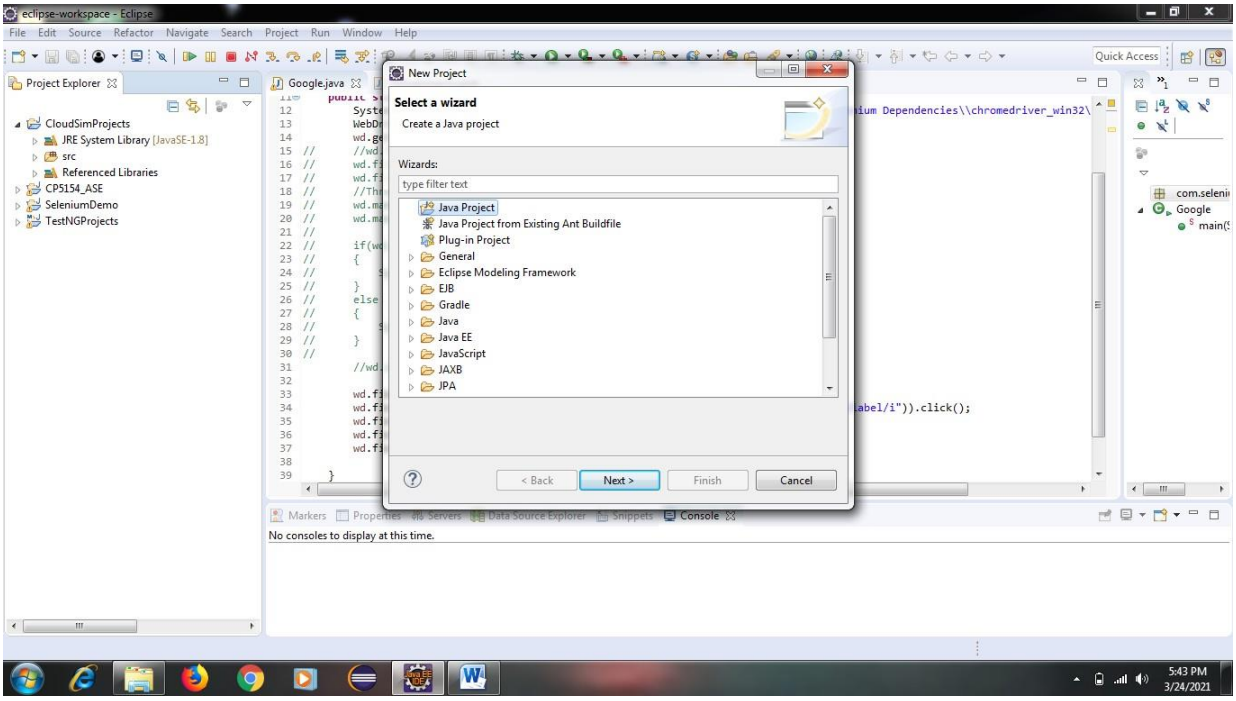
EXP.NO:5 DATE:	SIMULATE A CLOUD SCENARIO USING CLOUDSIMAND RUN A SCHEDULING ALGORITHM

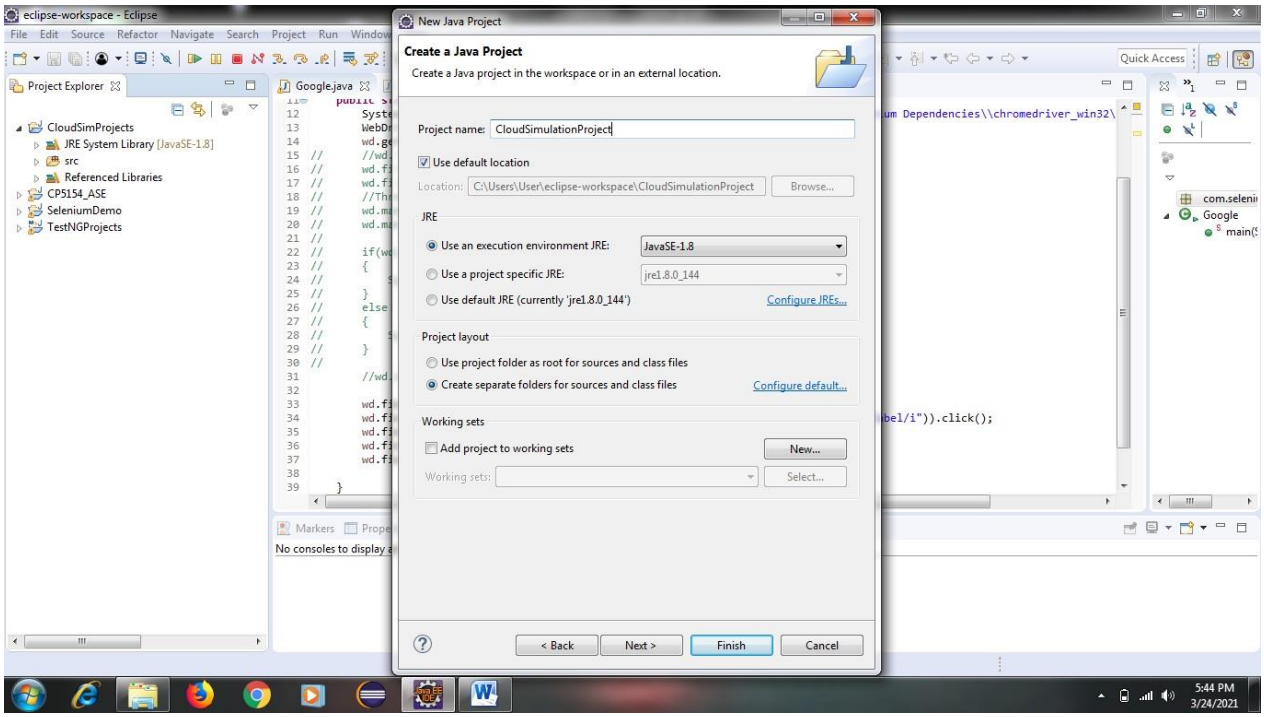
AIM:

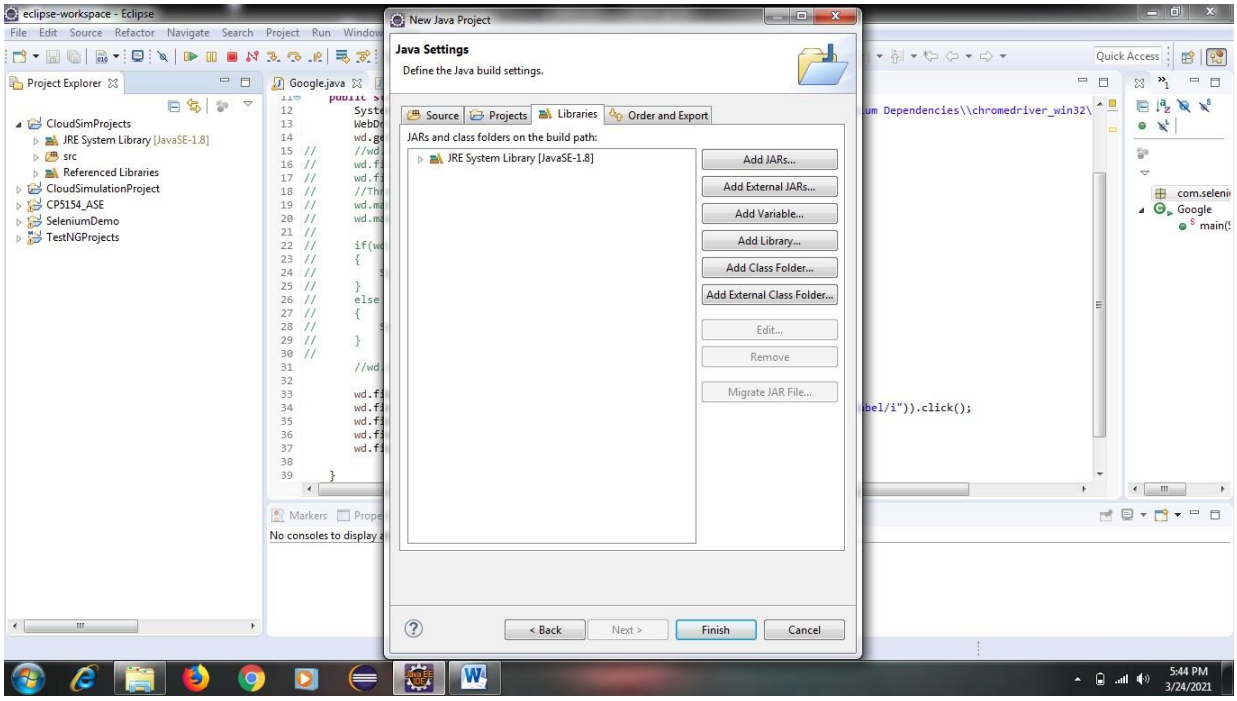
To simulate a cloud scenario using Cloudsim and run a scheduling algorithm(SJF)

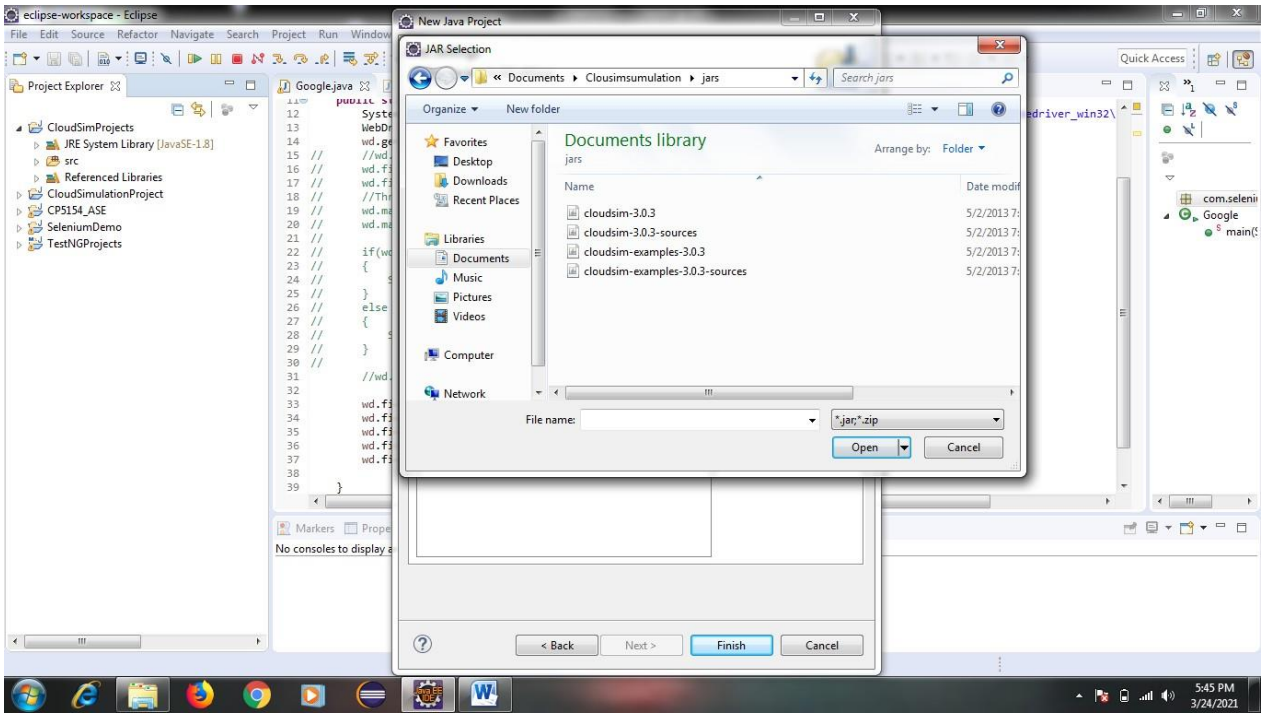
PROCEDURE:

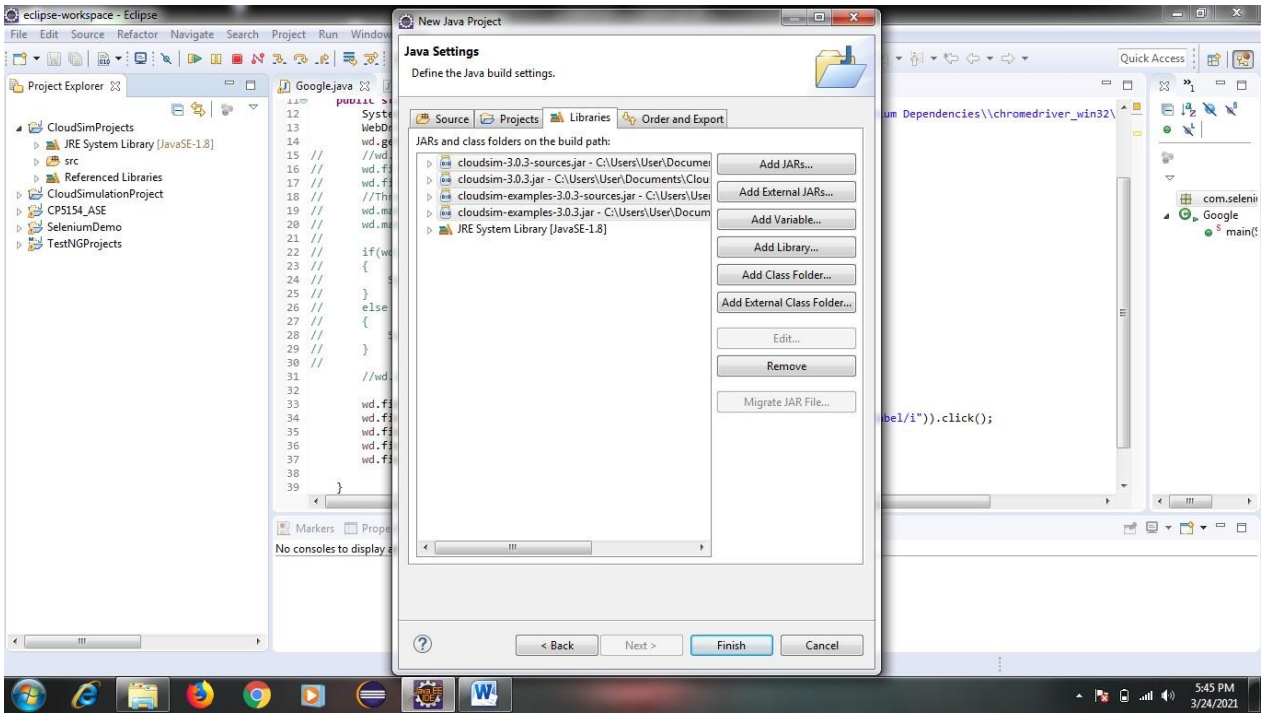
1. Download Java, Eclipse IDE and install it.
2. Download Cloudsim jar Files
3. Create a new java project and add Cloudsim jar files.



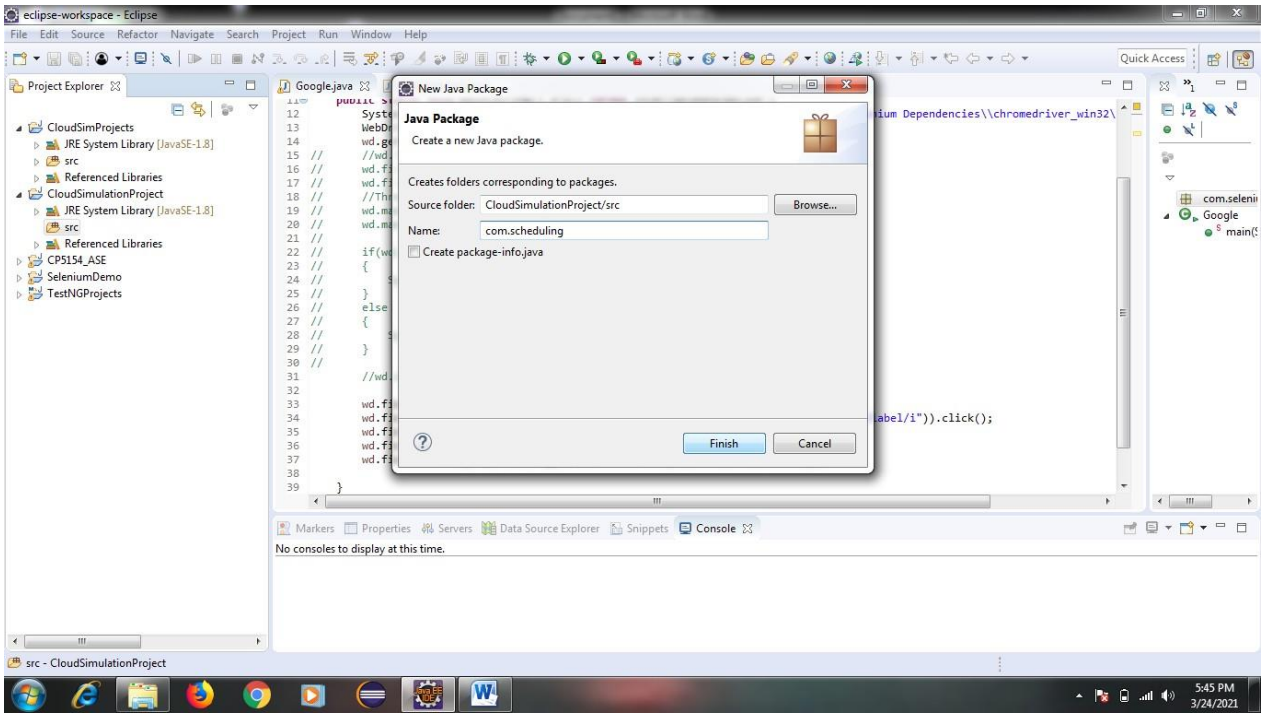

eclipse-workspace - Eclipse
New Java Project
Create a Java Project
Create a Java project in the workspace or in an external location.
Project name: CloudSimulationProject
Use default location
Location: C:\Users\User\eclipse-workspace\CloudSimulationProject
Browse...
JRE
Use an execution environment JRE: JavaSE-1.8
Use a project specific JRE: jre1.8.0_144
Use default JRE (currently 'jre1.8.0_144')
Configure JREs...
Project layout
Use project folder as root for sources and class files
Create separate folders for sources and class files
Configure default...
Working sets
Add project to working sets
New...
Working sets:
Select...
< Back
Next >
Finish
Cancel
Project Explorer
CloudSimProjects
JRE System Library [JavaSE-1.8]
src
Referenced Libraries
CP5154_ASE
SeleniumDemo
TestNGProjects
Google.java
Quick Access
5:44 PM
3/24/2021

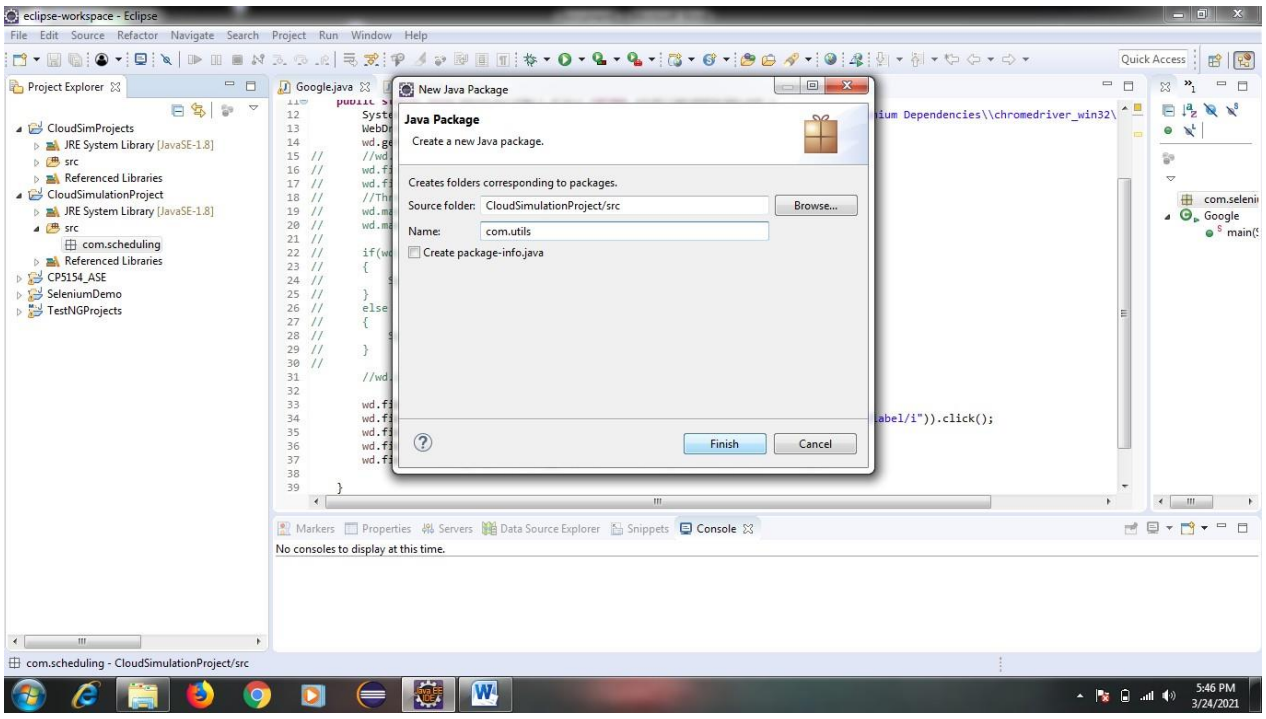

eclipse-workspace - Eclipse
New Java Project
Java Settings
Define the Java build settings.
Source
Projects
Libraries
Order and Export
JARs and class folders on the build path:
JRE System Library [JavaSE-1.8]
Add JARs...
Add External JARs...
Add Variable...
Add Library...
Add Class Folder...
Add External Class Folder...
Edit...
Remove
Migrate JAR File...
< Back
Next >
Finish
Cancel
Project Explorer
CloudSimProjects
JRE System Library [JavaSE-1.8]
src
Referenced Libraries
CloudSimulationProject
CP5154_ASE
SeleniumDemo
TestNGProjects
Google.java
Quick Access
5:44 PM
3/24/2021



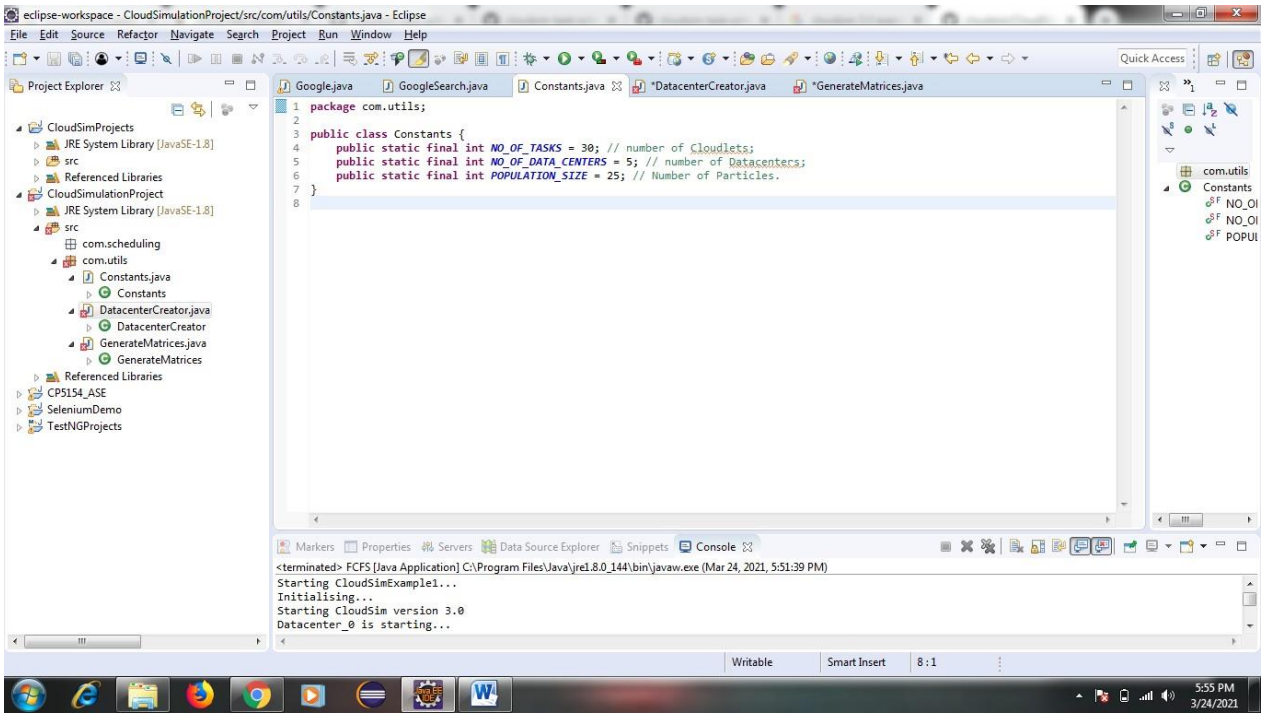


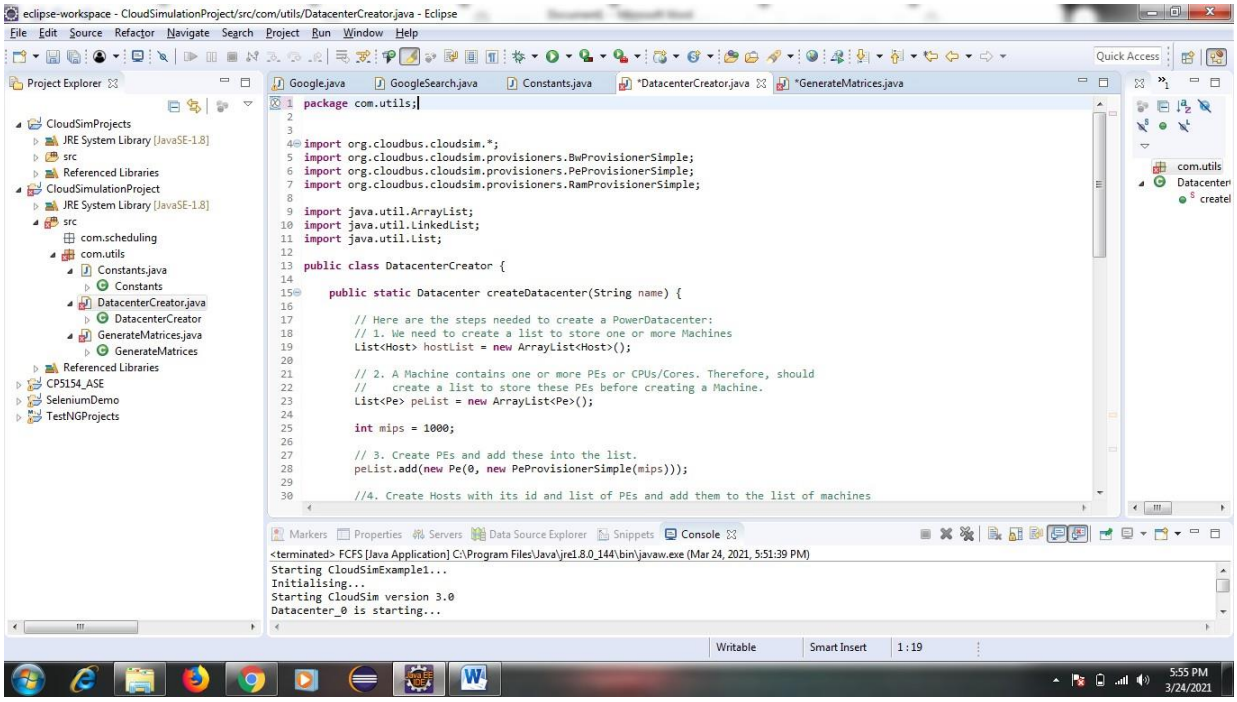
4. Create a Java packages.

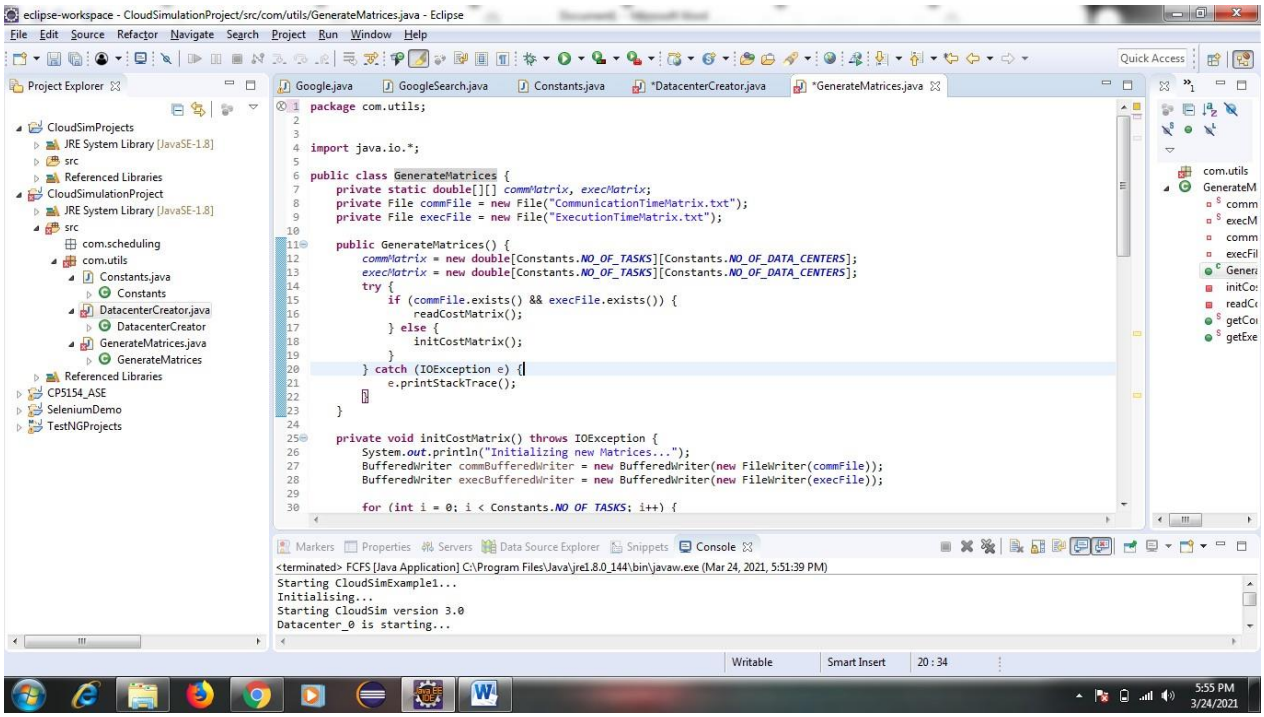




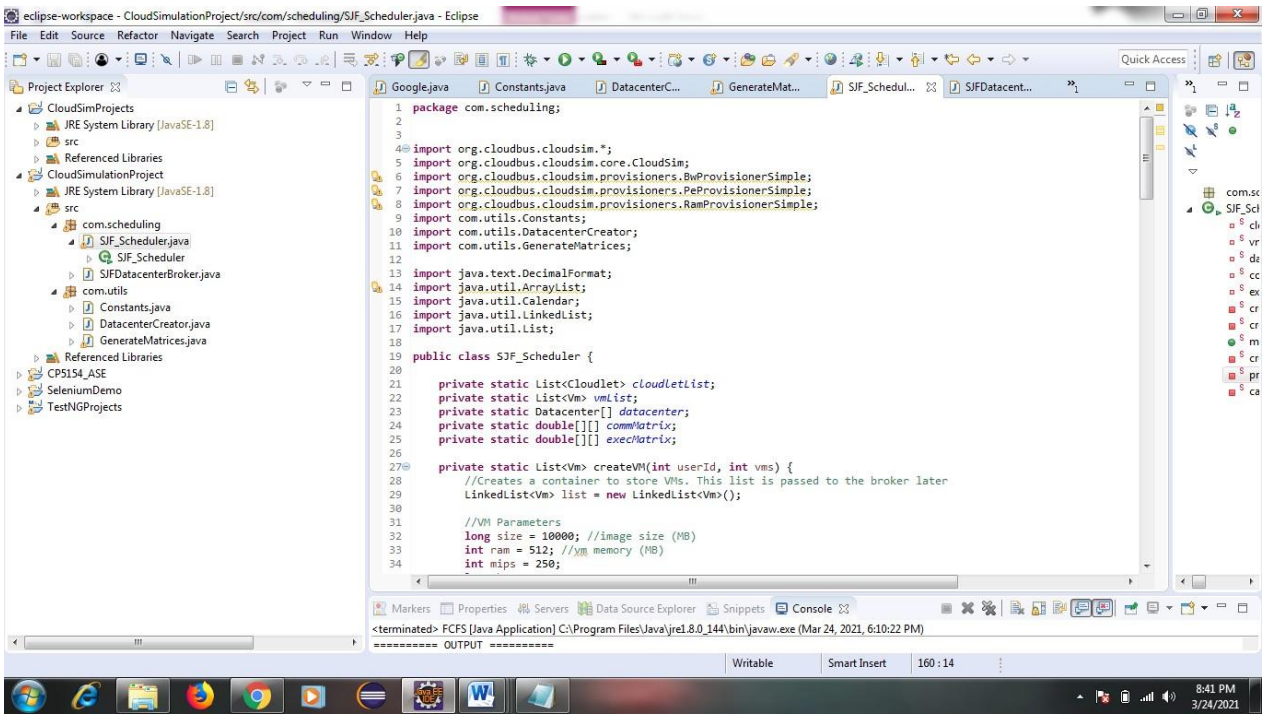
5. Under the Java package, create a java classes which consists of Data Center , Broker, Host, VMs, Cloudlets , scheduling algorithm and submit all cloudlets to broker.


eclipse-workspace - CloudSimulationProject/src/com/utils/Constants.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
CloudSimProjects
JRE System Library [JavaSE-1.8]
src
Referenced Libraries
CloudSimulationProject
JRE System Library [JavaSE-1.8]
src
com.scheduling
com.utils
Constants.java
Constants
DatacenterCreator.java
DatacenterCreator
GenerateMatrices.java
GenerateMatrices
Referenced Libraries
CP5154_ASE
SeleniumDemo
TestNGProjects
Google.java
GoogleSearch.java
Constants.java
*DatacenterCreator.java
*GenerateMatrices.java
1 package com.utils;
2
3 public class Constants {
4 public static final int NO_OF_TASKS = 30; // number of Cloudlets;
5 public static final int NO_OF_DATA_CENTERS = 5; // number of Datacenters;
6 public static final int POPULATION_SIZE = 25; // Number of Particles.
7 }
8
Markers Properties Servers Data Source Explorer Snippets Console
<terminated> FCFS [Java Application] C:\Program Files\Java\jre1.8.0_144\bin\javaw.exe (Mar 24, 2021, 5:51:39 PM)
Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Writable
Smart Insert
8 : 1
5:55 PM
3/24/2021


eclipse-workspace - CloudSimulationProject/src/com/utils/DatacenterCreator.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
CloudSimProjects
JRE System Library [JavaSE-1.8]
src
Referenced Libraries
CloudSimulationProject
JRE System Library [JavaSE-1.8]
src
com.scheduling
com.utils
Constants.java
Constants
DatacenterCreator.java
DatacenterCreator
GenerateMatrices.java
GenerateMatrices
Referenced Libraries
CP5154_ASE
SeleniumDemo
TestNGProjects
Google.java
GoogleSearch.java
Constants.java
*DatacenterCreator.java
*GenerateMatrices.java
1 package com.utils;
2
3
4 import org.cloudbus.cloudsim.*;
5 import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
6 import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
7 import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;
8
9 import java.util.ArrayList;
10 import java.util.LinkedList;
11 import java.util.List;
12
13 public class DatacenterCreator {
14
15 public static Datacenter createDatacenter(String name) {
16
17 // Here are the steps needed to create a PowerDatacenter:
18 // 1. We need to create a list to store one or more Machines
19 List<Host> hostList = new ArrayList<Host>();
20
21 // 2. A Machine contains one or more PEs or CPUs/Cores. Therefore, should
22 // create a list to store these PEs before creating a Machine.
23 List<Pe> peList = new ArrayList<Pe>();
24
25 int mips = 1000;
26
27 // 3. Create PEs and add these into the list.
28 peList.add(new Pe(0, new PeProvisionerSimple(mips)));
29
30 //4. Create Hosts with its id and list of PEs and add them to the list of machines
Markers Properties Servers Data Source Explorer Snippets Console
<terminated> FCFS [Java Application] C:\Program Files\Java\jre1.8.0_144\bin\javaw.exe (Mar 24, 2021, 5:51:39 PM)
Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Writable
Smart Insert
1 : 19
5:55 PM
3/24/2021



6. Type the SJF Program



SOURCE CODE:

Constants.java

```
package com.utils;

public class Constants {

    public static final int NO_OF_TASKS = 30; // number of Cloudlets;
```

```
    public static final int NO_OF_DATA_CENTERS = 5; // number of
Datacenters;

    public static final int POPULATION_SIZE = 25; // Number of Particles.

}
```

GenerateMatrices.java

```
package com.utils;

import java.io.*;

public class GenerateMatrices {

    private static double[][] commMatrix, execMatrix;

    private File commFile = new File("CommunicationTimeMatrix.txt");

    private File execFile = new File("ExecutionTimeMatrix.txt");


    public GenerateMatrices() {

        commMatrix = new
double[Constants.NO_OF_TASKS][Constants.NO_OF_DATA_CENTERS];

        execMatrix = new
double[Constants.NO_OF_TASKS][Constants.NO_OF_DATA_CENTERS];

        try {

            if (commFile.exists() && execFile.exists()) {

                readCostMatrix();

            } else {

                initCostMatrix();

            }

        } catch (IOException e) {

            e.printStackTrace();

        }

    }


    private void initCostMatrix() throws IOException {

        System.out.println("Initializing new Matrices...");

        BufferedWriter commBufferedWriter = new BufferedWriter(new
FileWriter(commFile));

        BufferedWriter execBufferedWriter = new BufferedWriter(new
FileWriter(execFile));
```

```
        for (int i = 0; i < Constants.NO_OF_TASKS; i++) {
            for (int j = 0; j < Constants.NO_OF_DATA_CENTERS; j++) {
                commMatrix[i][j] = Math.random() * 600 + 20;
                execMatrix[i][j] = Math.random() * 500 + 10;
                commBufferedWriter.write(String.valueOf(commMatrix[i][j]) +
' ');
                execBufferedWriter.write(String.valueOf(execMatrix[i][j]) +
' ');
            }
            commBufferedWriter.write('\n');
            execBufferedWriter.write('\n');
        }
        commBufferedWriter.close();
        execBufferedWriter.close();
    }

    private void readCostMatrix() throws IOException {
        System.out.println("Reading the Matrices...");
        BufferedReader commBufferedReader = new BufferedReader(new
FileReader(commFile));

        int i = 0, j = 0;
        do {
            String line = commBufferedReader.readLine();
            for (String num : line.split(" ")) {
                commMatrix[i][j++] = new Double(num);
            }
            ++i;
            j = 0;
        } while (commBufferedReader.ready());


        BufferedReader execBufferedReader = new BufferedReader(new
FileReader(execFile));
```

```
        i = j = 0;
        do {
            String line = execBufferedReader.readLine();
            for (String num : line.split(" ")) {
                execMatrix[i][j++] = new Double(num);
            }
            ++i;
            j = 0;
        } while (execBufferedReader.ready());
    }

    public static double[][] getCommMatrix() {
        return commMatrix;
    }

    public static double[][] getExecMatrix() {
        return execMatrix;
    }
}
```

DatacenterCreator.java

```
package com.utils;
import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;
import java.util.ArrayList;
import java.util.LinkedList;
import java.util.List;
public class DatacenterCreator {

    public static Datacenter createDatacenter(String name) {
```

```
        // Here are the steps needed to create a PowerDatacenter:
        // 1. We need to create a list to store one or more Machines
        List<Host> hostList = new ArrayList<Host>();

        // 2. A Machine contains one or more PEs or CPUs/Cores. Therefore,
should
        //    create a list to store these PEs before creating a Machine.
        List<Pe> peList = new ArrayList<Pe>();

        int mips = 1000;

        // 3. Create PEs and add these into the list.
        peList.add(new Pe(0, new PeProvisionerSimple(mips)));

        //4. Create Hosts with its id and list of PEs and add them to the
list of machines
        int hostId = 0;
        int ram = 2048; //host memory (MB)
        long storage = 1000000; //host storage
        int bw = 10000;

        hostList.add(
                new Host(
                        hostId,
                        new RamProvisionerSimple(ram),
                        new BwProvisionerSimple(bw),
                        storage,
                        peList,
                        new VmSchedulerTimeShared(peList)
                )
        ); // This is our first machine
```

```
        // 5. Create a DatacenterCharacteristics object that stores the
        //    properties of a data center: architecture, OS, list of
        //    Machines, allocation policy: time- or space-shared, time zone
        //    and its price (G$/Pe time unit).
        String arch = "x86";      // system architecture
        String os = "Linux";          // operating system
        String vmm = "Xen";
        double time_zone = 10.0;         // time zone this resource located
        double cost = 3.0;              // the cost of using processing in
this resource
        double costPerMem = 0.05;        // the cost of using memory in
this resource
        double costPerStorage = 0.1;    // the cost of using storage in
this resource
        double costPerBw = 0.1;            // the cost of using bw in this
resource
        LinkedList<Storage> storageList = new LinkedList<Storage>();
//we are not adding SAN devices by now


        DatacenterCharacteristics characteristics = new
DatacenterCharacteristics(
                arch, os, vmm, hostList, time_zone, cost, costPerMem,
costPerStorage, costPerBw);




        // 6. Finally, we need to create a PowerDatacenter object.
        Datacenter datacenter = null;
        try {
            datacenter = new Datacenter(name, characteristics, new
VmAllocationPolicySimple(hostList), storageList, 0);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return datacenter;
    }
}
```

SJF_Scheduler.java :

```
package com.scheduling;

import org.cloudbus.cloudsim.*;

import org.cloudbus.cloudsim.core.CloudSim;

import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;

import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;

import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;

import com.utils.Constants;

import com.utils.DatacenterCreator;

import com.utils.GenerateMatrices;

import java.text.DecimalFormat;

import java.util.ArrayList;

import java.util.Calendar;

import java.util.LinkedList;

import java.util.List;

public class SJF_Scheduler {

    private static List<Cloudlet> cloudletList;

    private static List<Vm> vmList;

    private static Datacenter[] datacenter;

    private static double[][] commMatrix;

    private static double[][] execMatrix;

    private static List<Vm> createVM(int userId, int vms) {

        //Creates a container to store VMs. This list is passed to the
broker later

        LinkedList<Vm> list = new LinkedList<Vm>();


        //VM Parameters

        long size = 10000; //image size (MB)

        int ram = 512; //vm memory (MB)

        int mips = 250;

        long bw = 1000;

        int pesNumber = 1; //number of cpus
```

```
        String vmm = "Xen"; //VMM name

        //create VMs

        Vm[] vm = new Vm[vms];


        for (int i = 0; i < vms; i++) {

            vm[i] = new Vm(datacenter[i].getId(), userId, mips, pesNumber,
ram, bw, size, vmm, new CloudletSchedulerSpaceShared());

            list.add(vm[i]);

        }


        return list;

    }

    private static List<Cloudlet> createCloudlet(int userId, int cloudlets,
int idShift) {

        // Creates a container to store Cloudlets

        LinkedList<Cloudlet> list = new LinkedList<Cloudlet>();


        //cloudlet parameters

        long fileSize = 300;

        long outputSize = 300;

        int pesNumber = 1;

        UtilizationModel utilizationModel = new UtilizationModelFull();


        Cloudlet[] cloudlet = new Cloudlet[cloudlets];


        for (int i = 0; i < cloudlets; i++) {

            int dcId = (int) (Math.random() *
Constants.NO_OF_DATA_CENTERS);

            long length = (long) (1e3 * (commMatrix[i][dcId] +
execMatrix[i][dcId]));

            cloudlet[i] = new Cloudlet(idShift + i, length, pesNumber,
fileSize, outputSize, utilizationModel, utilizationModel,
utilizationModel);

            // setting the owner of these Cloudlets

            cloudlet[i].setUserId(userId);

            cloudlet[i].setVmId(dcId + 2);
```

```
            list.add(cloudlet[i]);
        }
        return list;
    }


public static void main(String[] args) {
        Log.printLine("Starting SJF Scheduler...");

        new GenerateMatrices();
        execMatrix = GenerateMatrices.getExecMatrix();
        commMatrix = GenerateMatrices.getCommMatrix();

        try {
            int num_user = 1;   // number of grid users
            Calendar calendar = Calendar.getInstance();
            boolean trace_flag = false;  // mean trace events

            CloudSim.init(num_user, calendar, trace_flag);

            // Second step: Create Datacenters
            datacenter = new Datacenter[Constants.NO_OF_DATA_CENTERS];
            for (int i = 0; i < Constants.NO_OF_DATA_CENTERS; i++) {
                datacenter[i] =
DatacenterCreator.createDatacenter("Datacenter_" + i);
            }

            //Third step: Create Broker
            SJFDatacenterBroker broker = createBroker("Broker_0");
            int brokerId = broker.getId();

            //Fourth step: Create VMs and Cloudlets and send them to broker
            vmList = createVM(brokerId, Constants.NO_OF_DATA_CENTERS);
```

```
            cloudletList = createCloudlet(brokerId, Constants.NO_OF_TASKS,
0);

            broker.submitVmList(vmList);
            broker.submitCloudletList(cloudletList);

            // Fifth step: Starts the simulation
            CloudSim.startSimulation();

            // Final step: Print results when simulation is over
            List<Cloudlet> newList = broker.getCloudletReceivedList();

//newList.addAll(globalBroker.getBroker().getCloudletReceivedList());

            CloudSim.stopSimulation();

            printCloudletList(newList);

            Log.printLine(SJF_Scheduler.class.getName() + " finished!");
        } catch (Exception e) {
            e.printStackTrace();
            Log.printLine("The simulation has been terminated due to an
unexpected error");
        }
    }

    private static SJFDatacenterBroker createBroker(String name) throws
Exception {
        return new SJFDatacenterBroker(name);
    }

    /**
     * Prints the Cloudlet objects
     *
     * @param list list of Cloudlets
```

```
     */
    private static void printCloudletList(List<Cloudlet> list) {
        int size = list.size();
        Cloudlet cloudlet;

        String indent = "    ";
        Log.printLine();
        Log.printLine("========== OUTPUT ==========");
        Log.printLine("Cloudlet ID" + indent + "STATUS" +
                indent + "Data center ID" +
                indent + "VM ID" +
                indent + indent + "Time" +
                indent + "Start Time" +
                indent + "Finish Time");

        DecimalFormat dft = new DecimalFormat("###.##");
        dft.setMinimumIntegerDigits(2);
        for (int i = 0; i < size; i++) {
            cloudlet = list.get(i);
            Log.print(indent + dft.format(cloudlet.getCloudletId()) +
indent + indent);

            if (cloudlet.getCloudletStatus() == Cloudlet.SUCCESS) {
                Log.print("SUCCESS");

                Log.printLine(indent + indent +
dft.format(cloudlet.getResourceId()) +
                        indent + indent + indent +
dft.format(cloudlet.getVmId()) +
                        indent + indent +
dft.format(cloudlet.getActualCPUTime()) +
                        indent + indent +
dft.format(cloudlet.getExecStartTime()) +
                        indent + indent + indent +
dft.format(cloudlet.getFinishTime()));
            }
```

```
        }
        double makespan = calcMakespan(list);
        Log.printLine("Makespan using SJF: " + makespan);
    }

    private static double calcMakespan(List<Cloudlet> list) {
        double makespan = 0;
        double[] dcWorkingTime = new double[Constants.NO_OF_DATA_CENTERS];

        for (int i = 0; i < Constants.NO_OF_TASKS; i++) {
            int dcId = list.get(i).getVmId() %
Constants.NO_OF_DATA_CENTERS;
            if (dcWorkingTime[dcId] != 0) --dcWorkingTime[dcId];
            dcWorkingTime[dcId] += execMatrix[i][dcId] +
commMatrix[i][dcId];
            makespan = Math.max(makespan, dcWorkingTime[dcId]);
        }
        return makespan;
    }
}
```

SJFDatacenterBroker.java

```
package com.scheduling;
import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.core.CloudSimTags;
import org.cloudbus.cloudsim.core.SimEvent;
import java.util.ArrayList;
import java.util.List;

public class SJFDatacenterBroker extends DatacenterBroker {

    SJFDatacenterBroker(String name) throws Exception {
        super(name);
```

```
    }

    public void scheduleTaskstoVms() {
        int reqTasks = cloudletList.size();
        int reqVms = vmList.size();
        Vm vm = vmList.get(0);

        for (int i = 0; i < reqTasks; i++) {
            bindCloudletToVm(i, (i % reqVms));
            System.out.println("Task" + cloudletList.get(i).getCloudletId()
+ " is bound with VM" + vmList.get(i % reqVms).getId());
        }

        //System.out.println("reqTasks: "+ reqTasks);

        ArrayList<Cloudlet> list = new ArrayList<Cloudlet>();
        for (Cloudlet cloudlet : getCloudletReceivedList()) {
            list.add(cloudlet);
        }

        //setCloudletReceivedList(null);

        Cloudlet[] list2 = list.toArray(new Cloudlet[list.size()]);

        //System.out.println("size :"+list.size());

        Cloudlet temp = null;

        int n = list.size();

        for (int i = 0; i < n; i++) {
            for (int j = 1; j < (n - i); j++) {
```

```
                if (list2[j - 1].getCloudletLength() / (vm.getMips() *
vm.getNumberOfPes()) > list2[j].getCloudletLength() / (vm.getMips() *
vm.getNumberOfPes())) {
                    //swap the elements!
                    //swap(list2[j-1], list2[j]);
                    temp = list2[j - 1];
                    list2[j - 1] = list2[j];
                    list2[j] = temp;
                }
                // printNumbers(list2);
            }
        }

        ArrayList<Cloudlet> list3 = new ArrayList<Cloudlet>();

        for (int i = 0; i < list2.length; i++) {
            list3.add(list2[i]);
        }
        //printNumbers(list);

        setCloudletReceivedList(list);

        //System.out.println("\n\tSJFS Broker Schedules\n");
        //System.out.println("\n");
    }

    public void printNumber(Cloudlet[] list) {
        for (int i = 0; i < list.length; i++) {
            System.out.print(" " + list[i].getCloudletId());
            System.out.println(list[i].getCloudletStatusString());
        }
        System.out.println();
    }
```

```
    public void printNumbers(ArrayList<Cloudlet> list) {
        for (int i = 0; i < list.size(); i++) {
            System.out.print(" " + list.get(i).getCloudletId());
        }
        System.out.println();
    }

    @Override
    protected void processCloudletReturn(SimEvent ev) {
        Cloudlet cloudlet = (Cloudlet) ev.getData();
        getCloudletReceivedList().add(cloudlet);
        Log.printLine(CloudSim.clock() + ": " + getName() + ": Cloudlet " +
cloudlet.getCloudletId()
                + " received");
        cloudletsSubmitted--;
        if (getCloudletList().size() == 0 && cloudletsSubmitted == 0) {
            scheduleTaskstoVms();
            cloudletExecution(cloudlet);
        }
    }

    protected void cloudletExecution(Cloudlet cloudlet) {

        if (getCloudletList().size() == 0 && cloudletsSubmitted == 0) { //
all cloudlets executed
            Log.printLine(CloudSim.clock() + ": " + getName() + ": All
Cloudlets executed. Finishing...");
            clearDatacenters();
            finishExecution();
        } else { // some cloudlets haven't finished yet
            if (getCloudletList().size() > 0 && cloudletsSubmitted == 0) {
                // all the cloudlets sent finished. It means that some
bount
                // cloudlet is waiting its VM be created
                clearDatacenters();
```

```
                createVmsInDatacenter(0);

            }

        }

    }


    @Override

    protected void processResourceCharacteristics(SimEvent ev) {

        DatacenterCharacteristics characteristics =
(DatacenterCharacteristics) ev.getData();

        getDatacenterCharacteristicsList().put(characteristics.getId(),
characteristics);



        if (getDatacenterCharacteristicsList().size() ==
getDatacenterIdsList().size()) {

            distributeRequestsForNewVmsAcrossDatacenters();

        }

    }



    protected void distributeRequestsForNewVmsAcrossDatacenters() {

        int numberOfVmsAllocated = 0;

        int i = 0;



        final List<Integer> availableDatacenters = getDatacenterIdsList();



        for (Vm vm : getVmList()) {

            int datacenterId = availableDatacenters.get(i++ %
availableDatacenters.size());

            String datacenterName = CloudSim.getEntityName(datacenterId);



            if (!getVmsToDatacentersMap().containsKey(vm.getId())) {

                Log.printLine(CloudSim.clock() + ": " + getName() + ":
Trying to Create VM #" + vm.getId() + " in " + datacenterName);

                sendNow(datacenterId, CloudSimTags.VM_CREATE_ACK, vm);

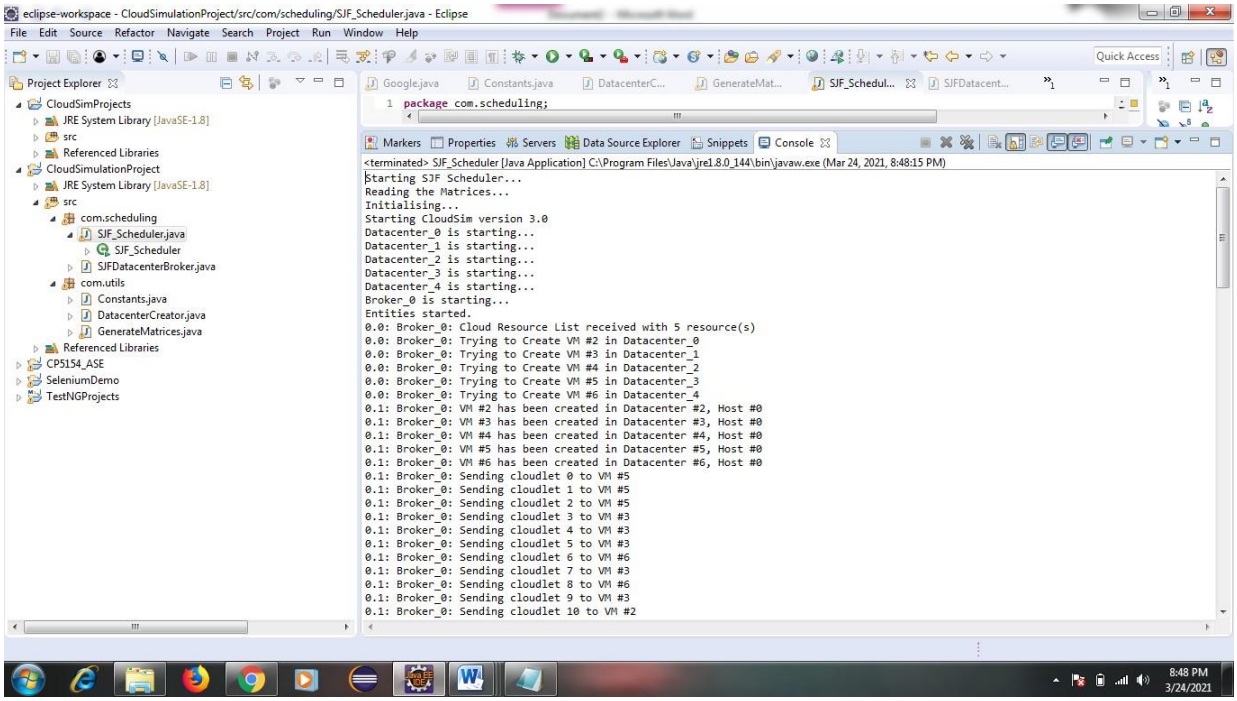
                numberOfVmsAllocated++;

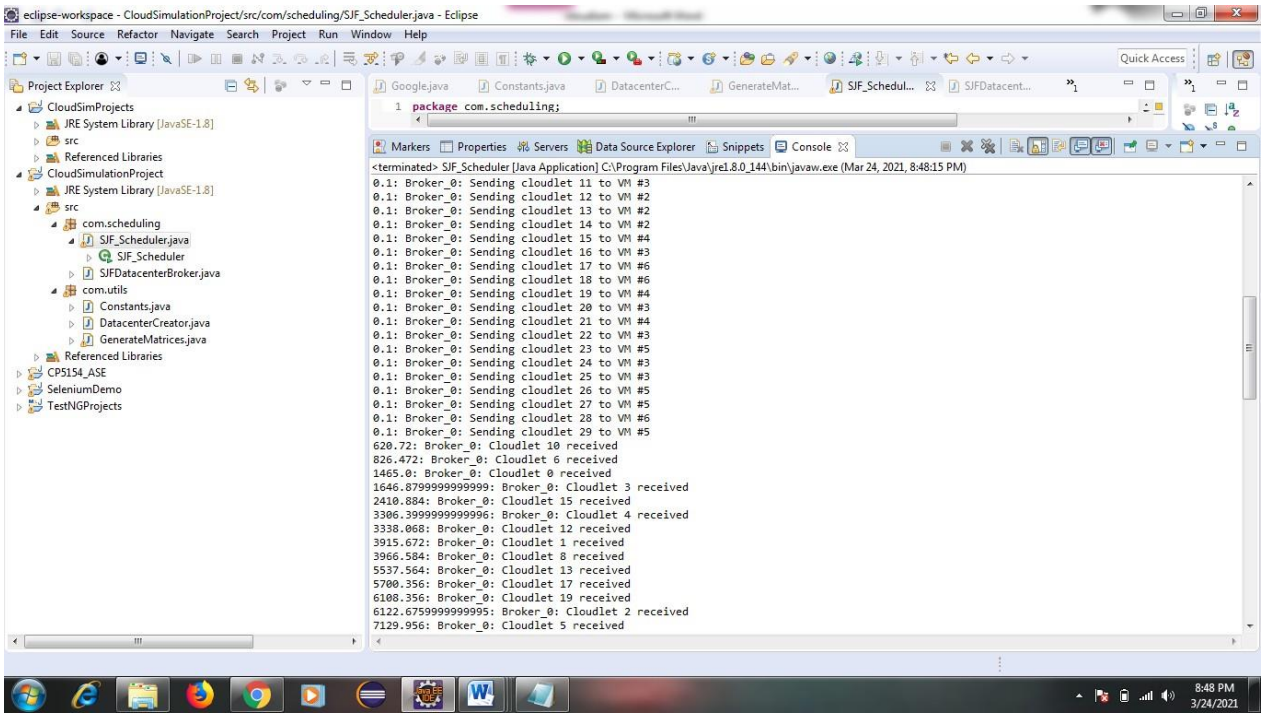
            }
```

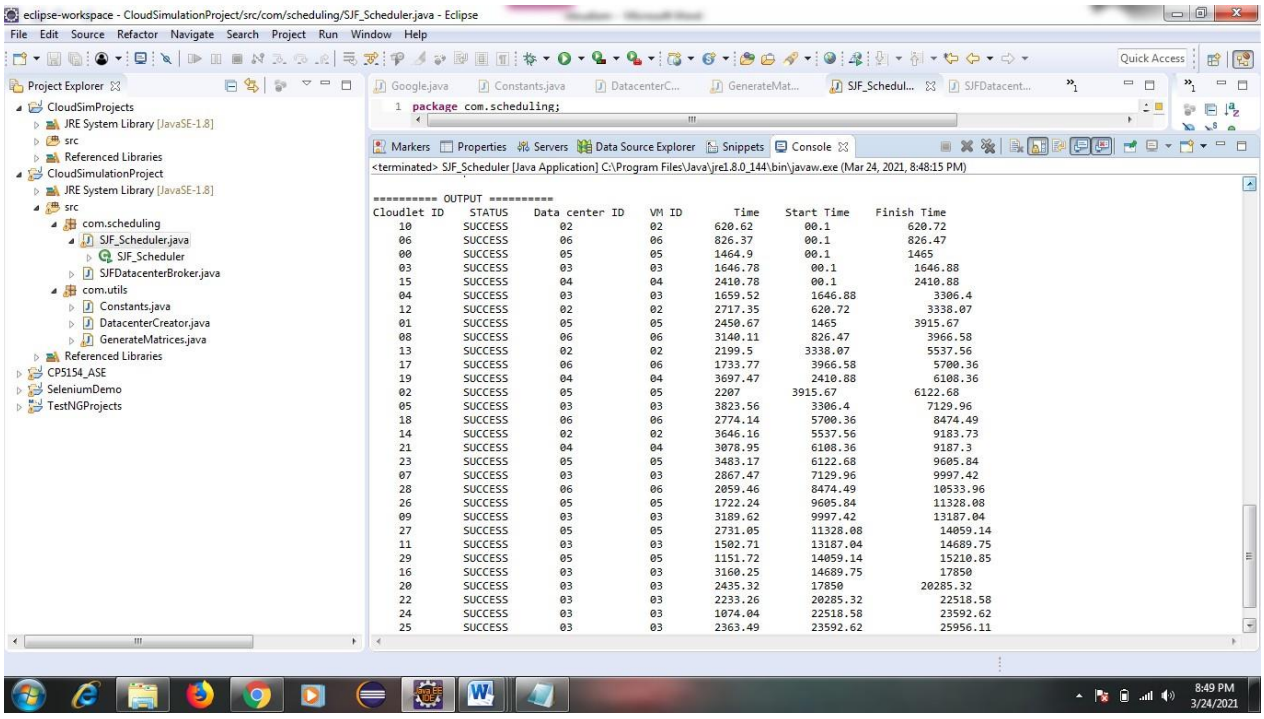
```
        }

        setVmsRequested(numberOfVmsAllocated);
        setVmsAcks(0);
    }
}
```

OUTPUT / SCREENSHOTS:




eclipse-workspace - CloudSimulationProject/src/com/scheduling/SJF_Scheduler.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
CloudSimProjects
JRE System Library [JavaSE-1.8]
src
Referenced Libraries
CloudSimulationProject
JRE System Library [JavaSE-1.8]
src
com.scheduling
SJF_Scheduler.java
SJF_Scheduler
SJFDatacenterBroker.java
com.utils
Constants.java
DatacenterCreator.java
GenerateMatrices.java
Referenced Libraries
CP5154_ASE
SeleniumDemo
TestNGProjects
1 package com.scheduling;
Markers Properties Servers Data Source Explorer Snippets Console
<terminated> SJF_Scheduler [Java Application] C:\Program Files\Java\jre1.8.0_144\bin\javaw.exe (Mar 24, 2021, 8:48:15 PM)
0.1: Broker_0: Sending cloudlet 11 to VM #3
0.1: Broker_0: Sending cloudlet 12 to VM #2
0.1: Broker_0: Sending cloudlet 13 to VM #2
0.1: Broker_0: Sending cloudlet 14 to VM #2
0.1: Broker_0: Sending cloudlet 15 to VM #4
0.1: Broker_0: Sending cloudlet 16 to VM #3
0.1: Broker_0: Sending cloudlet 17 to VM #6
0.1: Broker_0: Sending cloudlet 18 to VM #6
0.1: Broker_0: Sending cloudlet 19 to VM #4
0.1: Broker_0: Sending cloudlet 20 to VM #3
0.1: Broker_0: Sending cloudlet 21 to VM #4
0.1: Broker_0: Sending cloudlet 22 to VM #3
0.1: Broker_0: Sending cloudlet 23 to VM #5
0.1: Broker_0: Sending cloudlet 24 to VM #3
0.1: Broker_0: Sending cloudlet 25 to VM #3
0.1: Broker_0: Sending cloudlet 26 to VM #5
0.1: Broker_0: Sending cloudlet 27 to VM #5
0.1: Broker_0: Sending cloudlet 28 to VM #6
0.1: Broker_0: Sending cloudlet 29 to VM #5
620.72: Broker_0: Cloudlet 10 received
826.472: Broker_0: Cloudlet 6 received
1465.0: Broker_0: Cloudlet 0 received
1646.8799999999999: Broker_0: Cloudlet 3 received
2410.884: Broker_0: Cloudlet 15 received
3306.3999999999996: Broker_0: Cloudlet 4 received
3338.068: Broker_0: Cloudlet 12 received
3915.672: Broker_0: Cloudlet 1 received
3966.584: Broker_0: Cloudlet 8 received
5537.564: Broker_0: Cloudlet 13 received
5700.356: Broker_0: Cloudlet 17 received
6108.356: Broker_0: Cloudlet 19 received
6122.6759999999995: Broker_0: Cloudlet 2 received
7129.956: Broker_0: Cloudlet 5 received
8:48 PM
3/24/2021


eclipse-workspace - CloudSimulationProject/src/com/scheduling/SJF_Scheduler.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
CloudSimProjects
JRE System Library [JavaSE-1.8]
src
Referenced Libraries
CloudSimulationProject
JRE System Library [JavaSE-1.8]
src
com.scheduling
SJF_Scheduler.java
SJF_Scheduler
SJFDatacenterBroker.java
com.utils
Constants.java
DatacenterCreator.java
GenerateMatrices.java
Referenced Libraries
CP5154_ASE
SeleniumDemo
TestNGProjects
1 package com.scheduling;
Markers Properties Servers Data Source Explorer Snippets Console
<terminated> SJF_Scheduler [Java Application] C:\Program Files\Java\jre1.8.0_144\bin\javaw.exe (Mar 24, 2021, 8:48:15 PM)
========== OUTPUT ==========
Cloudlet ID STATUS Data center ID VM ID Time Start Time Finish Time
10 SUCCESS 02 02 620.62 00.1 620.72
06 SUCCESS 06 06 826.37 00.1 826.47
00 SUCCESS 05 05 1464.9 00.1 1465
03 SUCCESS 03 03 1646.78 00.1 1646.88
15 SUCCESS 04 04 2410.78 00.1 2410.88
04 SUCCESS 03 03 1659.52 1646.88 3306.4
12 SUCCESS 02 02 2717.35 620.72 3338.07
01 SUCCESS 05 05 2450.67 1465 3915.67
08 SUCCESS 06 06 3140.11 826.47 3966.58
13 SUCCESS 02 02 2199.5 3338.07 5537.56
17 SUCCESS 06 06 1733.77 3966.58 5700.36
19 SUCCESS 04 04 3697.47 2410.88 6108.36
02 SUCCESS 05 05 2207 3915.67 6122.68
05 SUCCESS 03 03 3823.56 3306.4 7129.96
18 SUCCESS 06 06 2774.14 5700.36 8474.49
14 SUCCESS 02 02 3646.16 5537.56 9183.73
21 SUCCESS 04 04 3078.95 6108.36 9187.3
23 SUCCESS 05 05 3483.17 6122.68 9605.84
07 SUCCESS 03 03 2867.47 7129.96 9997.42
28 SUCCESS 06 06 2059.46 8474.49 10533.96
26 SUCCESS 05 05 1722.24 9605.84 11328.08
09 SUCCESS 03 03 3189.62 9997.42 13187.04
27 SUCCESS 05 05 2731.05 11328.08 14059.14
11 SUCCESS 03 03 1502.71 13187.04 14689.75
29 SUCCESS 05 05 1151.72 14059.14 15210.85
16 SUCCESS 03 03 3160.25 14689.75 17850
20 SUCCESS 03 03 2435.32 17850 20285.32
22 SUCCESS 03 03 2233.26 20285.32 22518.58
24 SUCCESS 03 03 1074.04 22518.58 23592.62
25 SUCCESS 03 03 2363.49 23592.62 25956.11
8:49 PM
3/24/2021

RESULT:

EXP.NO:6 DATE:	TRANSFER FILES FROM ONE VIRTUAL MACHINE TO ANOTHER VIRTUAL MACHINE

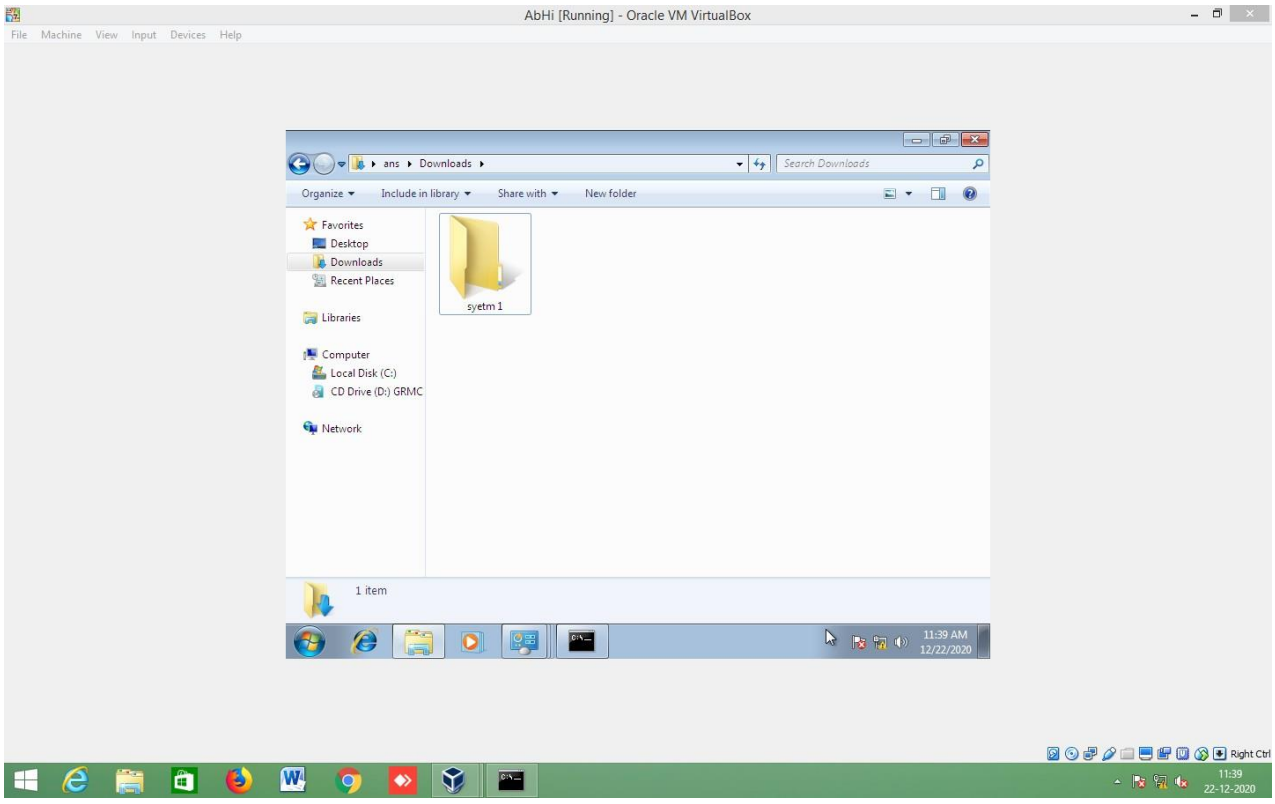
AIM:

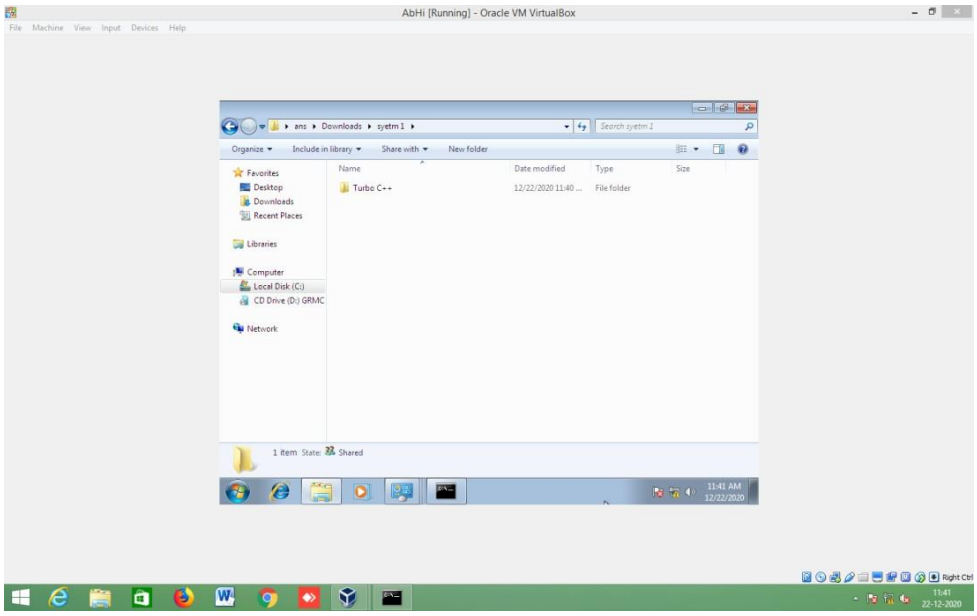
To write a procedure to transfer the files form one virtual machine to another virtual machine.

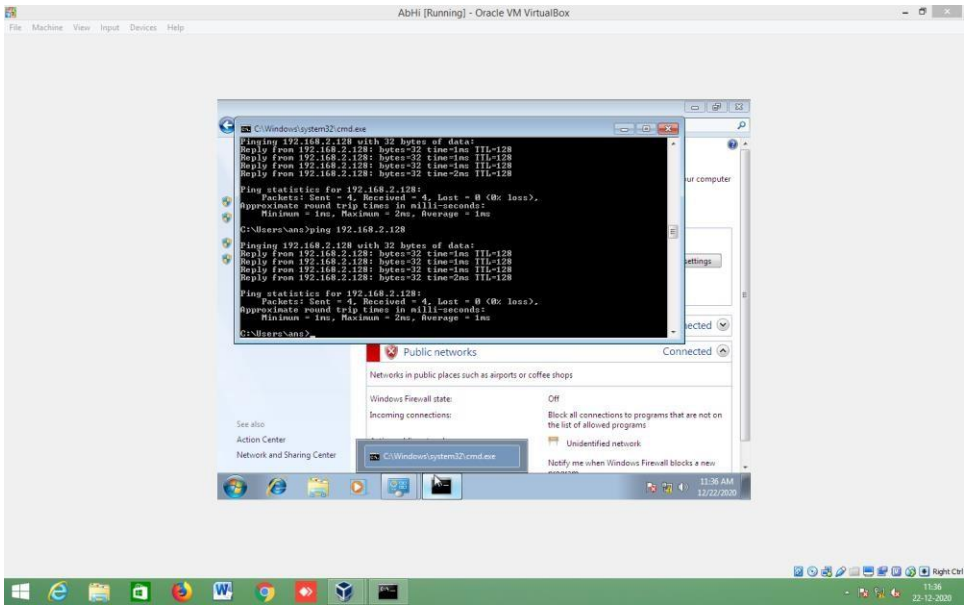
PROCEDURE:

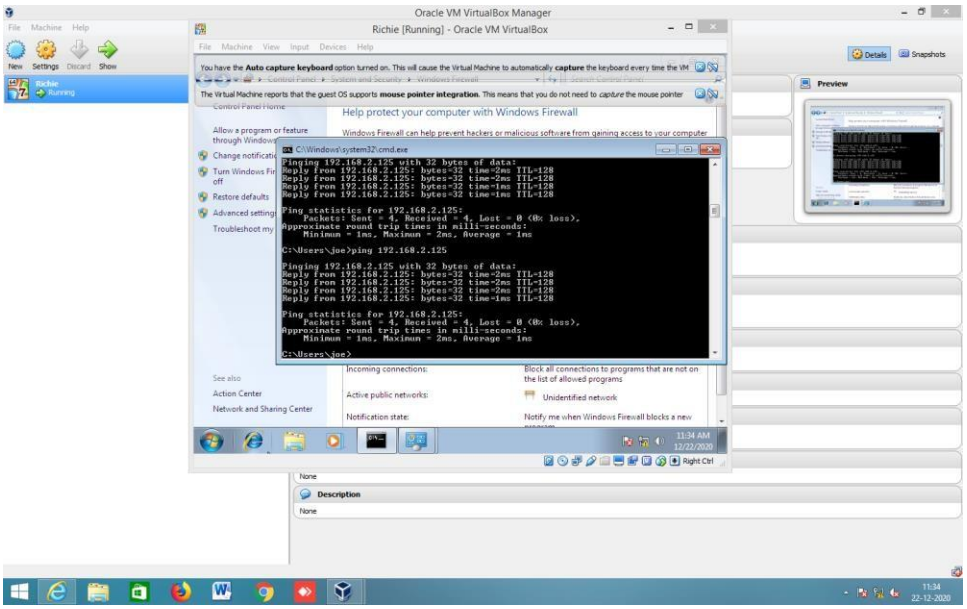
1. Create Virtual Machines in the Network using Oracle Virtualbox.
2. Assign IP address for the each VM.
3. Create a shared folder in One Virtual VM.
4. Ping from one VM to another VM for checking the connection.
5. Copy the files from one VM to another VM.

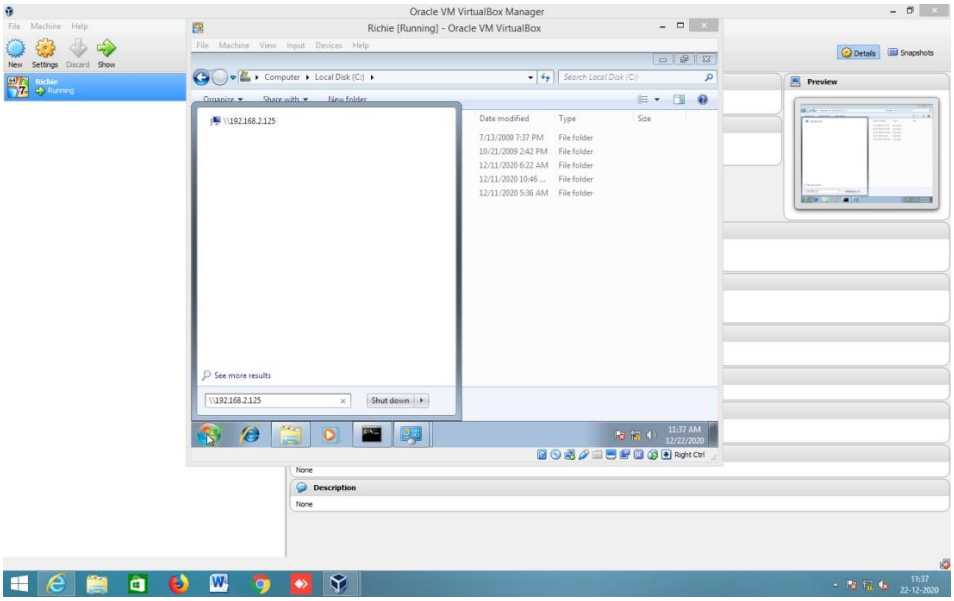
OUTPUT / SCREENSHOTS:

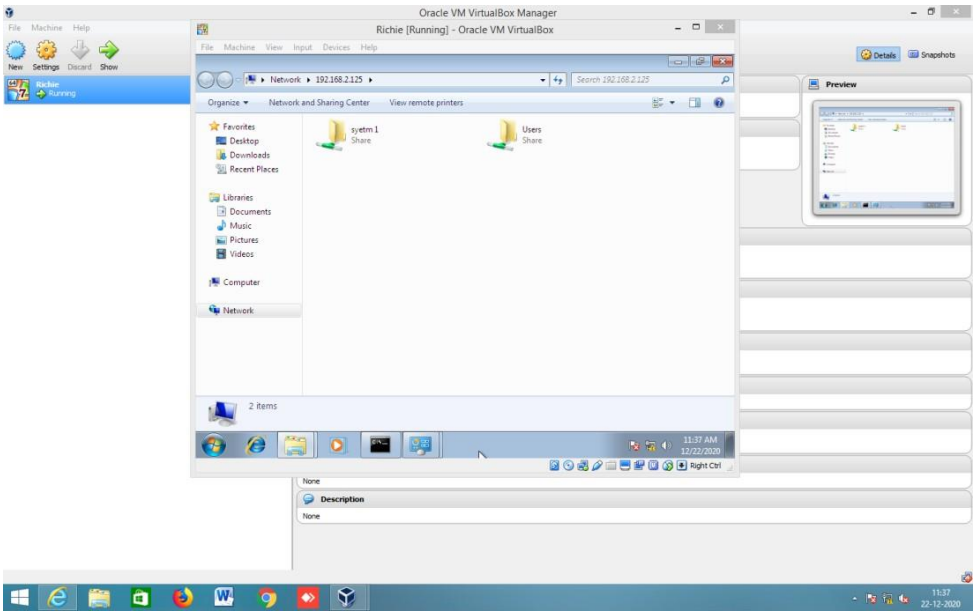


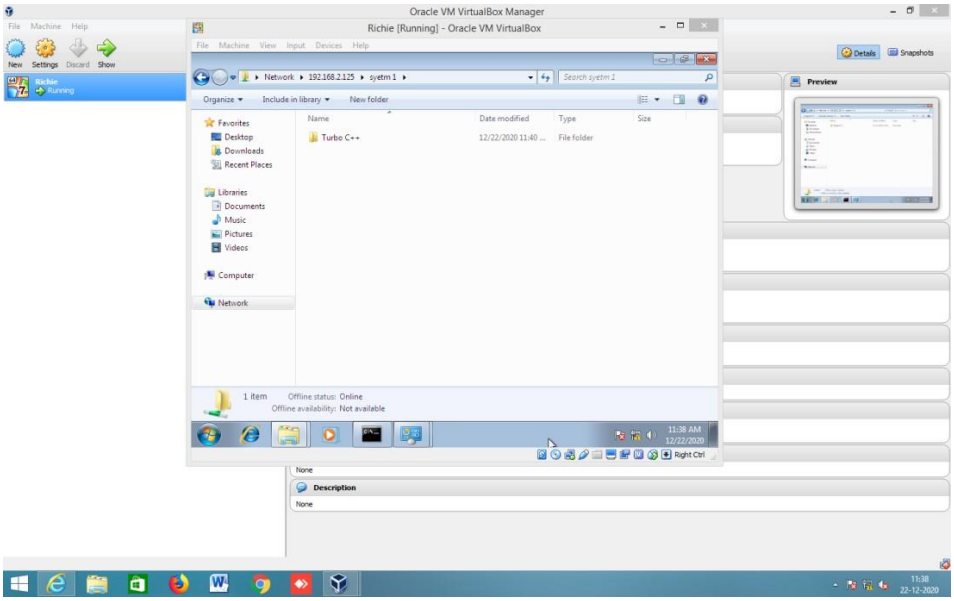

AbHi [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
ans ▸ Downloads ▸ syetm 1 ▸
Search syetm 1
Organize
Include in library
Share with
New folder
Favorites
Desktop
Downloads
Recent Places
Libraries
Computer
Local Disk (C:)
CD Drive (D:) GRMC
Network
Name
Date modified
Type
Size
Turbo C++
12/22/2020 11:40 ...
File folder
1 item State: Shared
11:41 AM
12/22/2020
Right Ctrl
11:41
22-12-2020

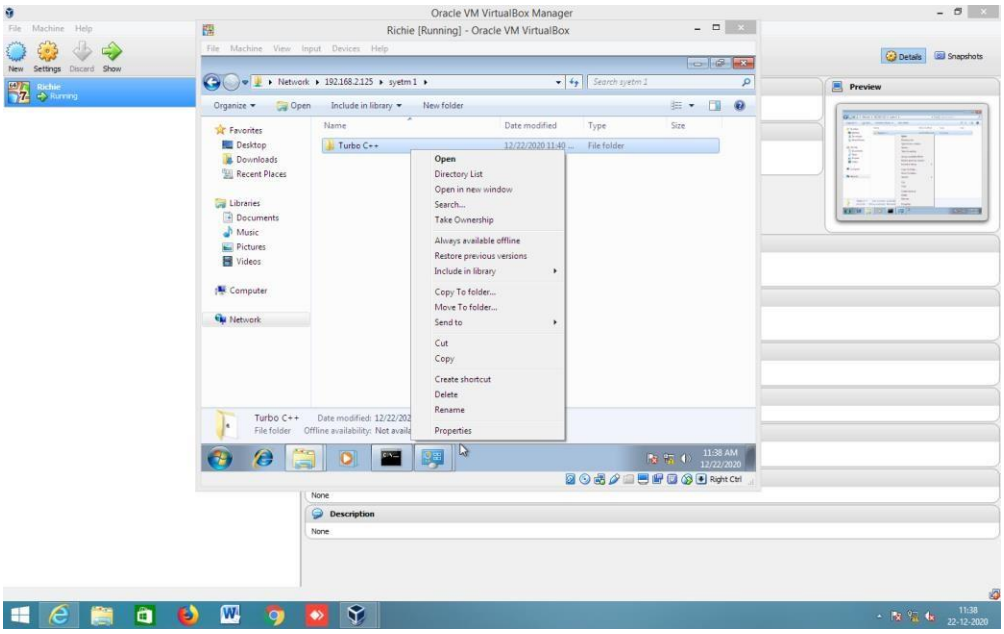

AbHi [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
C:\Windows\system32\cmd.exe
Pinging 192.168.2.128 with 32 bytes of data:
Reply from 192.168.2.128: bytes=32 time=1ms TTL=128
Reply from 192.168.2.128: bytes=32 time=1ms TTL=128
Reply from 192.168.2.128: bytes=32 time=1ms TTL=128
Reply from 192.168.2.128: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.2.128:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\ans>ping 192.168.2.128
Pinging 192.168.2.128 with 32 bytes of data:
Reply from 192.168.2.128: bytes=32 time=1ms TTL=128
Reply from 192.168.2.128: bytes=32 time=1ms TTL=128
Reply from 192.168.2.128: bytes=32 time=1ms TTL=128
Reply from 192.168.2.128: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.2.128:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\ans>
Public networks
Connected
Networks in public places such as airports or coffee shops
Windows Firewall state:
Off
Incoming connections:
Block all connections to programs that are not on the list of allowed programs
Unidentified network
Notify me when Windows Firewall blocks a new
See also
Action Center
Network and Sharing Center
11:36 AM
12/22/2020
Right Ctrl
11:36
22-12-2020

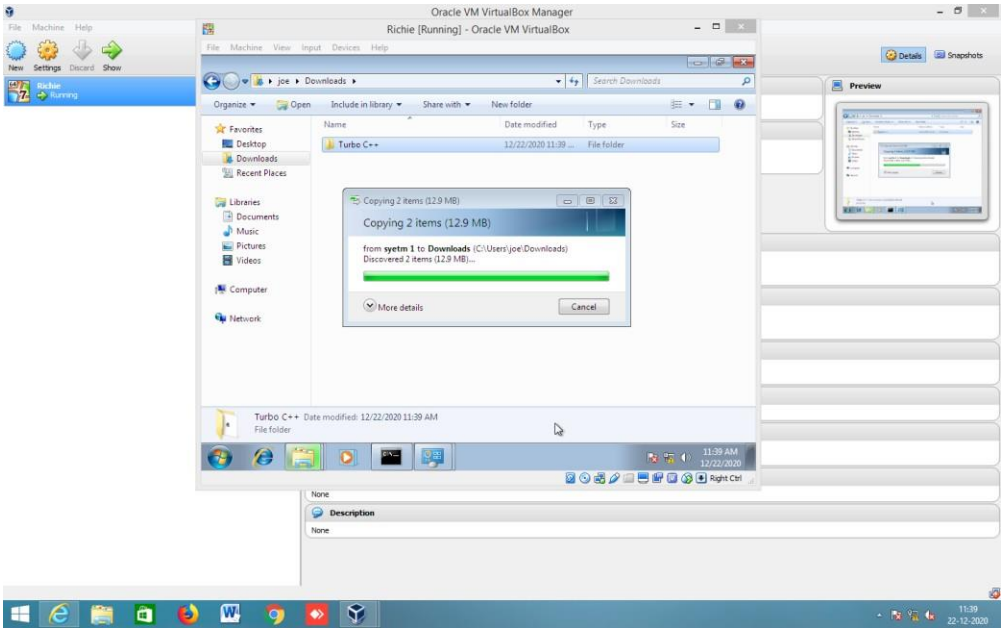

Oracle VM VirtualBox Manager
Richie [Running] - Oracle VM VirtualBox
Help protect your computer with Windows Firewall
C:\Windows\system32\cmd.exe
Pinging 192.168.2.125 with 32 bytes of data:
Reply from 192.168.2.125: bytes=32 time=2ms TTL=128
Reply from 192.168.2.125: bytes=32 time=2ms TTL=128
Reply from 192.168.2.125: bytes=32 time=1ms TTL=128
Reply from 192.168.2.125: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.2.125:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\joe>ping 192.168.2.125
Pinging 192.168.2.125 with 32 bytes of data:
Reply from 192.168.2.125: bytes=32 time=2ms TTL=128
Reply from 192.168.2.125: bytes=32 time=2ms TTL=128
Reply from 192.168.2.125: bytes=32 time=2ms TTL=128
Reply from 192.168.2.125: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.2.125:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\joe>
Description
None


Oracle VM VirtualBox Manager
Richie [Running] - Oracle VM VirtualBox
\\192.168.2.125
See more results
Shut down
Description
None


Oracle VM VirtualBox Manager
Richie [Running] - Oracle VM VirtualBox
Network and Sharing Center
syetm 1
Users
2 items
Description
None


Oracle VM VirtualBox Manager
Richie [Running] - Oracle VM VirtualBox
Turbo C++
1 item
Description
None


Oracle VM VirtualBox Manager
Richie [Running] - Oracle VM VirtualBox
Turbo C++
Open
Directory List
Open in new window
Search...
Take Ownership
Always available offline
Restore previous versions
Include in library
Copy To folder...
Move To folder...
Send to
Cut
Copy
Create shortcut
Delete
Rename
Properties


Oracle VM VirtualBox Manager
Richie [Running] - Oracle VM VirtualBox
Copying 2 items (12.9 MB)
from syetm 1 to Downloads (C:\Users\joe\Downloads)
Discovered 2 items (12.9 MB)...
More details
Cancel

RESULT:

EXP.NO:7 DATE:	INSTALL HADOOP SINGLE NODE CLUSTER AND RUN SIMPLE APPLICATIONS LIKE WORDCOUNT

AIM:

To install hadoop and create a single node cluster and run the word count program in hadoop.

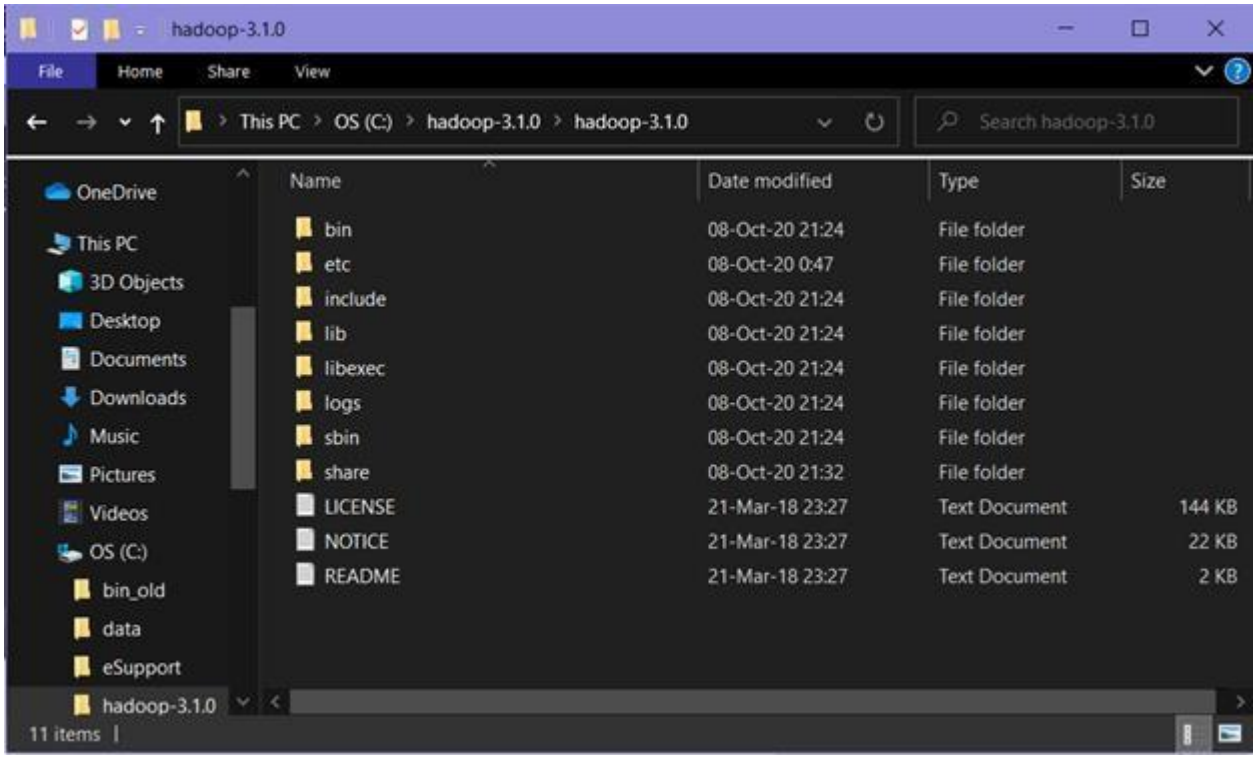
PROCEDURE

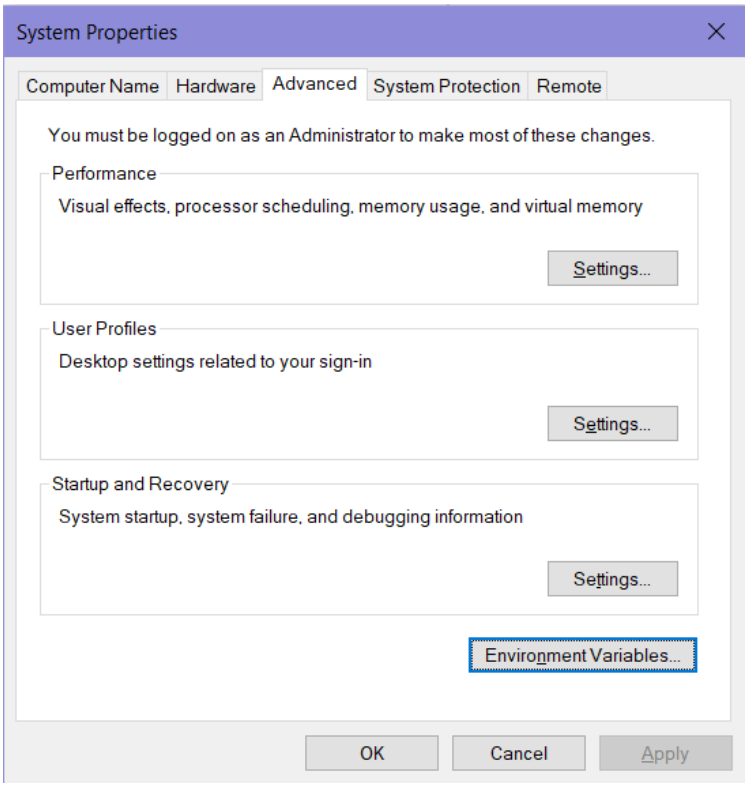
1. Download any java version, java 1.8 and save it in C folder "C:/java/jdk" click ok and it will start installing.
2. Then we have to save the jre file in the java folder.
3. The installion process starts.
4. Download Hadoop package, hadoop-3.1.0.
5. Download Hadoop configuration files.
6. Extract hadoop-3.1.0 tar file and save it in C folder.
7. Copy hadoop-3.1.0 and place it in C folder.
8. Path setting in environment variables for hadoop and java.
9. For that right click--->my computer--->choose properties--->select advanced system settings from the left side.
    a. A dialog box with system settings opens in that choose the environment variables and we must include the path.
    b. Choose new button and add path for both hadoop and java.
    c. The variable name is java home and find the bin directory path and copy it and place it in the variable name and click ok.
    d. Similarly add hadoop path, the variable name is hadoop home and find the bin directory path and edit the path settings click edit and click ok.
    e. Copy the path of hadoop bin directory and java bin directory in the path variable.
10. We can give the value by separating with the semicolon.
11. After editing and path setting everything is changed java home and hadoop home and path is also changed where we give the value of hadoop bin directory and java bin directory.
12. Next we must configure hadoop.
    a. For that open hadoop folder inside C drive and create a folder called data.

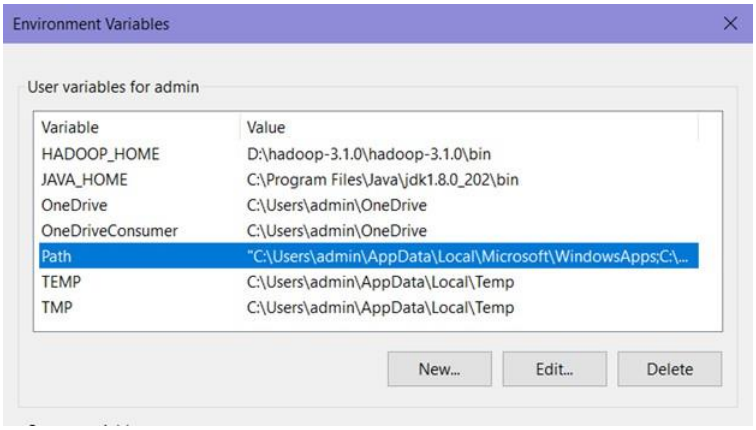
b. Inside data folder create two folders one is data node and the other is name node.

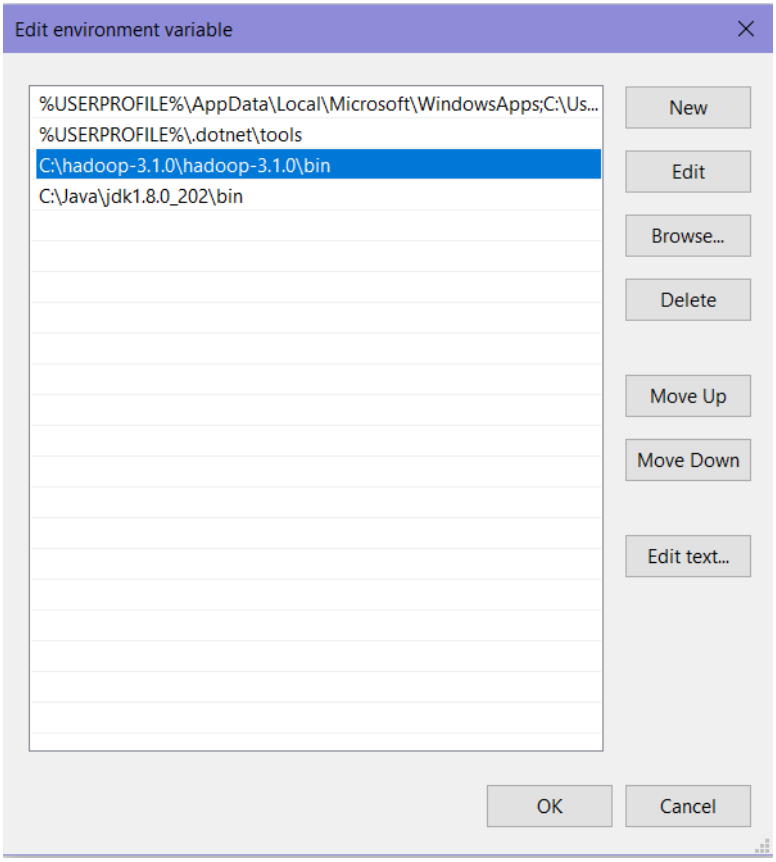
13. In C drive open hadoop folder hadoop 3.1.0 inside that you must choose the folder etc inside that we must choose hadoop.
14. Where we must choose the file coresite.xml.
15. We should specify the values one is name and corresponding value.
16. In the name we should give "fs.defaultfs" the file system denotes the address of the name node and the specify the value as hdfs:localhost:9000.
17. After typing save the particular file. Open the mapred-site.xml file and add the
18. Open the hdfssite.xml.

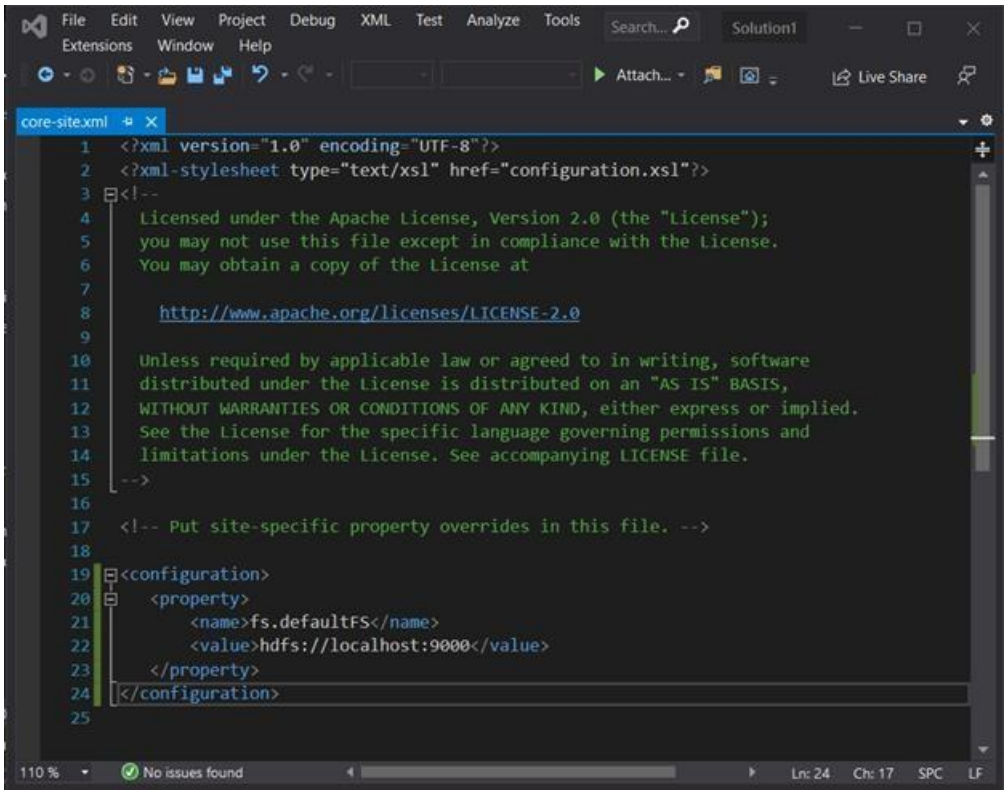
OUPUT/ SCREENSHOTS:

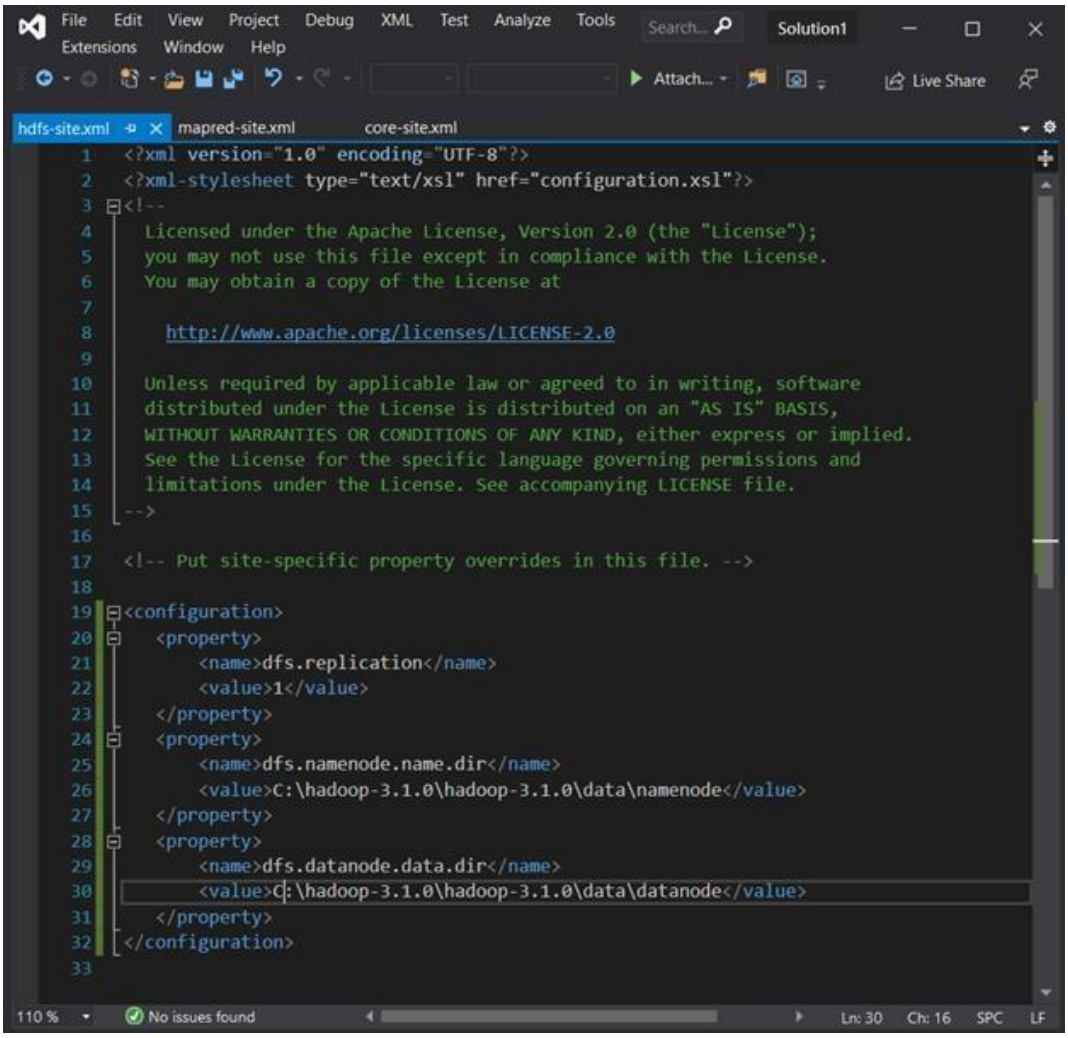


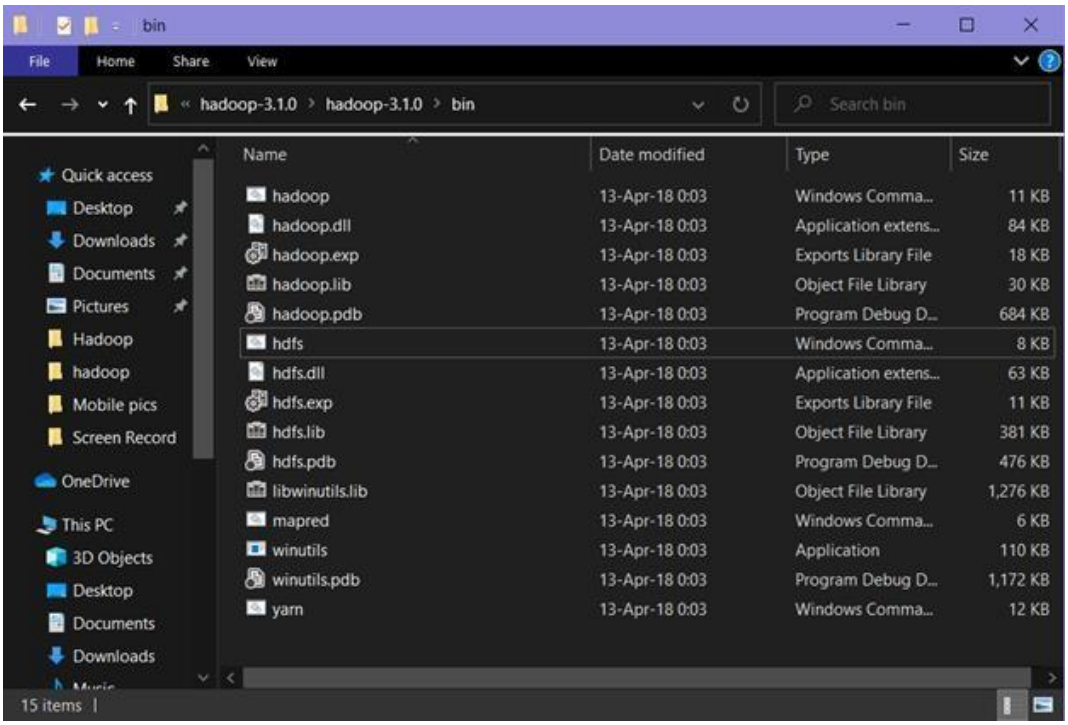

System Properties
Computer Name
Hardware
Advanced
System Protection
Remote
You must be logged on as an Administrator to make most of these changes.
Performance
Visual effects, processor scheduling, memory usage, and virtual memory
Settings...
User Profiles
Desktop settings related to your sign-in
Settings...
Startup and Recovery
System startup, system failure, and debugging information
Settings...
Environment Variables...
OK
Cancel
Apply


Environment Variables
User variables for admin
Variable
Value
HADOOP_HOME
D:\hadoop-3.1.0\hadoop-3.1.0\bin
JAVA_HOME
C:\Program Files\Java\jdk1.8.0_202\bin
OneDrive
C:\Users\admin\OneDrive
OneDriveConsumer
C:\Users\admin\OneDrive
Path
"C:\Users\admin\AppData\Local\Microsoft\WindowsApps;C:\...
TEMP
C:\Users\admin\AppData\Local\Temp
TMP
C:\Users\admin\AppData\Local\Temp
New...
Edit...
Delete

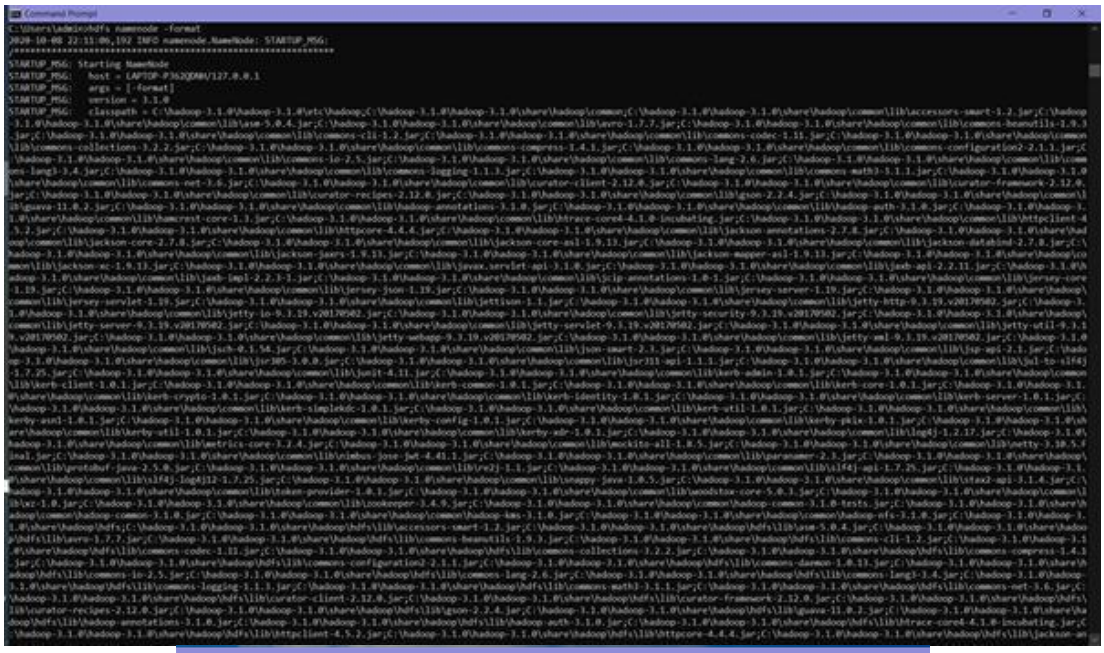

Edit environment variable
%USERPROFILE%\AppData\Local\Microsoft\WindowsApps;C:\Us...
%USERPROFILE%\.dotnet\tools
C:\hadoop-3.1.0\hadoop-3.1.0\bin
C:\Java\jdk1.8.0_202\bin
New
Edit
Browse...
Delete
Move Up
Move Down
Edit text...
OK
Cancel

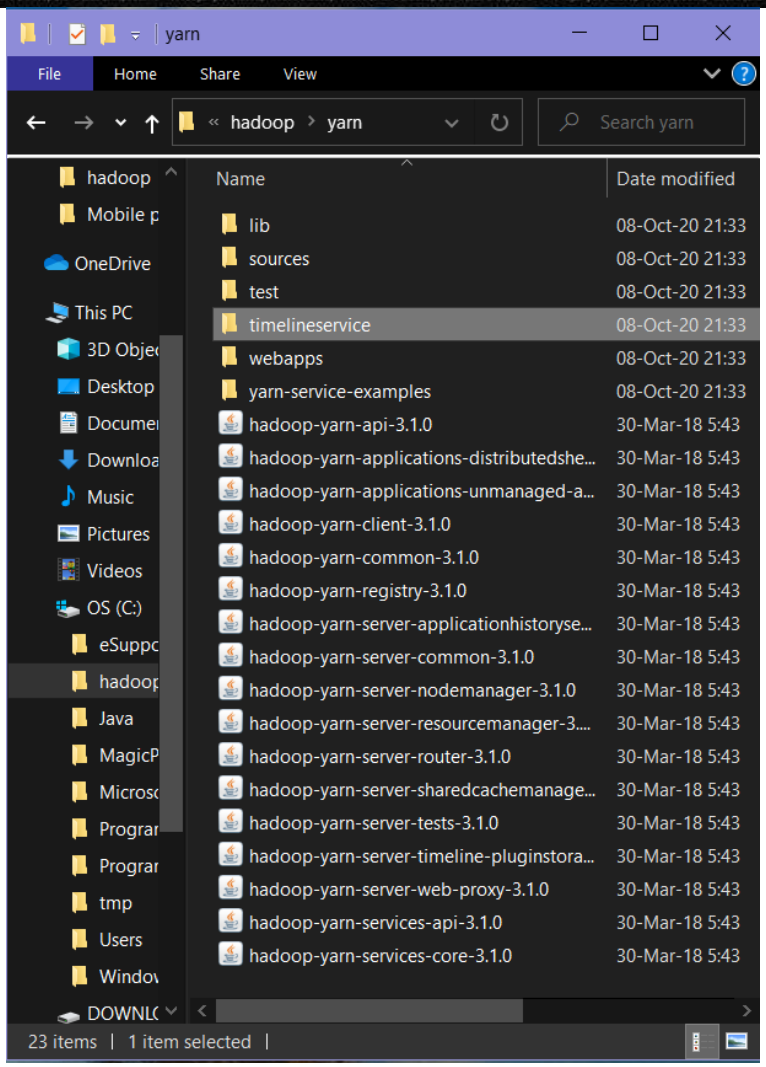

File Edit View Project Debug XML Test Analyze Tools
Extensions Window Help
Search...
Solution1
Attach...
Live Share
core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
Licensed under the Apache License, Version 2.0 (the "License");
you may not use this file except in compliance with the License.
You may obtain a copy of the License at
http://www.apache.org/licenses/LICENSE-2.0
Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->
<configuration>
<property>
<name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value>
</property>
</configuration>
110 %
No issues found
Ln: 24
Ch: 17
SPC
LF

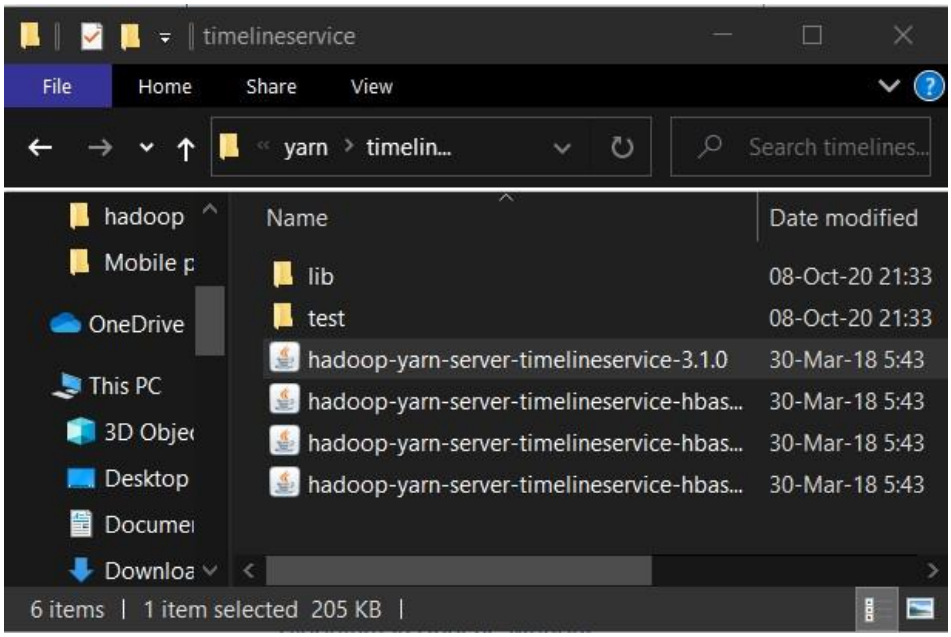

File Edit View Project Debug XML Test Analyze Tools
Extensions Window Help
Search...
Solution1
Attach...
Live Share
hdfs-site.xml
mapred-site.xml
core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
Licensed under the Apache License, Version 2.0 (the "License");
you may not use this file except in compliance with the License.
You may obtain a copy of the License at
http://www.apache.org/licenses/LICENSE-2.0
Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>C:\hadoop-3.1.0\hadoop-3.1.0\data\namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>C:\hadoop-3.1.0\hadoop-3.1.0\data\datanode</value>
</property>
</configuration>
110 %
No issues found
Ln: 30
Ch: 16
SPC
LF


bin
File
Home
Share
View
hadoop-3.1.0 › hadoop-3.1.0 › bin
Search bin
Quick access
Desktop
Downloads
Documents
Pictures
Hadoop
hadoop
Mobile pics
Screen Record
OneDrive
This PC
3D Objects
Desktop
Documents
Downloads
Name
Date modified
Type
Size
hadoop 13-Apr-18 0:03 Windows Comma... 11 KB
hadoop.dll 13-Apr-18 0:03 Application extens... 84 KB
hadoop.exp 13-Apr-18 0:03 Exports Library File 18 KB
hadoop.lib 13-Apr-18 0:03 Object File Library 30 KB
hadoop.pdb 13-Apr-18 0:03 Program Debug D... 684 KB
hdfs 13-Apr-18 0:03 Windows Comma... 8 KB
hdfs.dll 13-Apr-18 0:03 Application extens... 63 KB
hdfs.exp 13-Apr-18 0:03 Exports Library File 11 KB
hdfs.lib 13-Apr-18 0:03 Object File Library 381 KB
hdfs.pdb 13-Apr-18 0:03 Program Debug D... 476 KB
libwinutils.lib 13-Apr-18 0:03 Object File Library 1,276 KB
mapred 13-Apr-18 0:03 Windows Comma... 6 KB
winutils 13-Apr-18 0:03 Application 110 KB
winutils.pdb 13-Apr-18 0:03 Program Debug D... 1,172 KB
yarn 13-Apr-18 0:03 Windows Comma... 12 KB
15 items

```
C:\Users\admin>hadoop version
Hadoop 3.1.0
Source code repository https://github.com/apache/hadoop -r 16b70619a24cdcf5d3b0fcf4b58ca77238ccbe6d
Compiled by centos on 2018-03-30T00:00Z
Compiled with protoc 2.5.0
From source with checksum 14182d20c972b3e2105580a1ad6990
This command was run using /C:/hadoop-3.1.0/hadoop-3.1.0/share/hadoop/common/hadoop-common-3.1.0.jar
```



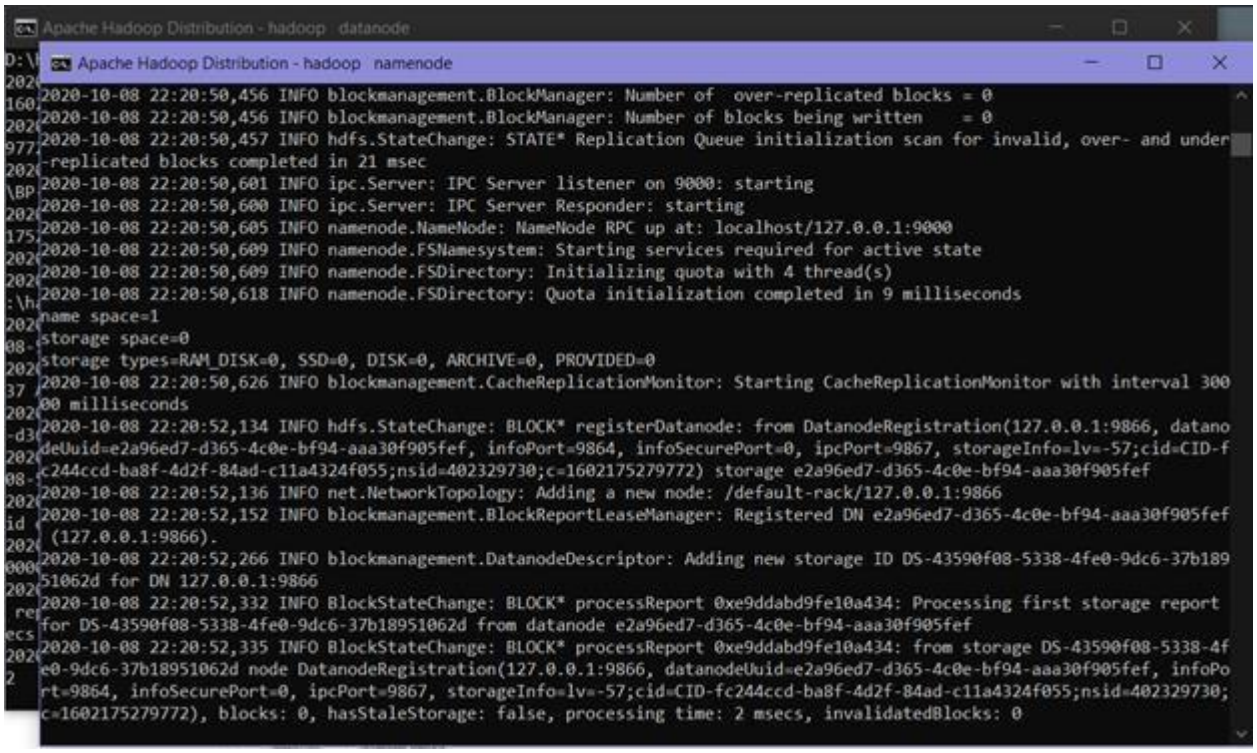

yarn
File
Home
Share
View
hadoop › yarn
Search yarn
hadoop
Mobile p
OneDrive
This PC
3D Objec
Desktop
Documer
Downloa
Music
Pictures
Videos
OS (C:)
eSuppc
hadoop
Java
MagicP
Microsc
Prograr
Prograr
tmp
Users
Windo
DOWNLC
Name
Date modified
lib 08-Oct-20 21:33
sources 08-Oct-20 21:33
test 08-Oct-20 21:33
timelineservice 08-Oct-20 21:33
webapps 08-Oct-20 21:33
yarn-service-examples 08-Oct-20 21:33
hadoop-yarn-api-3.1.0 30-Mar-18 5:43
hadoop-yarn-applications-distributedshe... 30-Mar-18 5:43
hadoop-yarn-applications-unmanaged-a... 30-Mar-18 5:43
hadoop-yarn-client-3.1.0 30-Mar-18 5:43
hadoop-yarn-common-3.1.0 30-Mar-18 5:43
hadoop-yarn-registry-3.1.0 30-Mar-18 5:43
hadoop-yarn-server-applicationhistoryse... 30-Mar-18 5:43
hadoop-yarn-server-common-3.1.0 30-Mar-18 5:43
hadoop-yarn-server-nodemanager-3.1.0 30-Mar-18 5:43
hadoop-yarn-server-resourcemanager-3.... 30-Mar-18 5:43
hadoop-yarn-server-router-3.1.0 30-Mar-18 5:43
hadoop-yarn-server-sharedcachemanage... 30-Mar-18 5:43
hadoop-yarn-server-tests-3.1.0 30-Mar-18 5:43
hadoop-yarn-server-timeline-pluginstora... 30-Mar-18 5:43
hadoop-yarn-server-web-proxy-3.1.0 30-Mar-18 5:43
hadoop-yarn-services-api-3.1.0 30-Mar-18 5:43
hadoop-yarn-services-core-3.1.0 30-Mar-18 5:43
23 items | 1 item selected |



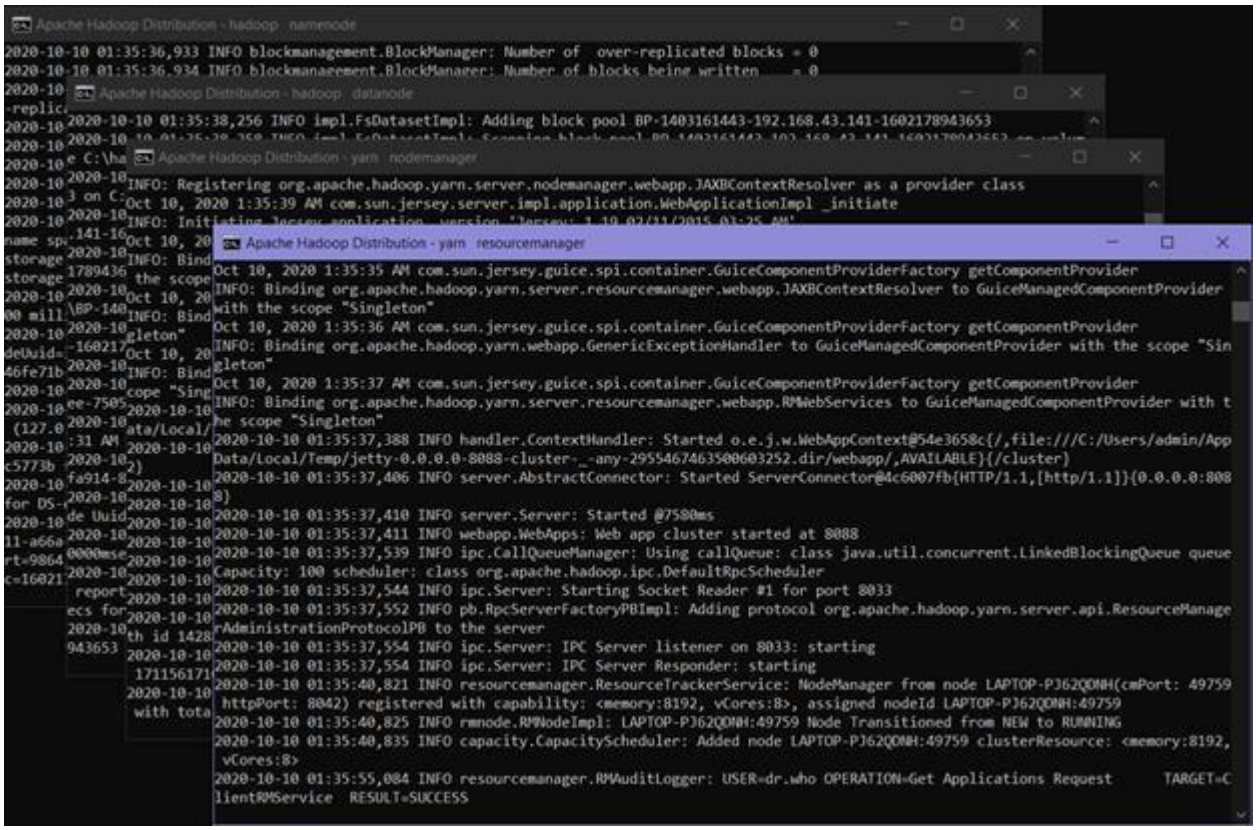

```
C:\Users\admin>cd C:\hadoop-3.1.0\hadoop-3.1.0\sbin

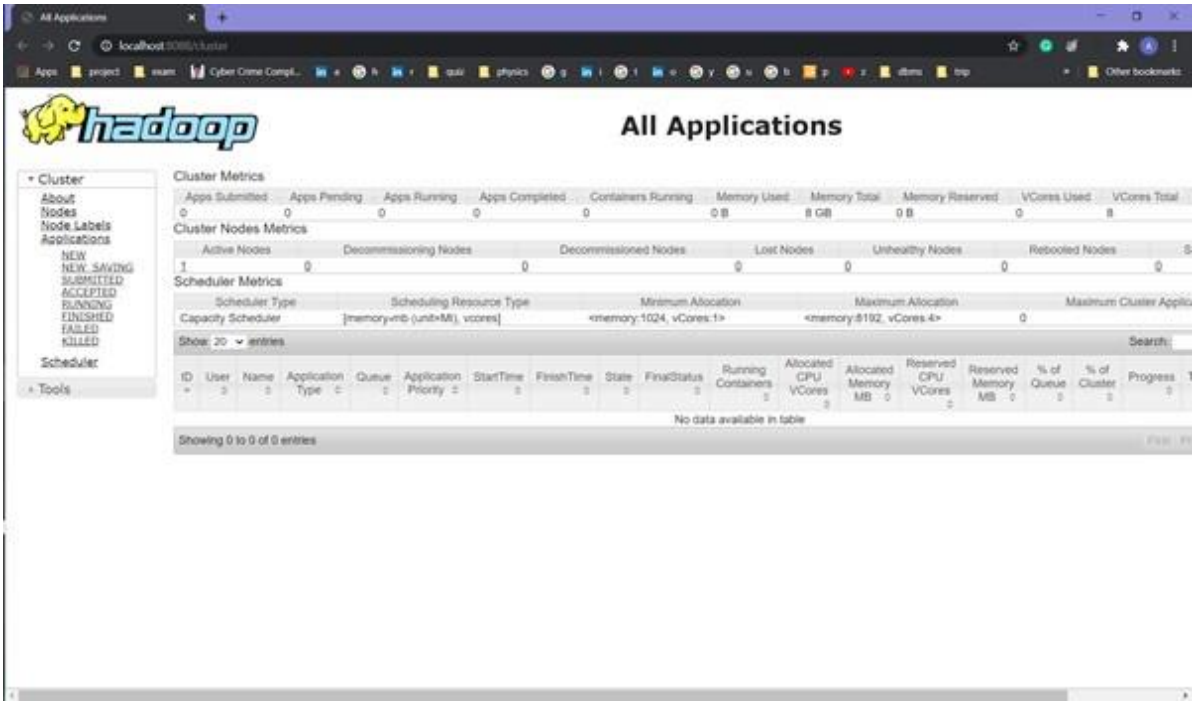
C:\hadoop-3.1.0\hadoop-3.1.0\sbin>
```

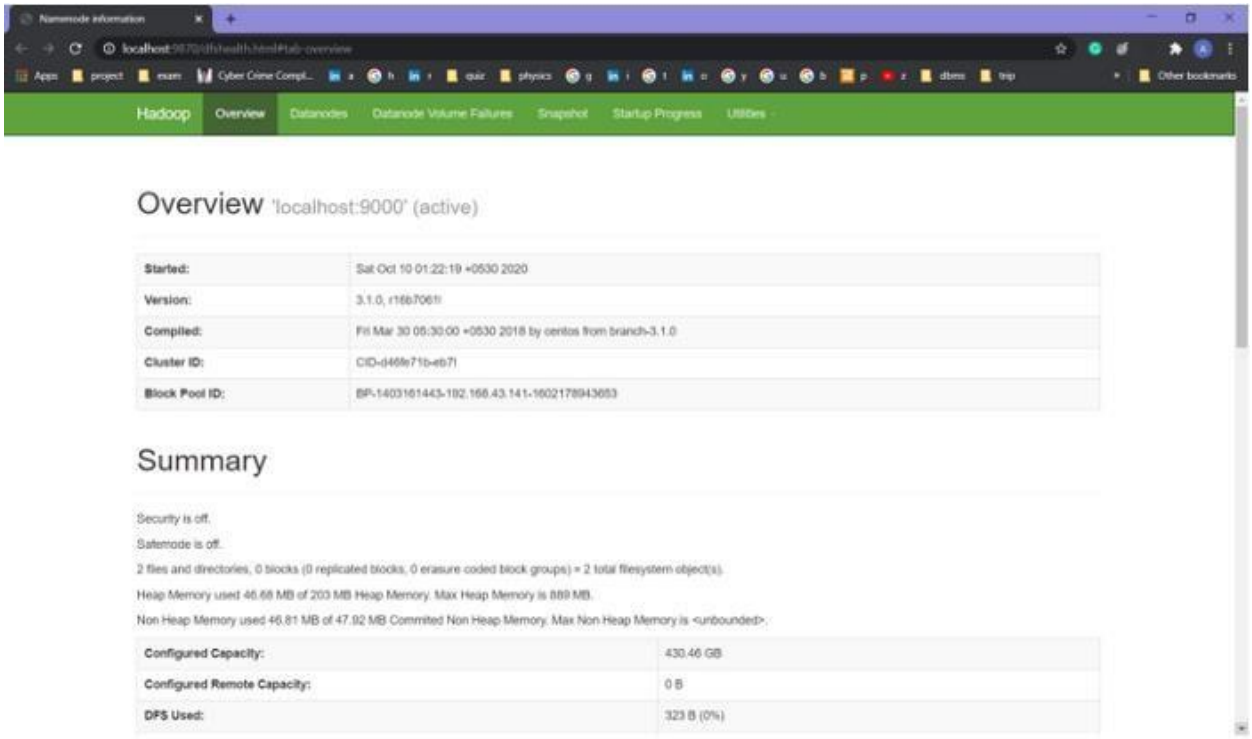
```
C:\hadoop-3.1.0\hadoop-3.1.0\sbin>start-dfs.cmd
```

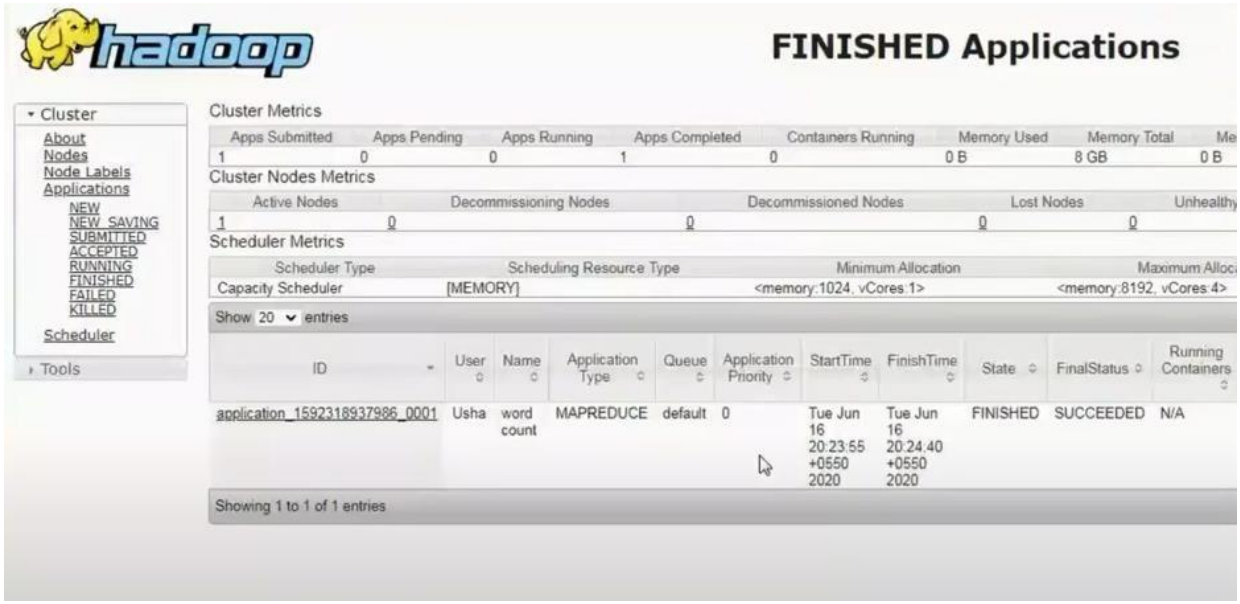


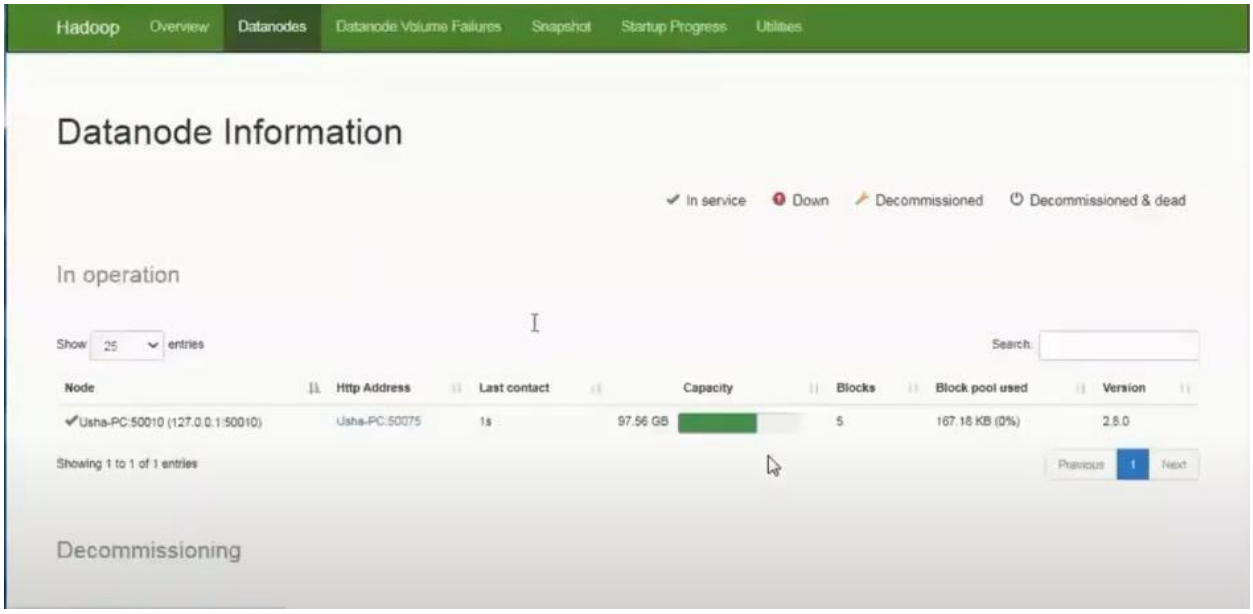
```
C:\hadoop-3.1.0\hadoop-3.1.0\sbin>start-yarn.cmd
starting yarn daemons
```






Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Overview 'localhost:9000' (active)
Started: Sat Oct 10 01:22:19 +0530 2020
Version: 3.1.0, r16b7061f
Compiled: Fri Mar 30 05:30:00 +0530 2018 by centos from branch-3.1.0
Cluster ID: CID-d468e71b-eb7f
Block Pool ID: BP-1403161443-192.168.43.141-1602178943653
Summary
Security is off.
Safemode is off.
2 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 2 total filesystem object(s).
Heap Memory used 46.68 MB of 203 MB Heap Memory. Max Heap Memory is 889 MB.
Non Heap Memory used 46.81 MB of 47.92 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.
Configured Capacity: 430.46 GB
Configured Remote Capacity: 0 B
DFS Used: 323 B (0%)


hadoop
FINISHED Applications
Cluster
About
Nodes
Node Labels
Applications
NEW
NEW_SAVING
SUBMITTED
ACCEPTED
RUNNING
FINISHED
FAILED
KILLED
Scheduler
Tools
Cluster Metrics
Apps Submitted 1
Apps Pending 0
Apps Running 0
Apps Completed 1
Containers Running 0
Memory Used 0 B
Memory Total 8 GB
Cluster Nodes Metrics
Active Nodes 1
Decommissioning Nodes 0
Decommissioned Nodes 0
Lost Nodes 0
Unhealthy 0
Scheduler Metrics
Scheduler Type: Capacity Scheduler
Scheduling Resource Type: [MEMORY]
Minimum Allocation: <memory:1024, vCores:1>
Maximum Alloc: <memory:8192, vCores:4>
Show 20 entries
ID: application_1592318937986_0001
User: Usha
Name: word count
Application Type: MAPREDUCE
Queue: default
Application Priority: 0
StartTime: Tue Jun 16 20:23:55 +0550 2020
FinishTime: Tue Jun 16 20:24:40 +0550 2020
State: FINISHED
FinalStatus: SUCCEEDED
Running Containers: N/A
Showing 1 to 1 of 1 entries




```
C:\>hadoop dfs -cat /output_dir/*
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
cat     7
deer    8
dog     4
elephant        2
lion    12
tiger   9
```

RESULT:

Exp.No.: Date:	CREATING AND EXECUTING YOUR FIRST CONTAINER USING DOCKER

AIM:

To create and executing your first container using docker.

PROCEDURE:

Sure, here's a step-by-step guide on how to create and execute your first container using Docker. I'll provide you with the basic commands you need:

1. **Install Docker**: Make sure you have Docker installed on your system. You can download it from [Docker's official website](https://www.docker.com/get-started).

2. **Verify Installation**: To confirm that Docker is installed and running, open a terminal and run:

```
docker --version
```

You should see the Docker version information.

3. **Pull an Image**: You'll need an image to create a container from. You can pull a popular Linux distribution like Ubuntu with this command:

```
docker pull ubuntu
```

4. **Create a Container**: Now, let's create a container from the Ubuntu image. Replace `your_container_name` with a name for your container:

```
docker run -it --name your_container_name ubuntu
```

- `-it` allows you to interact with the container in the terminal.
- `--name` assigns a name to your container.
- `ubuntu` is the image you want to use.

5. **Inside the Container**: You are now inside the container. You can run commands within it. For example:

```
apt update
```

```
apt install <package_name>
```

6. **Exit the Container**: When you're done, exit the container:

```
exit
```

7. **List Containers**: To see the list of containers (including the stopped ones), use:

```
docker ps -a
```

8. **Start and Stop Containers**: You can start and stop containers using the container names or IDs:

```
docker start your_container_name
docker stop your_container_name
```

9. **Remove a Container**: If you want to remove a container, use:

```
docker rm your_container_name
```

Remember to replace `your_container_name` with a name of your choice. These are the basic steps to get started with Docker containers. You can explore Docker further and create your own Dockerfiles to build custom containers.

RESULT:

EXP.NO: 9 DATE:	RUN A CONTAINER FROM DOCKER HUB

AIM:

To wrire a program to run a container from docker hub.

PROCEDURE:

```
import subprocess

def run_docker_container(image_name):
    try:
        command = f"docker run {image_name}"
        subprocess.run(command, shell=True, check=True)
    except subprocess.CalledProcessError as e:
        print(f"Error: {e}")
    except Exception as e:
        print(f"An error occurred: {e}")

if __name__ == "__main__":
    image_name = input("Enter the Docker image name from Docker Hub: ")
    run_docker_container(image_name)
```

RESULT: